

Genomics data analysis 基因组学数据分析

RNA-seq转录组分析

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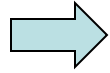
贺圆圆 <heyuanyuan@stu.pku.edu.cn>



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本节内容

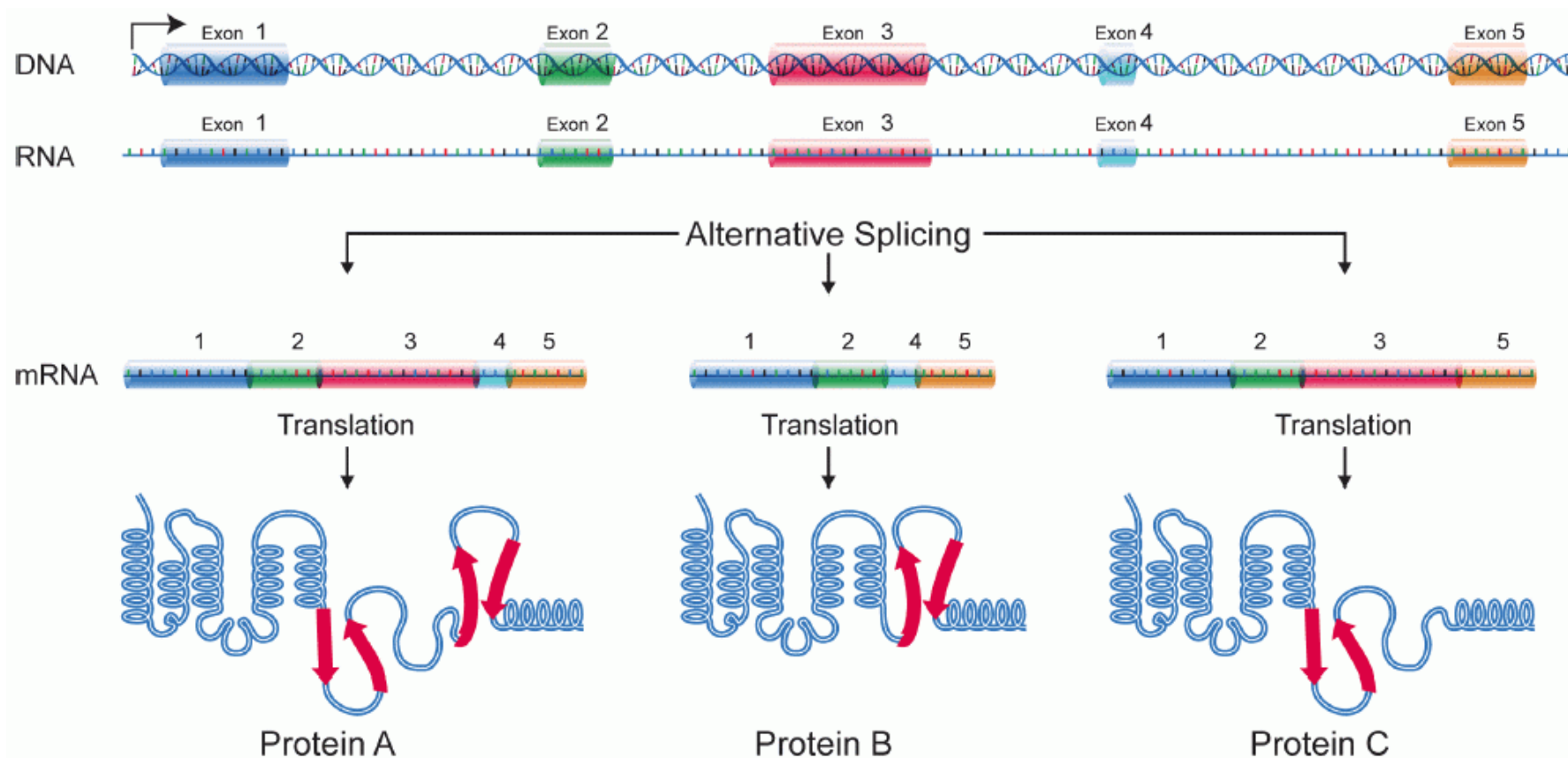


- RNA-seq测序技术
- RNA-seq分析流程
- RNA-seq分析举例
- 统计**模型**：数据归一化
- **助教演示**：Linux系统、RNA-seq分析

生物新技术驱动生物学研究



测序技术：检测基因表达量、序列变异



Source: Wikipedia (http://en.wikipedia.org/wiki/Alternative_splicing)

基因组诺奖

THE NOBEL PRIZE
IN PHYSIOLOGY OR MEDICINE 2022



Svante Pääbo

"for his discoveries concerning the genomes
of extinct hominins and human evolution"

THE NOBEL ASSEMBLY AT KAROLINSKA INSTITUTET

对已灭绝人种的基因组
和人类进化的发现

THE NOBEL PRIZE
IN CHEMISTRY 2020



Emmanuelle
Charpentier

Jennifer A.
Doudna

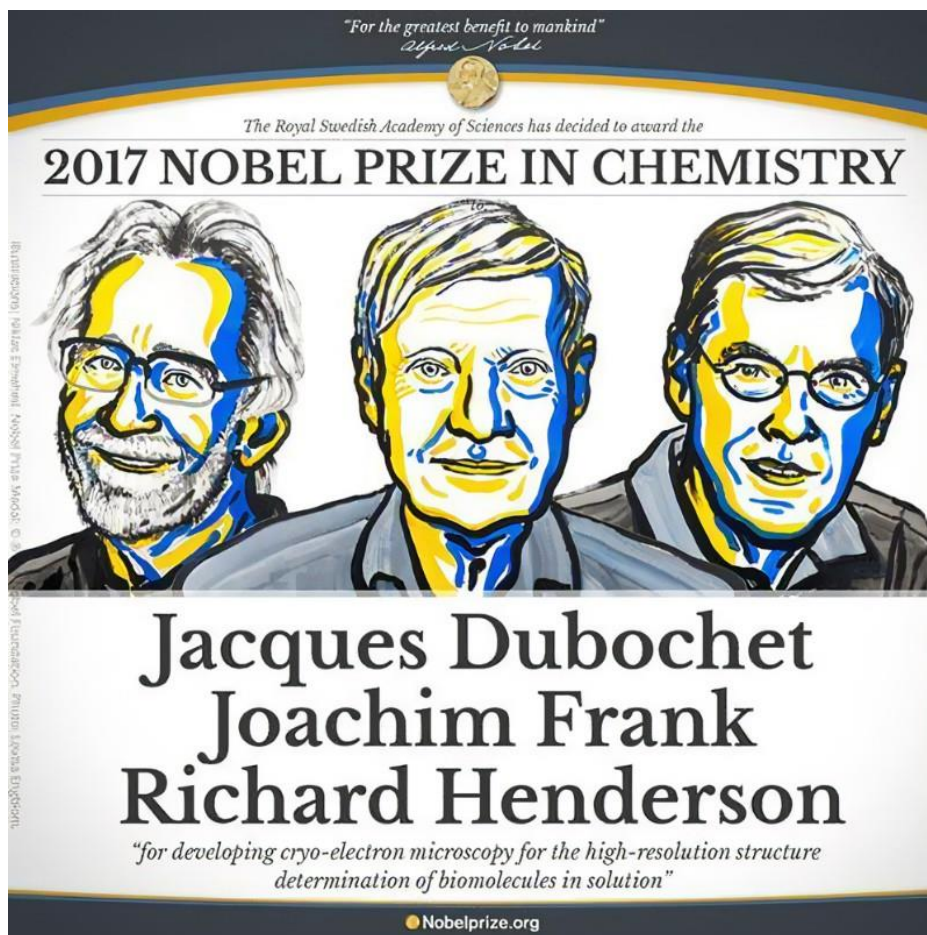
"for the development of a method
for genome editing"

发展基因组编辑方法

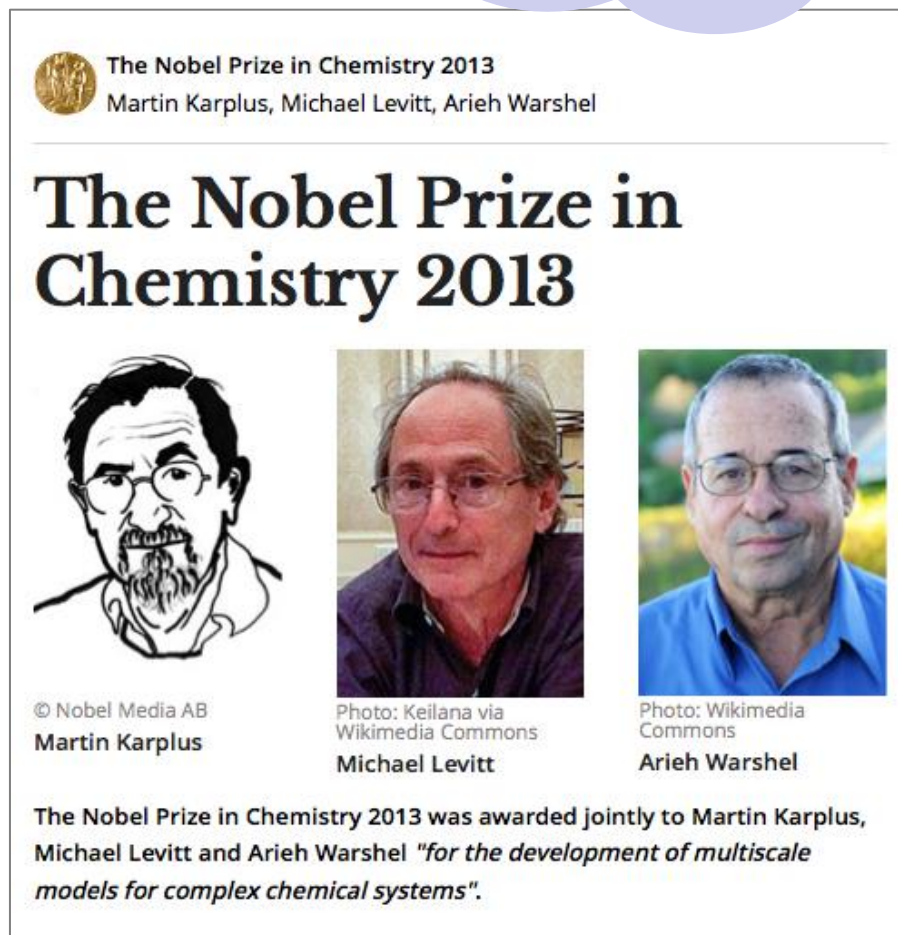
“这场电子显微学分辨率革命的突破有两个关键技术：直接电子相机和三维重构软件。”

算法、建模诺奖

“对于今天的化学家来说，计算机就像试管一样重要。模拟过程是如此的真实，以至于传统实验的结果也能被计算机预测出来。”

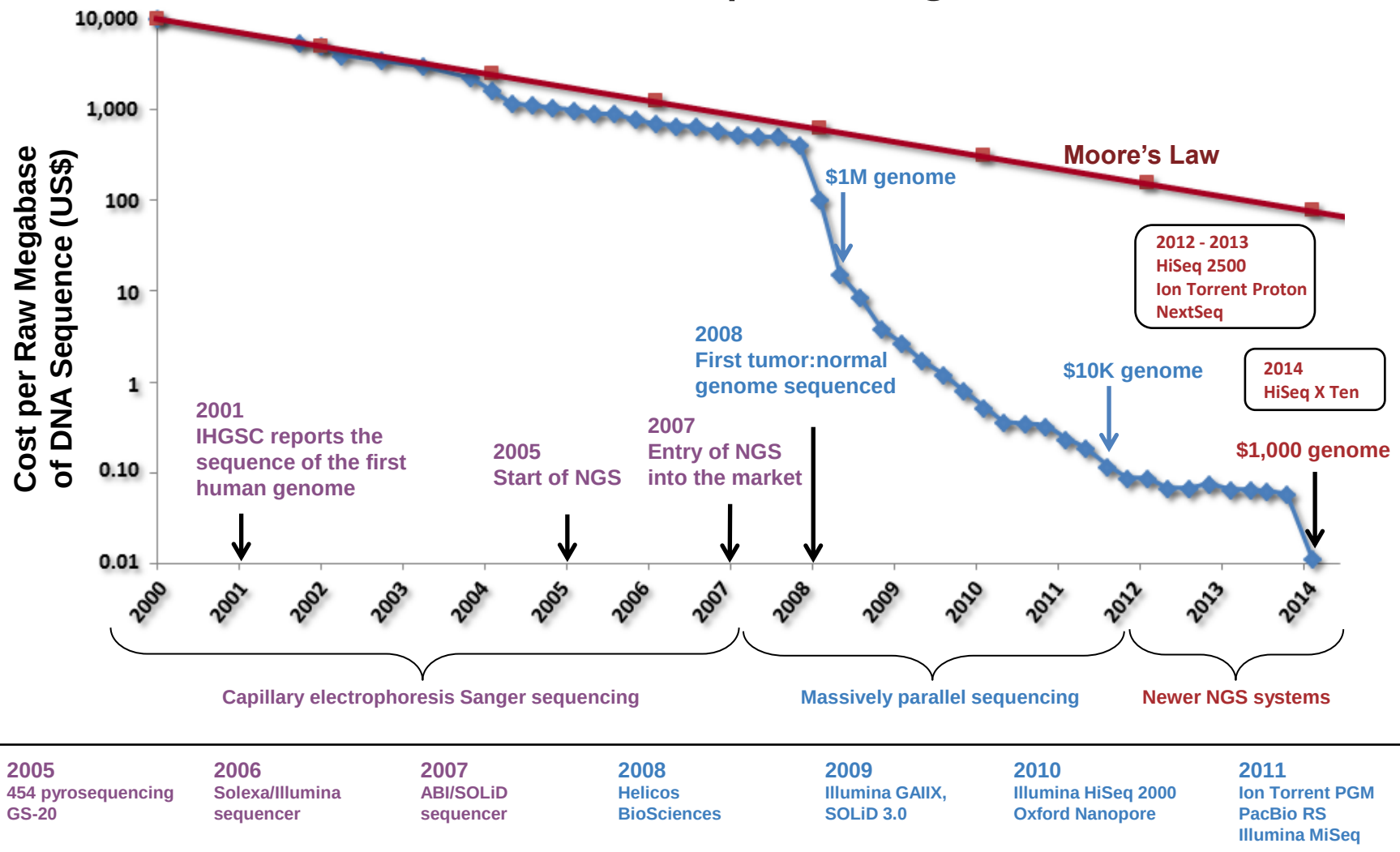


发展冷冻电镜解析生物分子的高分辨结构



发展复杂化学体系的多尺度模型

NGS (next generation sequencing) drives down the cost of sequencing



MacConaill LE. Existing and emerging technologies for tumor genomic profiling.
Journal of Clinical Oncology, 31(15), 1815-1824 (2013). (Source: Erick Lin)

华大测序仪：数据生产

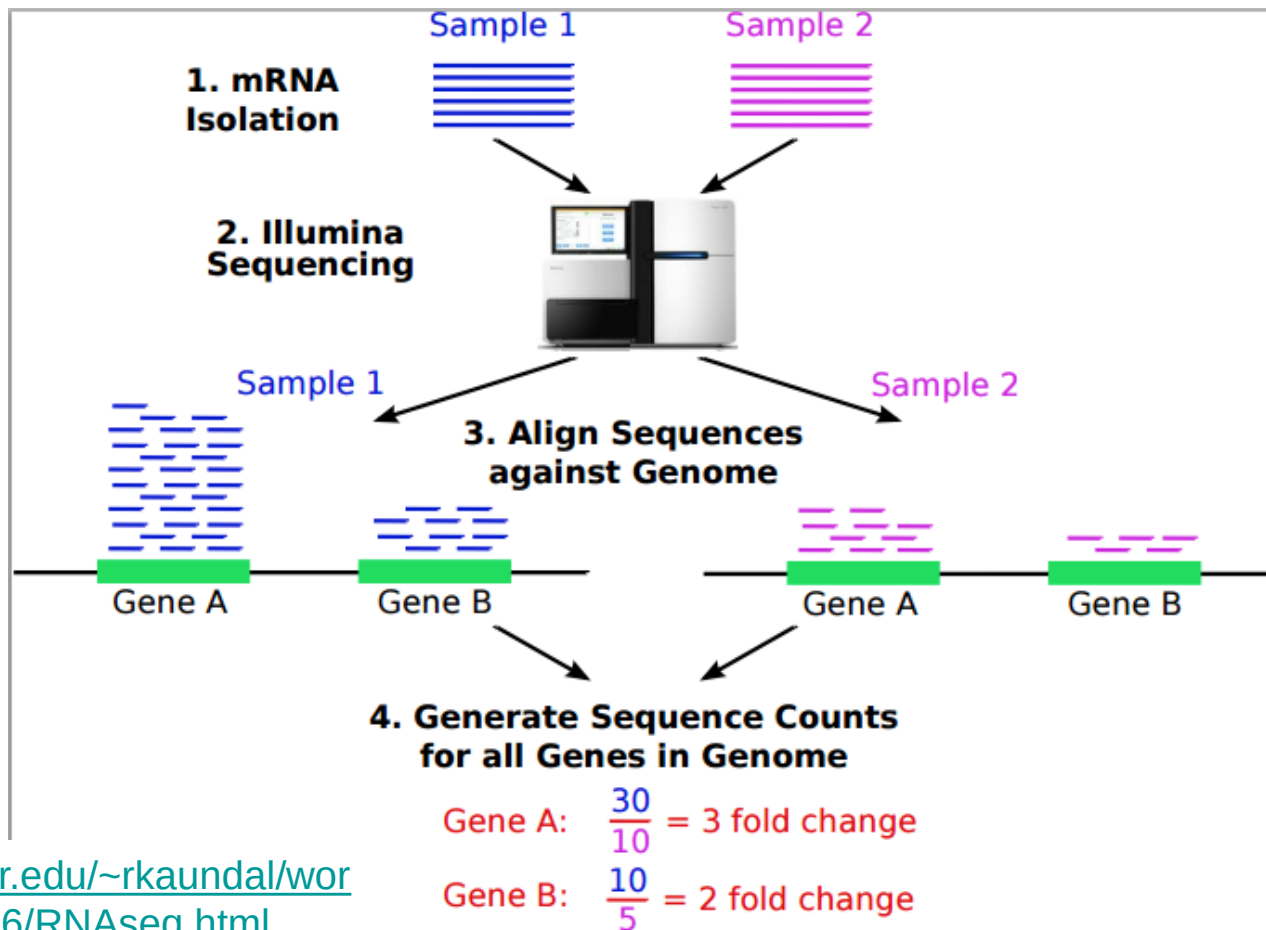
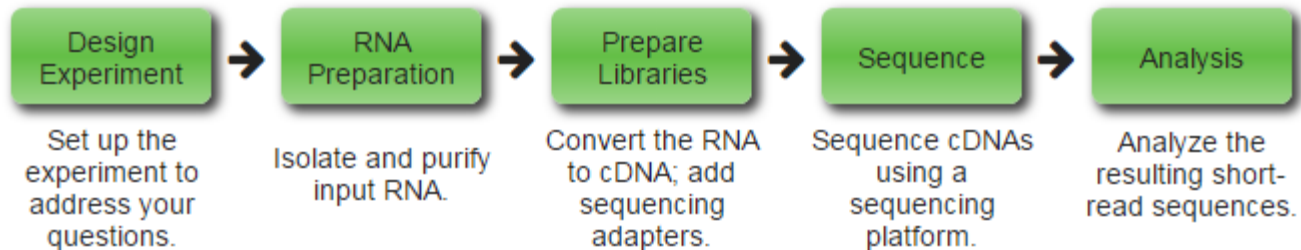


- 与DNBSEQ-T7的超高通量相匹配，华大智造还提供完整的测序前处理和测序后分析平台，支持超高通量的数据持续产出。
- 在这套解决方法中，无论是核酸提取、文库制备，还是下机数据的加速分析，均适配可选的自动化设备，以全方位、最大程度地释放DNBSEQ-T7强大的数据生产能力。
- 华大智造将推动全球基因测序行业大幅降低测序成本，进入百美元全基因组时代的后半程冲刺。

DNBSEQ-T7性能表现			
读长	数据产出	数据质量Q30 *	运行时间 **
PE100	1~4Tb	>85%	20-22h
PE150	1.5~6Tb	>80%	24-30h

<https://www.mgi-tech.com/news/18/>

Overview of RNA-seq experiment



RNA-seq可以回答的问题

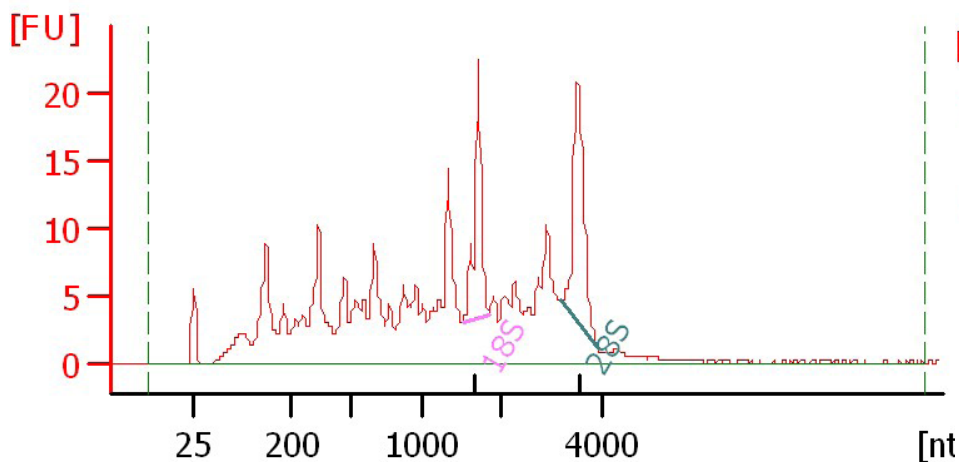
- 正常细胞和癌细胞中基因表达水平有什么差异？
- 在缺少一种抑癌基因的细胞系中，基因表达水平会发生什么变化？
- 诱变剂处理前后，在细胞系中基因表达有什么差异？
- 在大脑发育过程中，哪些基因被上调？
- 什么转录本存在于皮肤，但不存在于肌肉中？
- 在氧化应激过程中，基因剪接是如何改变的？
- 在人类胚胎干细胞样本中，我们能发现什么新的miRNA？

RNA分离

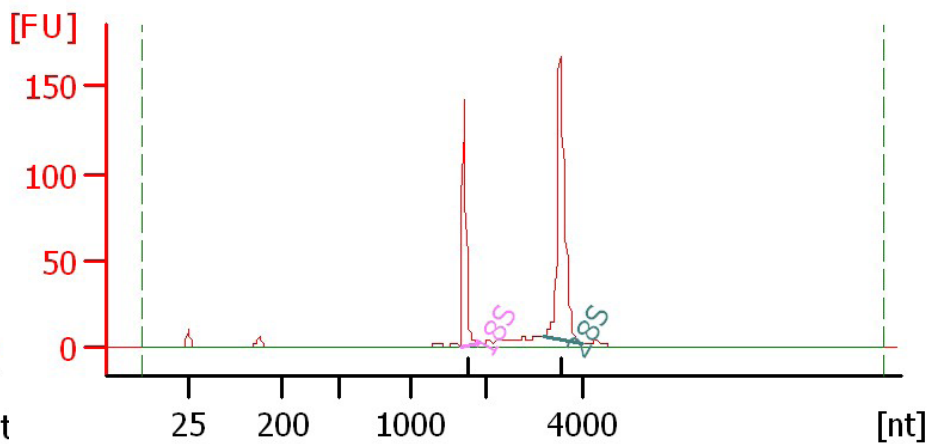
- **分离**：通常从新鲜或冷冻的细胞或组织样本中分离，使用商业试剂盒，如RNAEasy、TRIZOL、RiboPure
 - 容易使用，可产生大量的总RNA
 - 制备文库前，通常用DNA酶（DNase）处理分离的总RNA样品，以消化污染的DNA
 - 标准建库需要0.1~10 μg 总RNA，高灵敏度流程可使用少至 10 pg的RNA产生文库
- 生物体中总RNA = ($\sim 90\%$) rRNA + ($1 \sim 2\%$) mRNA+ ($8 \sim 9\%$) 其他RNA
 - 对于真核生物，可通过oligo dT磁珠富集mRNA来进行建库
 - 如果想研究lncRNA、全转录组或原核生物转录组，可以通过去除rRNA的方法进行建库

RNA质控

- **质控**：用Agilent Bioanalyzer测量样品中RNA的长度和丰度，提供关于浓度和核酸质量的信息

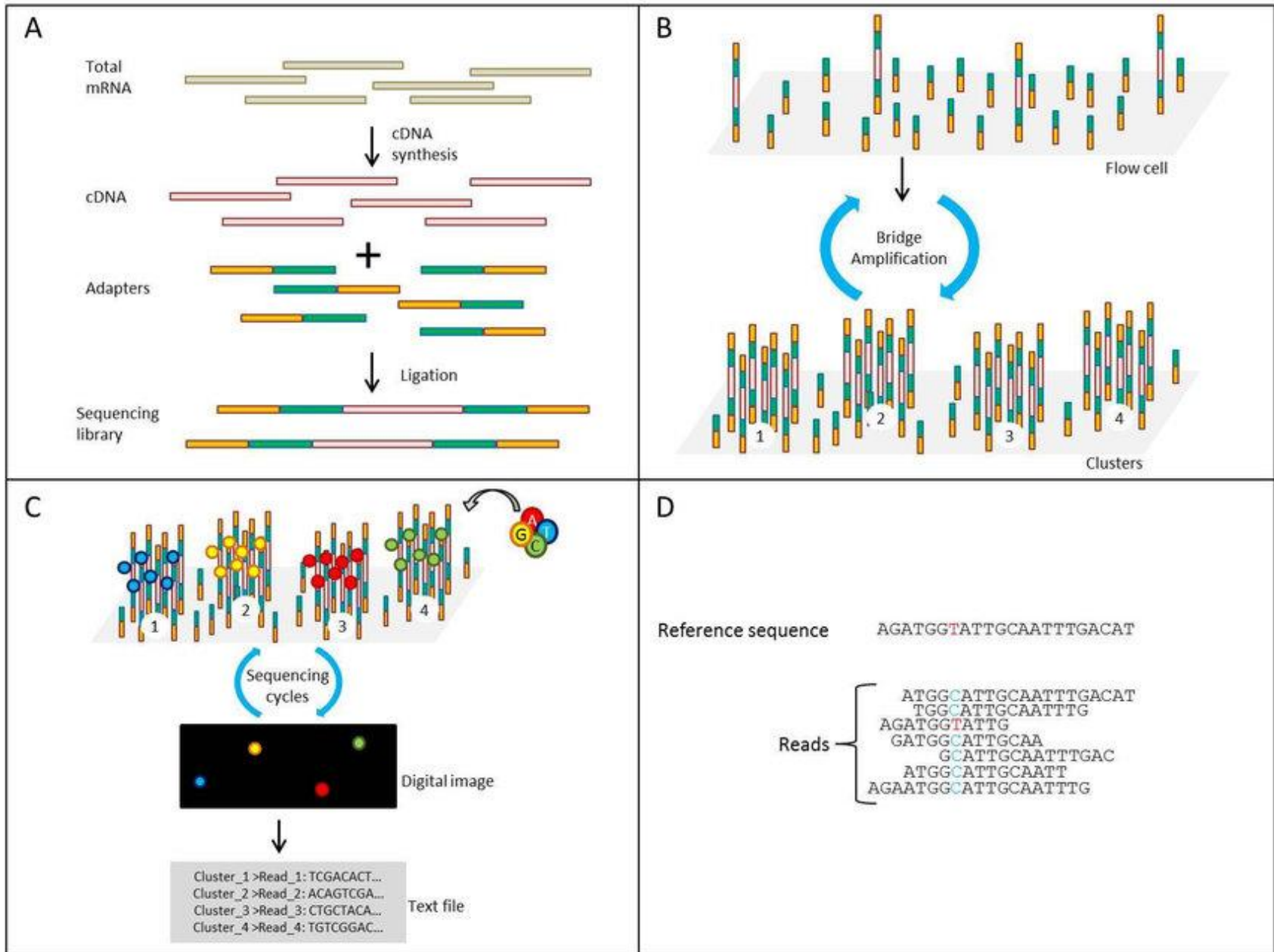


降解的RNA在低分子质量处显示为曳尾 (smear)

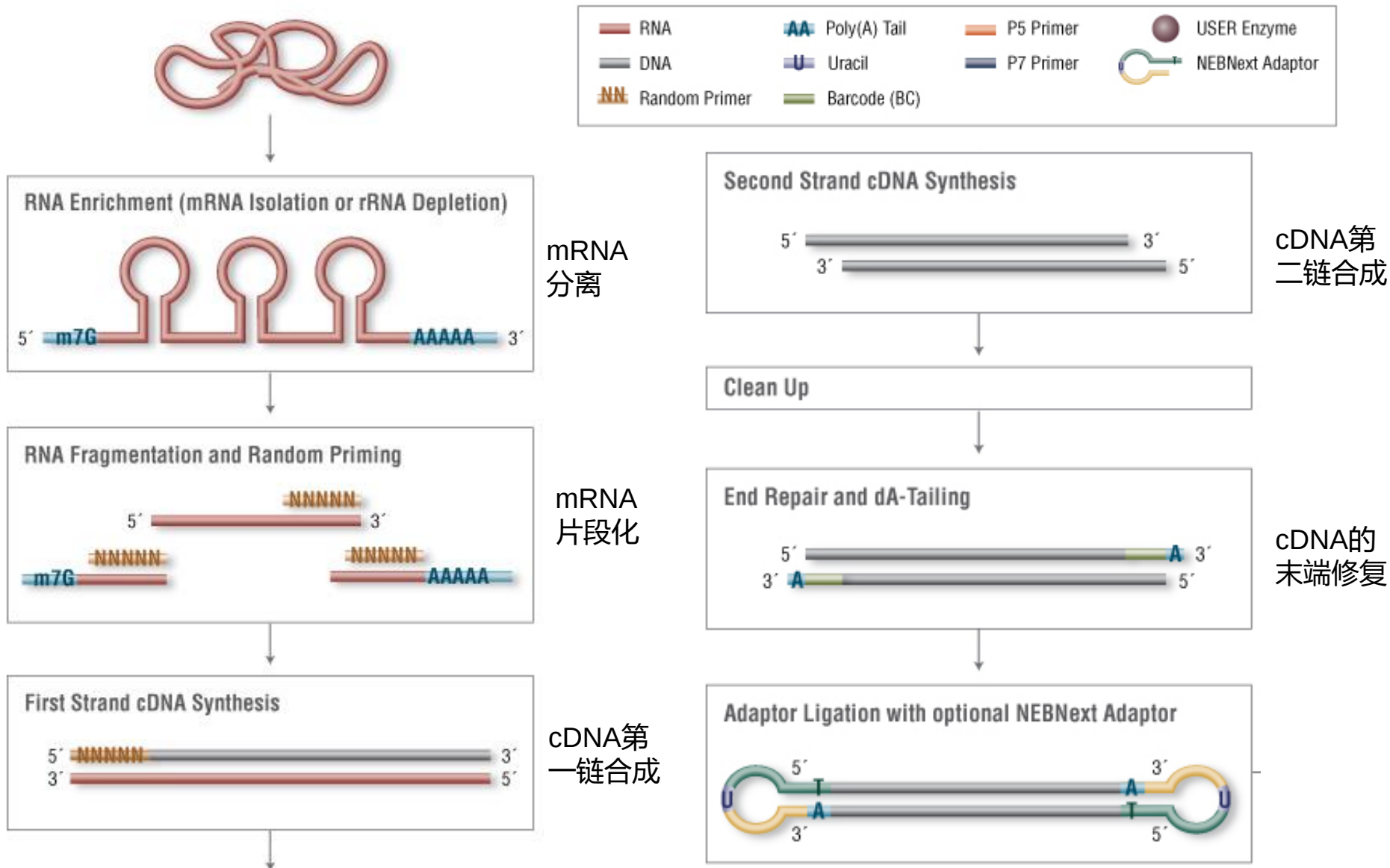


完整的总RNA显示陡峭的28S和18S rRNA峰

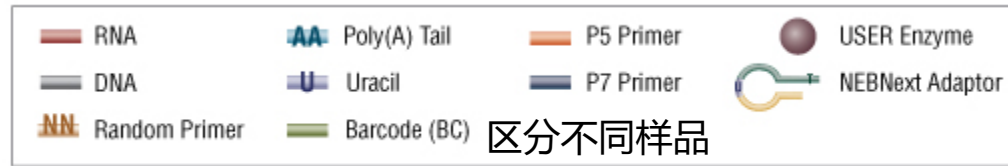
Overview of Illumina RNA-seq experiments



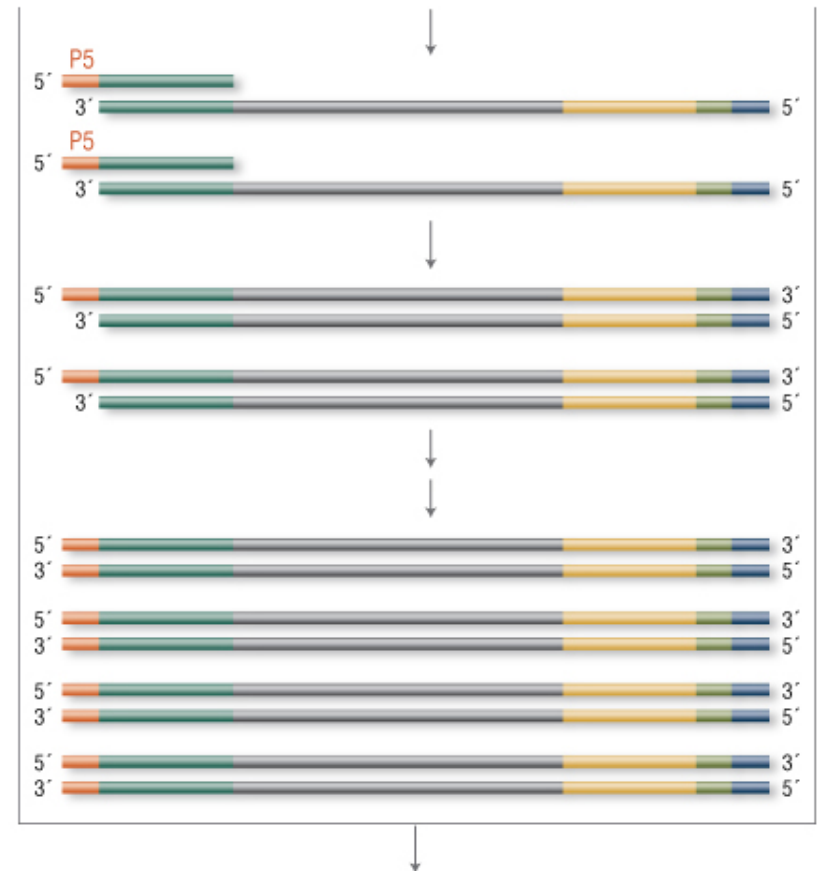
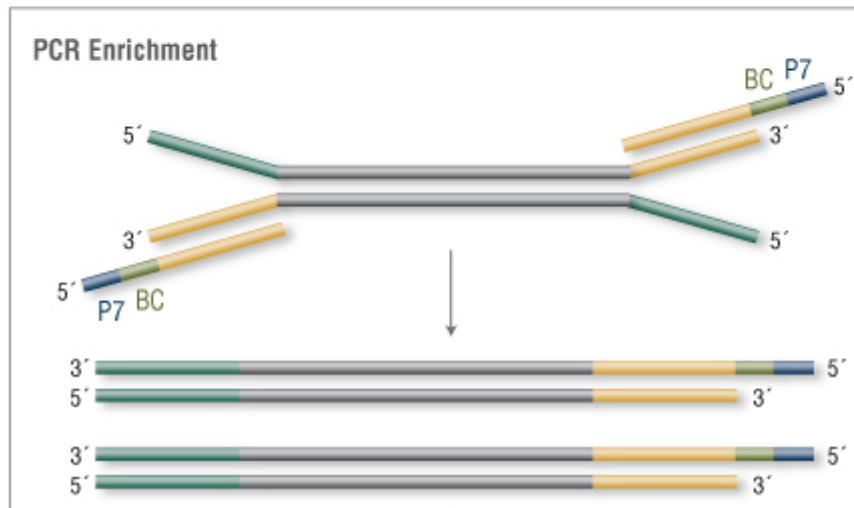
RNA-seq文库制备 (1)



RNA-seq文库制备 (2)

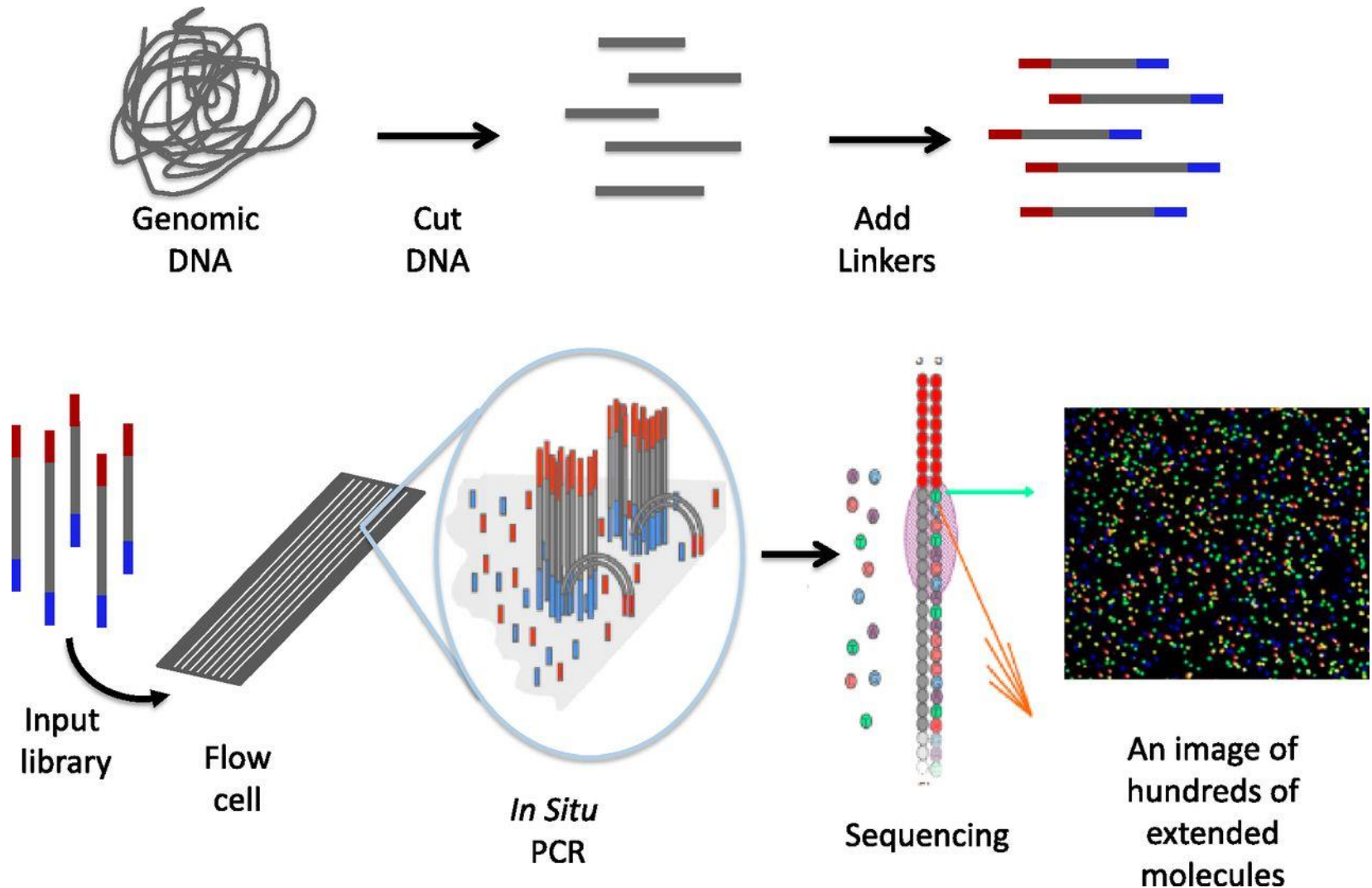


Clean Up/Size Selection

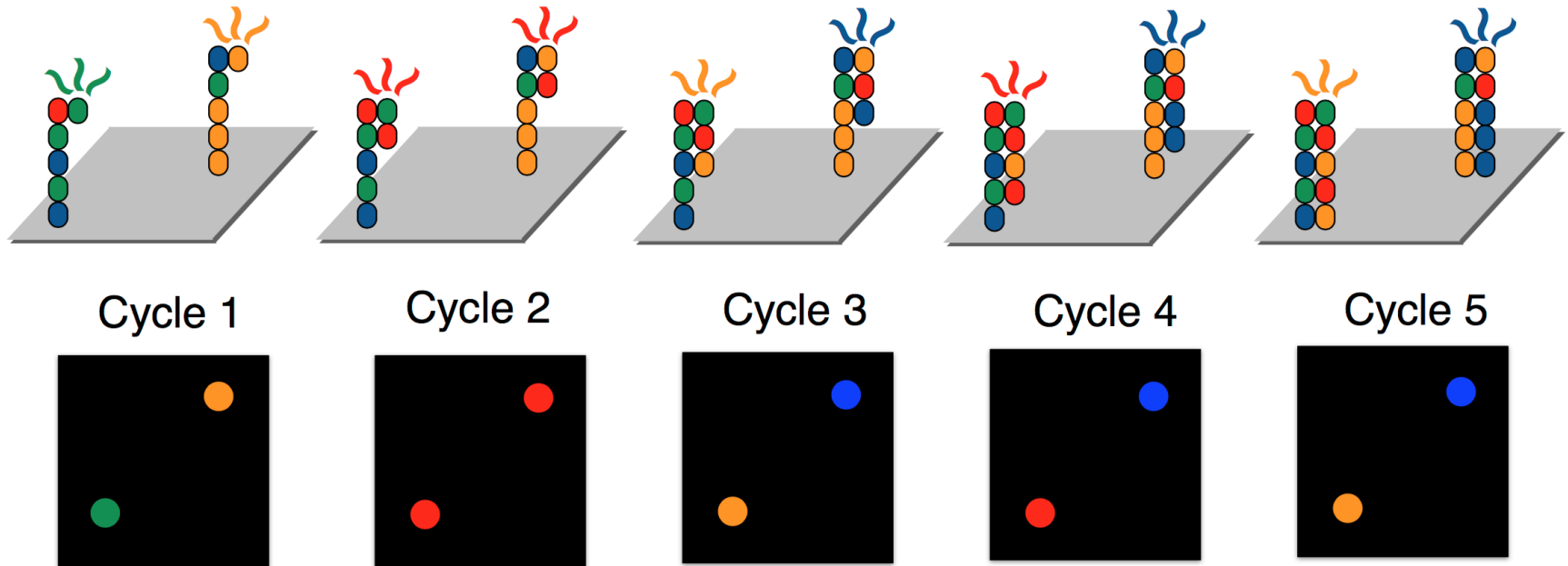


Clean Up

Illumina Sequencing by Synthesis



Sequencing by Synthesis

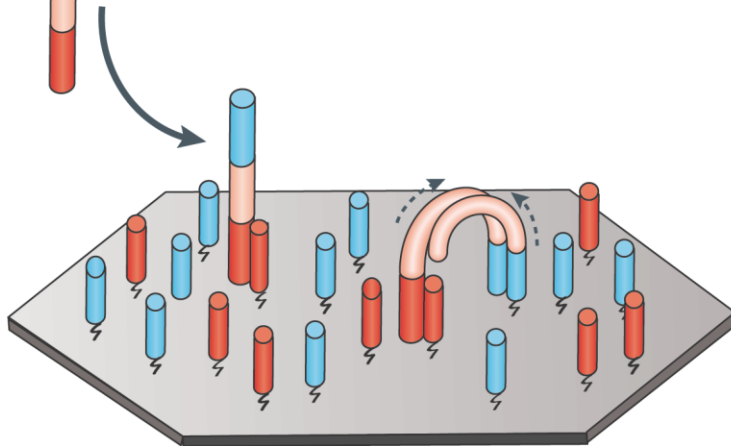


Template amplification

b Solid-phase bridge amplification (Illumina)

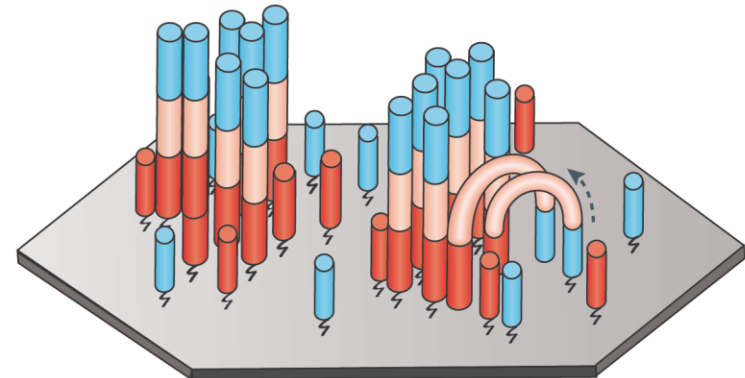
Template binding

Free templates hybridize with slide-bound adapters



Bridge amplification

Distal ends of hybridized templates interact with nearby primers where amplification can take place

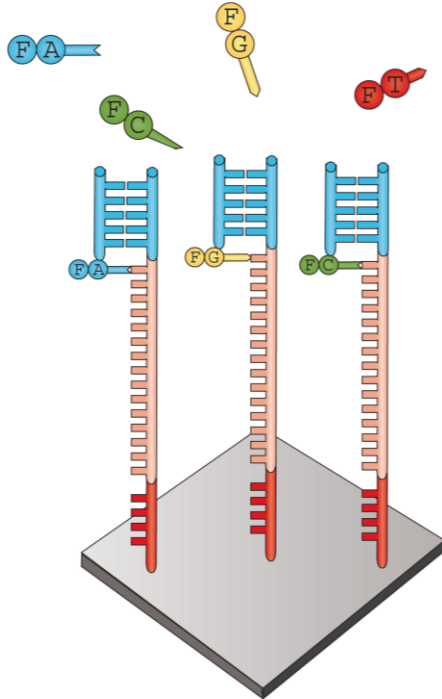


Cluster generation

After several rounds of amplification, 100–200 million clonal clusters are formed

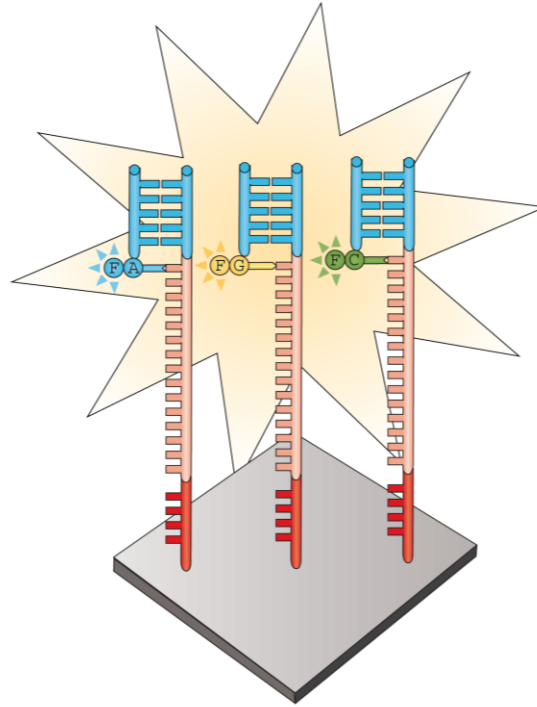
Cyclic reversible termination

Illumina



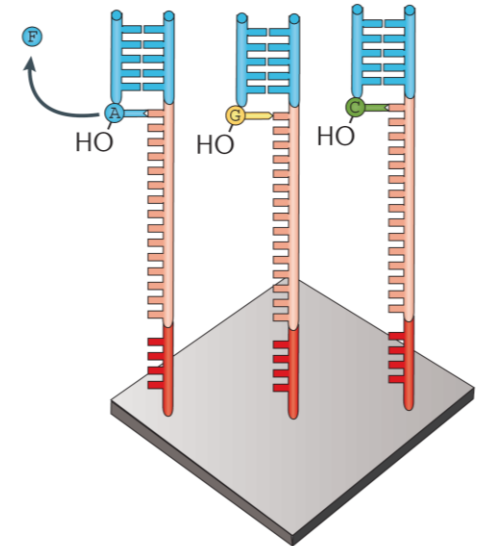
Nucleotide addition

Fluorophore-labelled, terminally blocked nucleotides hybridize to complementary base. Each cluster on a slide can incorporate a different base.



Imaging

Slides are imaged with either two or four laser channels. Each cluster emits a colour corresponding to the base incorporated during this cycle.



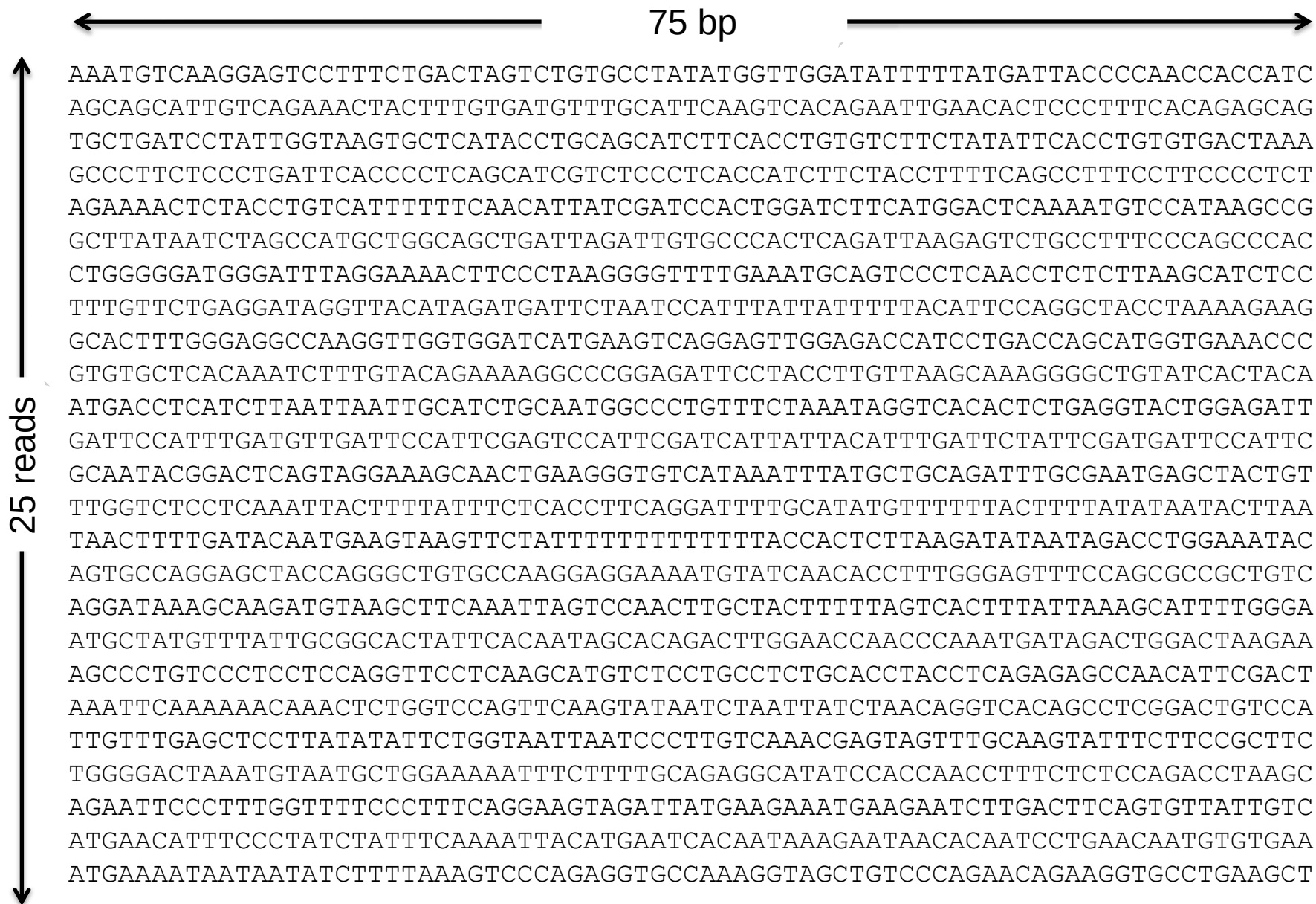
Cleavage

Fluorophores are cleaved and washed from flow cells and the 3'-OH group is regenerated. A new cycle begins with the addition of new nucleotides.

视频介绍: <https://www.bilibili.com/video/av16817060/>

本节内容

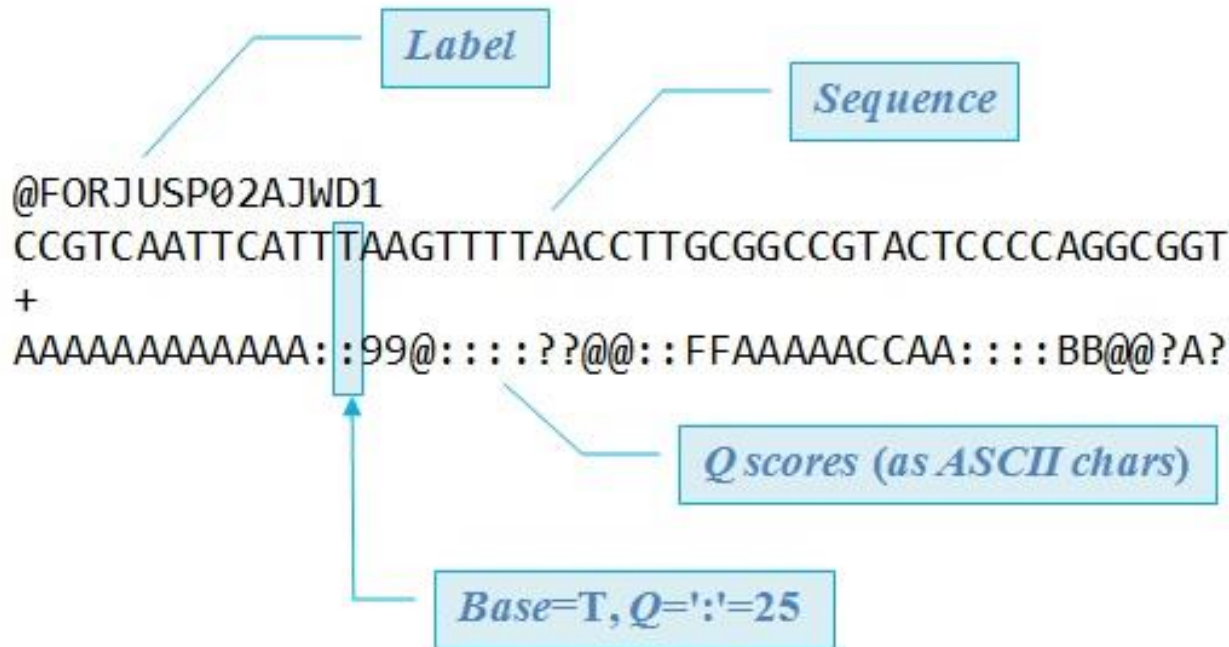
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On an Illumina HiSeq 2500, one lane of an eight-lane flow cell generates >600 million reads (**45 GB**), Source: Trevor Pugh

A FASTQ file containing reads

Sequencing Reads (FASTQ format)



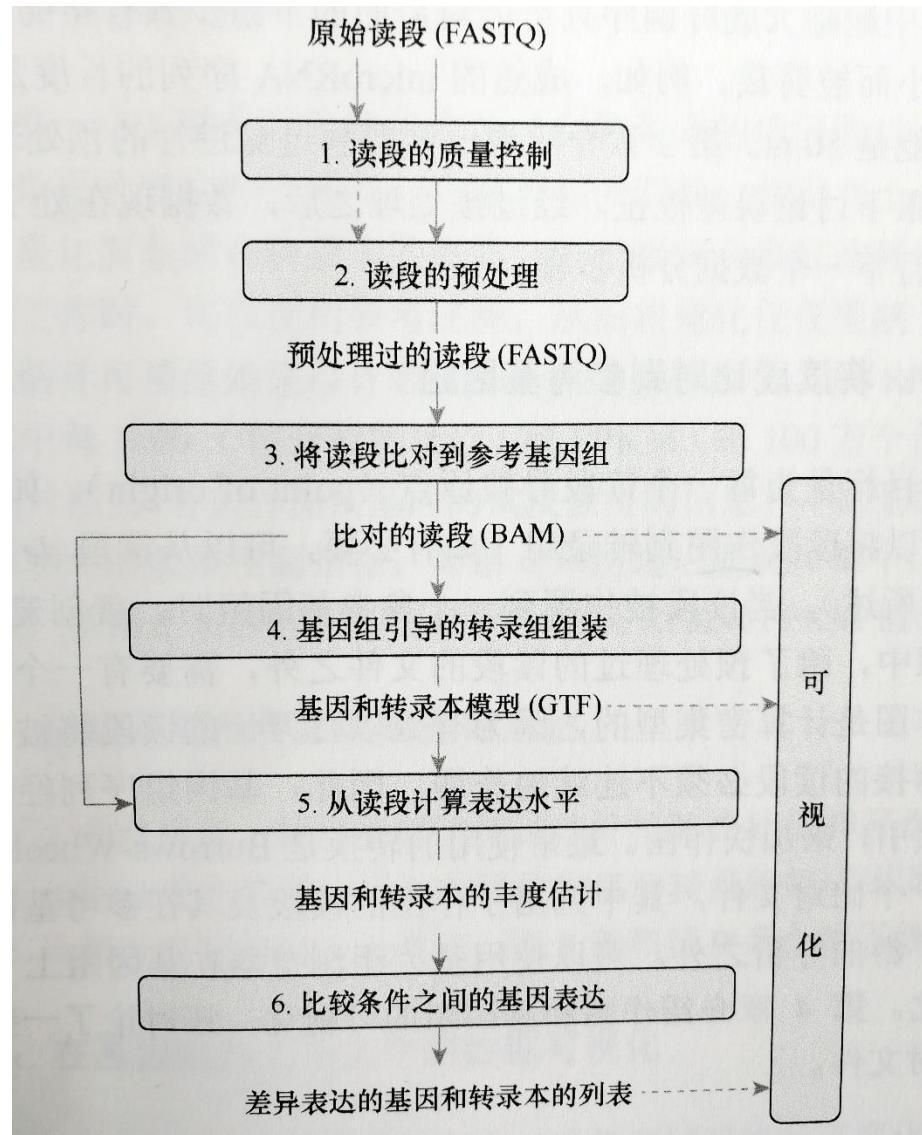
A Phred score of 40 can be represented as the ASCII char "I" (40+33= ASCII #73)

40 means 10^{-4} error probability.

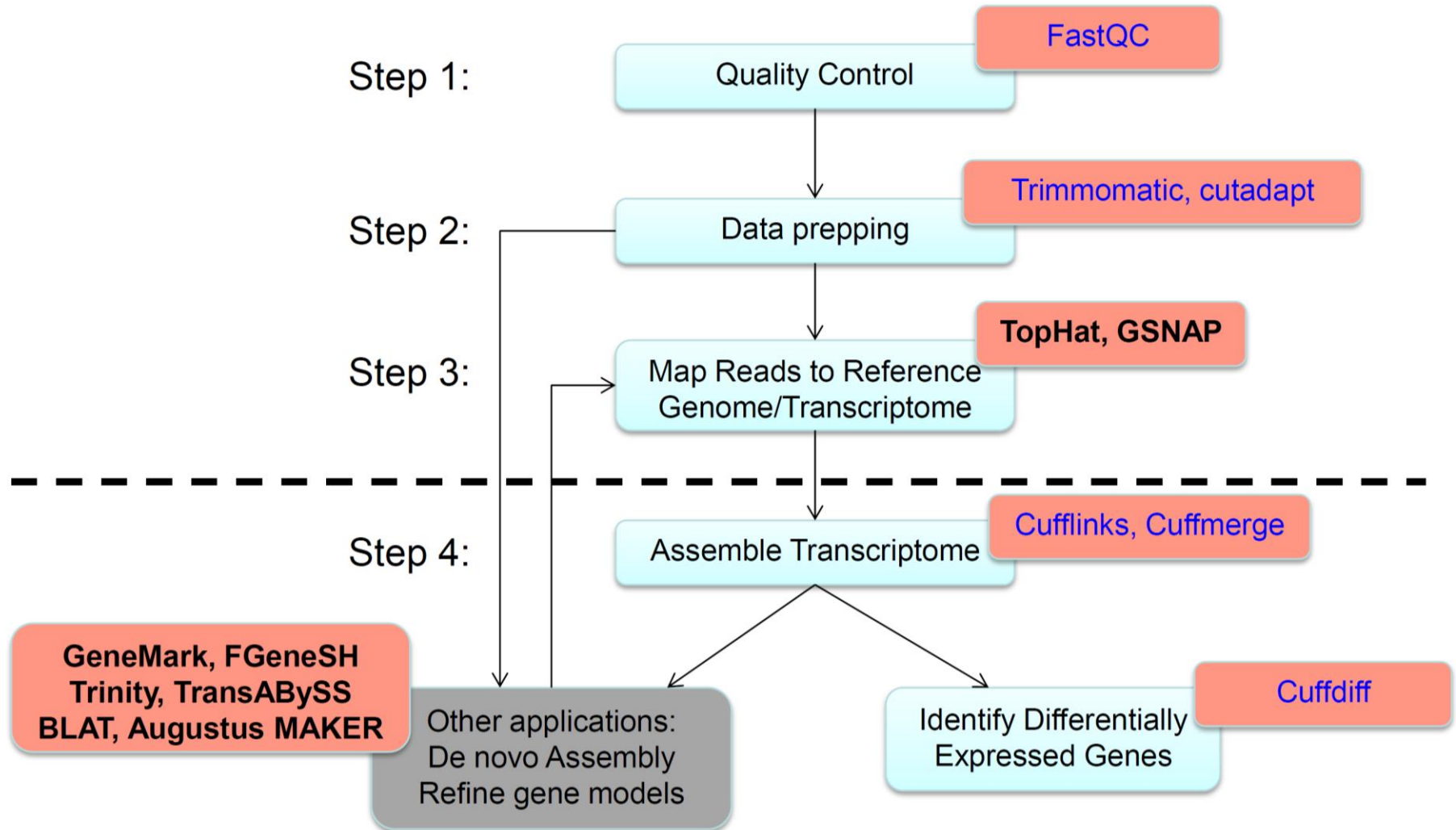
A FASTQ file normally uses four lines per sequence.

- Line 1 begins with a '@' character and is followed by a sequence identifier.
- Line 2 is the **raw sequence letters**.
- Line 3 begins with a '+' character, is optionally followed by the same sequence identifier.
- Line 4 encodes the **Phred quality values** for the sequence in line 2, each value represents the **error probability of a given base call**. (从图像转换为碱基的错误概率)

RNA-seq分析流程



RNA-seq分析软件



读段质量控制

- Evaluate read library quality
 - Identify contaminants
 - Identify poor/bad samples
- Software
 - FastQC (recommended)
 - Command-line, Java GUI, or Galaxy
 - SolexaQC
 - Command-line
 - Supports quality-based read trimming and filtering
 - SAMStat
 - Command-line
 - Also works with bam alignment files
- 对读段进行扫描，寻找可信度低的碱基、有偏差的核苷酸组成、接头、重复等。
- 指导**读段预处理**：删除低质量的碱基、假象（接头、建库序列、polyA尾、微生物）、根据大小裁剪（miRNA长~22nt，读段长度>50nt）

读段质量控制

- Trimming: remove bad bases from (end of) read
 - Adaptor sequence
 - Low quality bases
- Filtering: remove bad reads from library
 - Low quality reads
 - Contaminating sequence
 - Low complexity reads (repeats)
 - Short reads
 - Short (< 20bp) reads slow down mapping software
 - Only needed if trimming was performed
- Software
 - Galaxy, many options (NGS: QC and manipulation)
 - Tagdust
 - Many others: <http://seqanswers.com/wiki/Software/list>

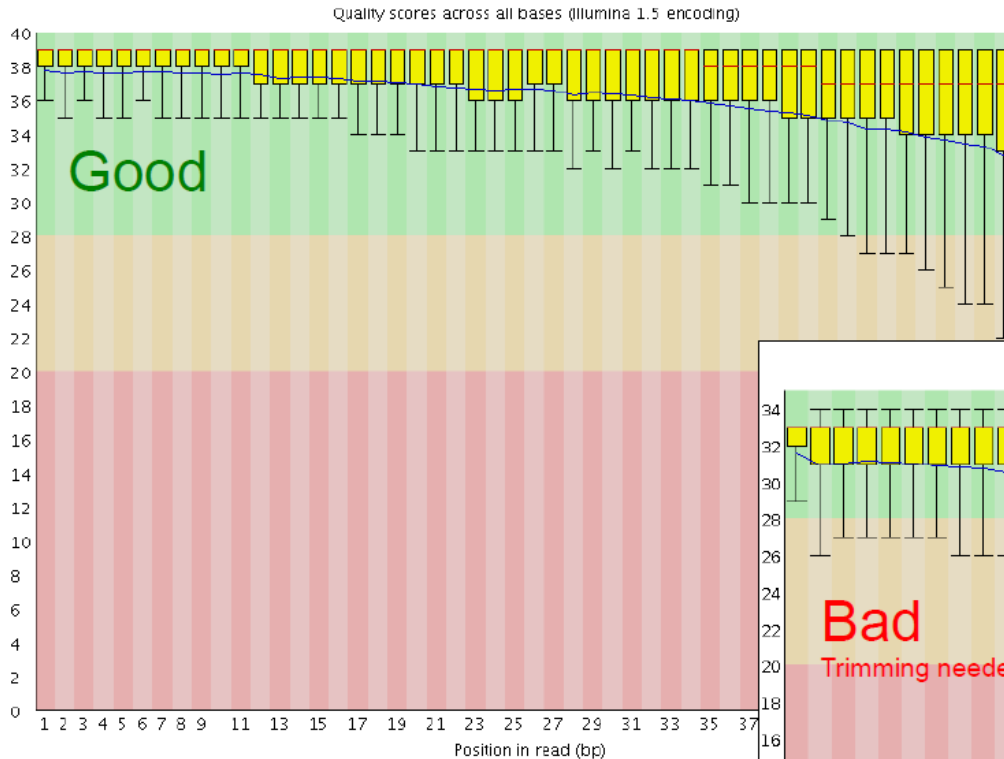
读段质量控制 (FastQC)

High level of sequencing adapter contamination, trimming needed

Overrepresented sequences

Sequence	Count	Percentage	Possible Source
GTATTACAGATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGATCTCG	820428	2.8366639370528275	Illumina Paired End PCR Primer 2 (100% over 43bp)
GTATACAGATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGATCTCGT	749728	2.5922157461699773	Illumina Paired End PCR Primer 2 (100% over 44bp)
CGGTTCAGCAGGAATGCCGAGATCGGAAGAGCGGTTCAGCAGGAATGCCG	648852	2.243432780066747	Illumina Paired End Adapter 2 (100% over 31bp)
GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGATCGGAAGAGCGGTTCAG	176765	0.6111723403310748	Illumina Paired End PCR Primer 2 (97% over 36bp)
ACGTCGTAGATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGATCTCG	143840	0.4973327832615156	Illumina Paired End PCR Primer 2 (100% over 43bp)
GTATTACAGATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGATCTCGT	124281	0.42970672717272257	Illumina Paired End PCR Primer 2 (100% over 44bp)
GTATCAGATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGATCTCGTA	99207	0.34301232917842867	Illumina Paired End PCR Primer 2 (100% over 45bp)
GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGATCTCGTATGCCGT	96289	0.33292322279941655	Illumina Paired End PCR Primer 2 (100% over 50bp)
CGGAAGAGCGGTTCAGCAGGAATGCCGAGATCGGAAGAGCGGTTCAGCAG	93842	0.3244626185124245	Illumina Paired End PCR Primer 2 (96% over 33bp)
CGTTACGAGATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGATCTCG	75370	0.26059491013918545	Illumina Paired End PCR Primer 2 (100% over 43bp)
CGTACGAGATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGATCTCGT	63691	0.22021428183196043	Illumina Paired End PCR Primer 2 (100% over 44bp)
ACGTAGATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGATCTCGTAT	56765	0.19626734873359242	Illumina Paired End PCR Primer 2 (100% over 46bp)
TACTGTAAGATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGATCTCG	42991	0.14864317078139472	Illumina Paired End PCR Primer 2 (100% over 43bp)

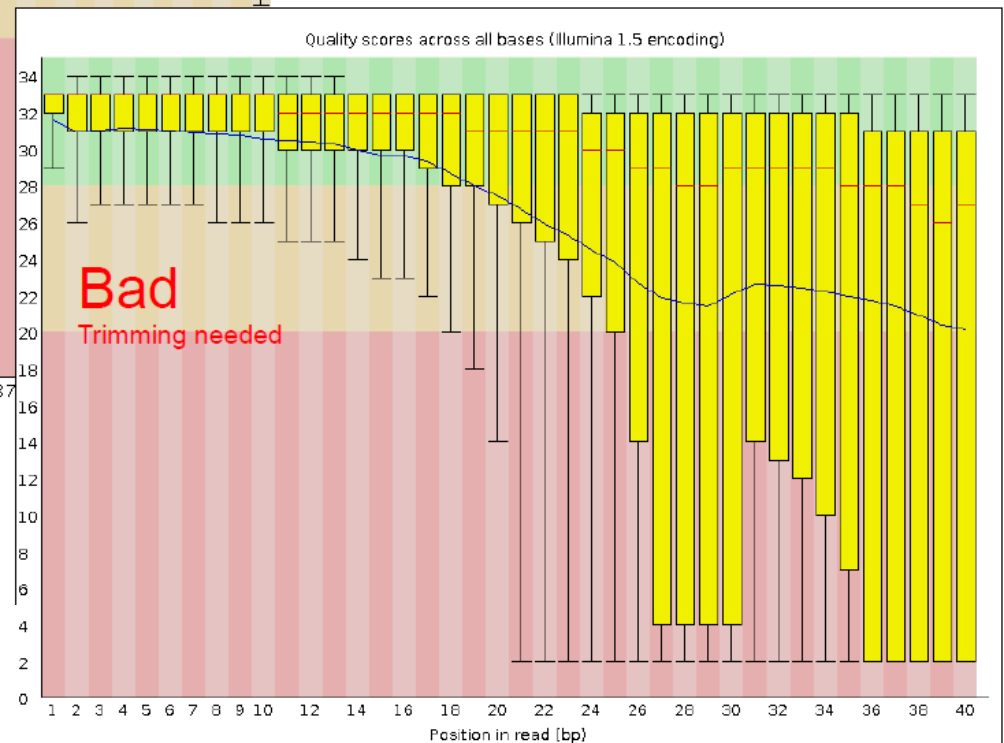
读段质量控制 (FastQC)



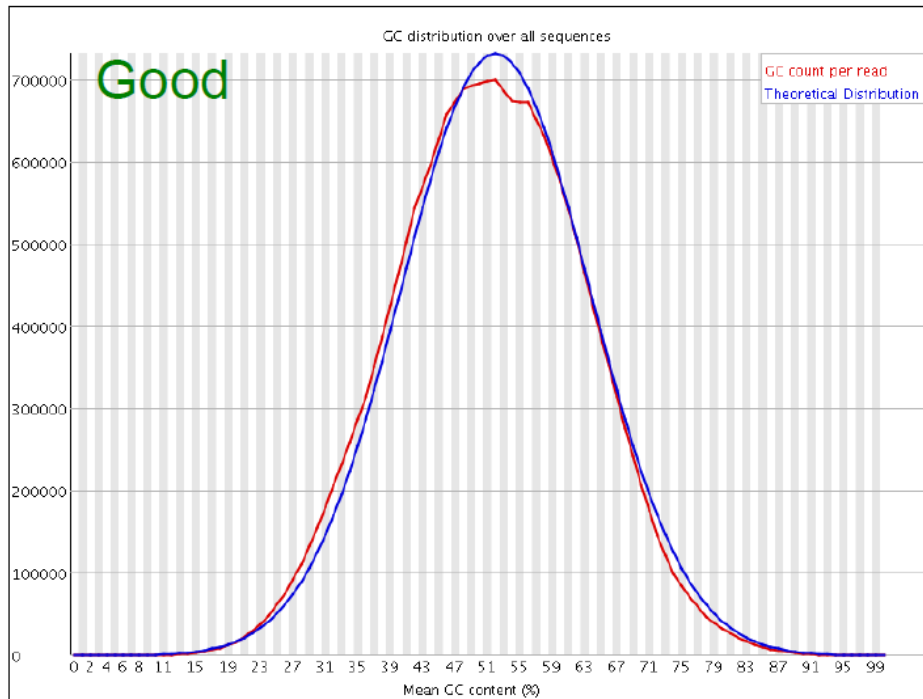
Phred 30 = 1 error / 1000 bases

Phred 20 = 1 error / 100 bases

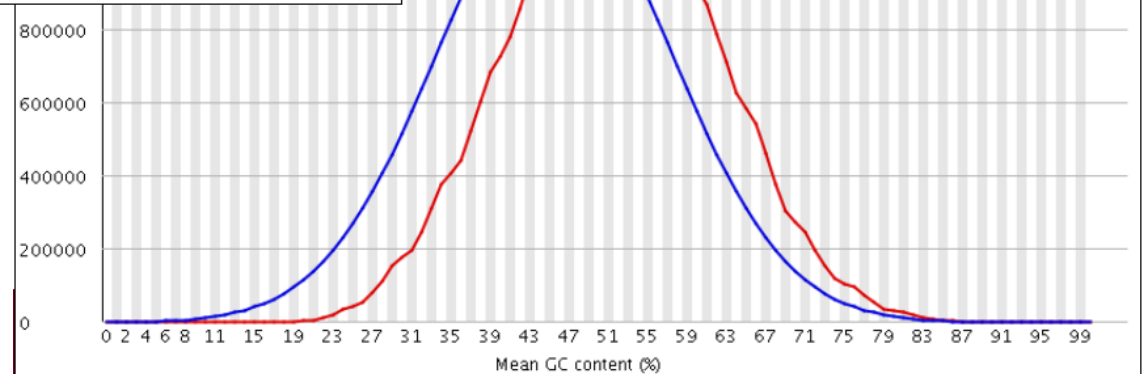
Quality scores across bases



读段质量控制 (FastQC)



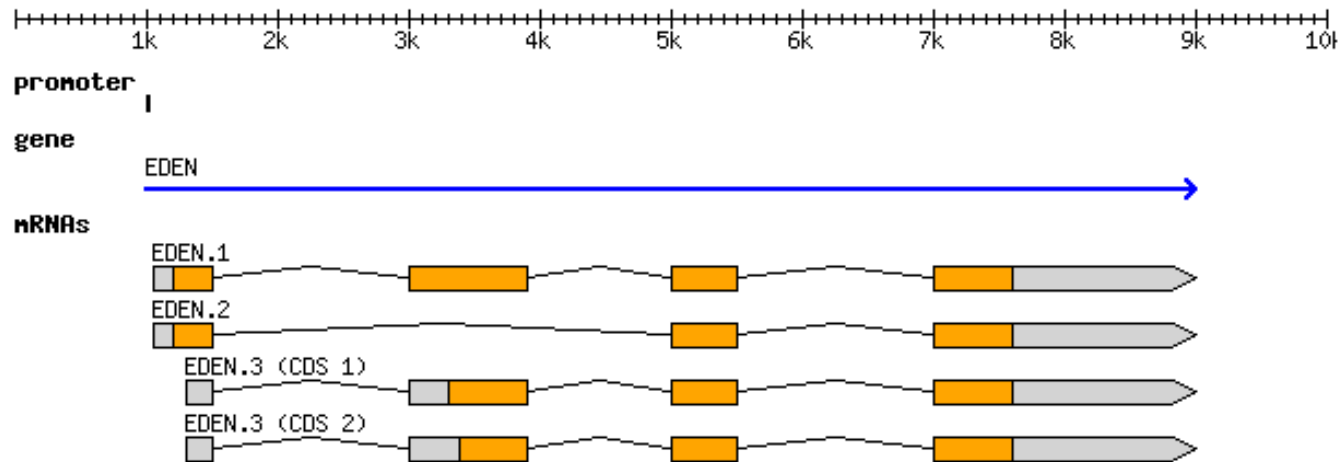
GC Distribution



将读段比对到参考基因组

- 为每个读段找到在参考基因组（或转录组）上的位置
- 计算密集型（需要巧妙算法）：基因组序列被转换（如Burrows-Wheeler变换）为索引，来加快比对
- 输出比对文件，列出比对的读段、它在参考基因组中的比对位置
- 使用基因组浏览器，在基因组上下文中将比对结果可视化

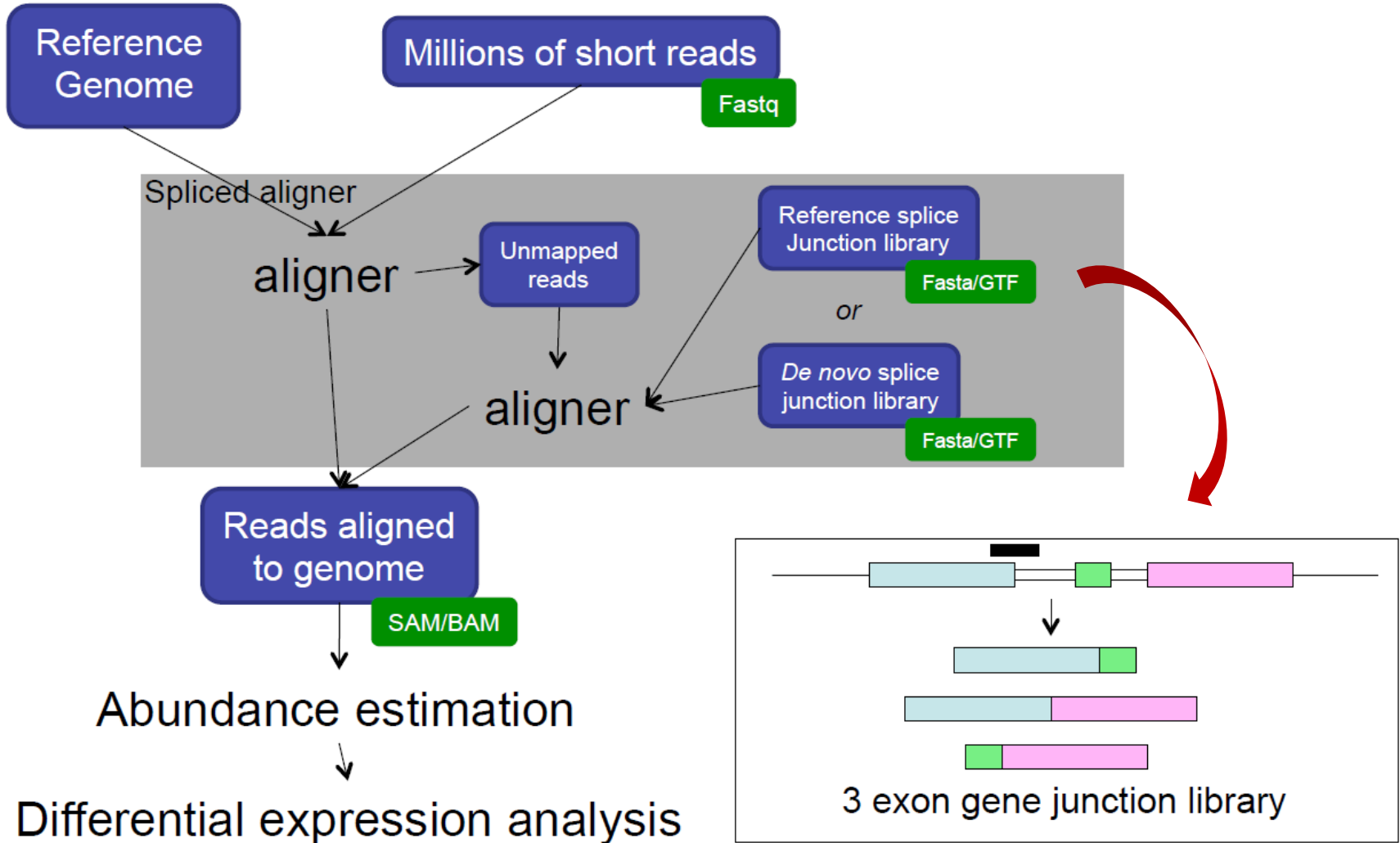
Gene structure annotation (GFF / GTF format)



```

0 ##gff-version 3.2.1
1 ##sequence-region ctg123 1 1497228
2 ctg123 . gene          1000  9000  .  +  .  ID=gene00001;Name=EDEN
3 ctg123 . TF_binding_site 1000  1012  .  +  .  ID=tfbs00001;Parent=gene00001
4 ctg123 . mRNA          1050  9000  .  +  .  ID=mRNA00001;Parent=gene00001;Name=EDEN.1
5 ctg123 . mRNA          1050  9000  .  +  .  ID=mRNA00002;Parent=gene00001;Name=EDEN.2
6 ctg123 . mRNA          1300  9000  .  +  .  ID=mRNA00003;Parent=gene00001;Name=EDEN.3
7 ctg123 . exon          1300  1500  .  +  .  ID=exon00001;Parent=mRNA00003
8 ctg123 . exon          1050  1500  .  +  .  ID=exon00002;Parent=mRNA00001,mRNA00002
9 ctg123 . exon          3000  3902  .  +  .  ID=exon00003;Parent=mRNA00001,mRNA00003
10 ctg123 . exon          5000  5500  .  +  .  ID=exon00004;Parent=mRNA00001,mRNA00002,mRNA00003
11 ctg123 . exon          7000  9000  .  +  .  ID=exon00005;Parent=mRNA00001,mRNA00002,mRNA00003
12 ctg123 . CDS           1201  1500  .  +  0  ID=cds00001;Parent=mRNA00001;Name=edenprotein.1
13 ctg123 . CDS           3000  3902  .  +  0  ID=cds00001;Parent=mRNA00001;Name=edenprotein.1
    
```

基于参考基因组的比对



Aligning reads to reference genome

1.1 An example

Suppose we have the following alignment with bases in lower cases clipped from the alignment. Read r001/1 and r001/2 constitute a read pair; r003 is a chimeric read; r004 represents a split alignment.

```
Coord      12345678901234 5678901234567890123456789012345
ref        AGCATGTTAGATAA**GATAGCTGTGCTAGTAGGCAGTCAGCGCCAT

+r001/1      TTAGATAAAGGATA*CTG
+r002        aaaAGATAA*GGATA
+r003        gcctaAGCTAA
+r004                ATAGCT.....TCAGC
-r003                ttagctTAGGC
-r001/2                        CAGCGGCAT
```

The corresponding SAM format is:¹

```
@HD VN:1.5 SO:coordinate
@SQ SN:ref LN:45
r001  99 ref  7 30 8M2I4M1D3M = 37 39 TTAGATAAAGGATACTG *
r002   0 ref  9 30 3S6M1P1I4M * 0 0 AAAAGATAAGGATA *
r003   0 ref  9 30 5S6M * 0 0 GCCTAAGCTAA * SA:Z:ref,29,-,6H5M,17,0;
r004   0 ref 16 30 6M14N5M * 0 0 ATAGCTTCAGC *
r003 2064 ref 29 17 6H5M * 0 0 TAGGC * SA:Z:ref,9,+,5S6M,30,1;
r001 147 ref 37 30 9M = 7 -39 CAGCGGCAT * NM:i:1
```

Alignment against reference genome (SAM/BAM)

```

Coor      12345678901234  5678901234567890123456789012345
ref       AGCATGTTAGATAA**GATAGCTGTGCTAGTAGGCAGTCAGCGCCAT
  
```

```

+r001/1      TTAGATAAAGGATA*CTG
+r002        aaaAGATAA*GGATA
+r003        gcctaAGCTAA
+r004                ATAGCT.....TCAGC
-r003                ttagctTAGGC
-r001/2                        CAGCGGCAT
  
```

This alignment contains the following features: (1) bases in lower cases are clipped from the alignment; (2) read r001/1 and r001/2 constitute a read pair; (3) r003 is a chimeric read; (4) r004 represents a **split alignment**.

@HD VN:1.5 SO:coordinate										
@SQ SN:ref LN:45										
r001	99	ref	7	30	8M2I4M1D3M	=	37	39	TTAGATAAAGGATACTG	*
r002	0	ref	9	30	3S6M1P1I4M	*	0	0	AAAAGATAAGGATA	*
r003	0	ref	9	30	5S6M	*	0	0	GCCTAAGCTAA	* SA:Z:ref,29,-,6H5M,17,0;
r004	0	ref	16	30	6M14N5M	*	0	0	ATAGCTTCAGC	*
r003	2064	ref	29	17	6H5M	*	0	0	TAGGC	* SA:Z:ref,9,+,5S6M,30,1;
r001	147	ref	37	30	9M	=	7	-39	CAGCGGCAT	* NM:i:1

Header section

Alignment section

Optional fields in the format of TAG:TYPE:VALUE

QUAL: read quality; * meaning such information is not available

SEQ: read sequence

TLEN: the number of bases covered by the reads from the same fragment. Plus/minus means the current read is the leftmost/rightmost read. E.g. compare first and last lines.

PNEXT: Position of the primary alignment of the NEXT read in the template. Set as 0 when the information is unavailable. It corresponds to POS column.

RNEXT: reference sequence name of the primary alignment of the NEXT read. For paired-end sequencing, NEXT read is the paired read, corresponding to the RNAME column.

CIGAR: summary of alignment, e.g. insertion, deletion

MAPQ: mapping quality

POS: 1-based position

RNAME: reference sequence name, e.g. chromosome/transcript id

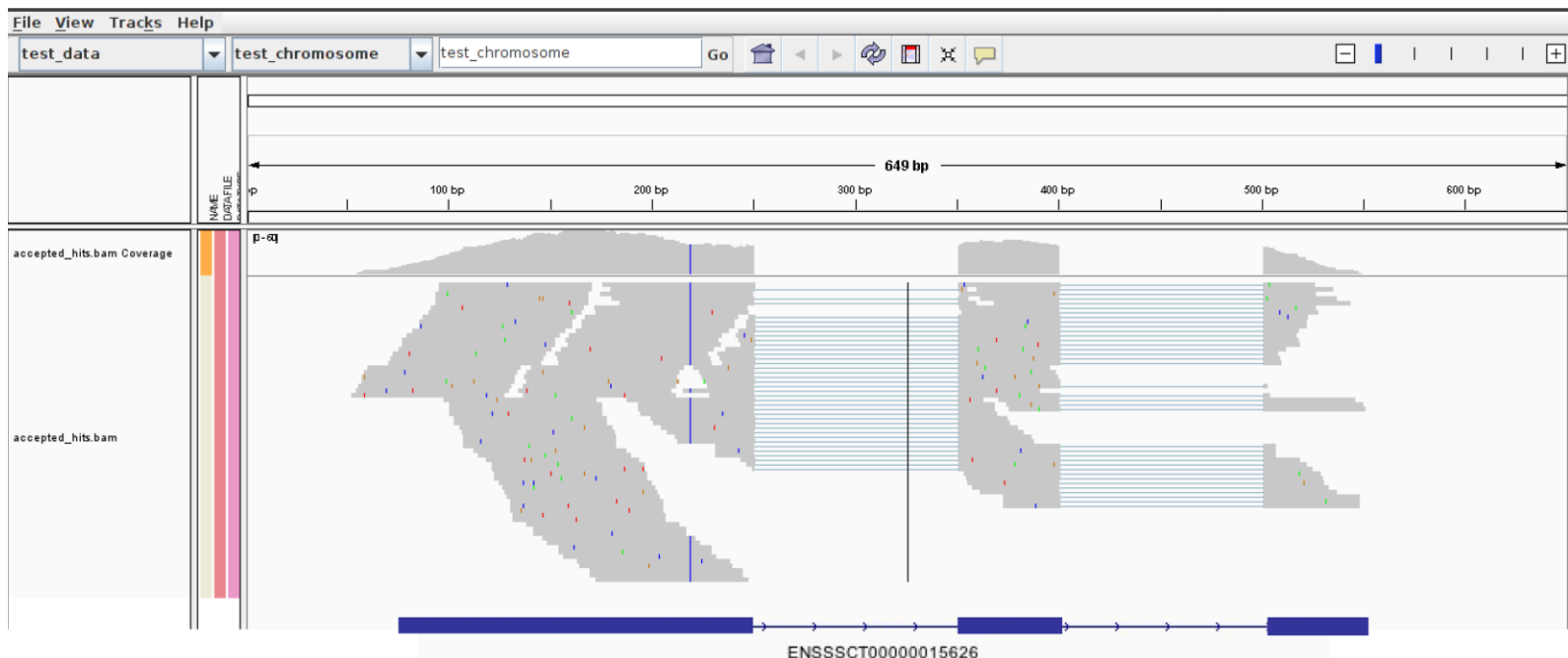
FLAG: indicates alignment information about the read, e.g. paired, aligned, etc.

QNAME: query template name, aka. read ID

思考：解释r001读段的CIGAR，
M匹配，I插入，D删除

可视化比对结果

- Integrative Genomics Viewer (IGV) <https://igv.org/>
 - Fast genome browser
 - Supports array-based and next-generation sequence data, and genomic annotations
 - Free Java program



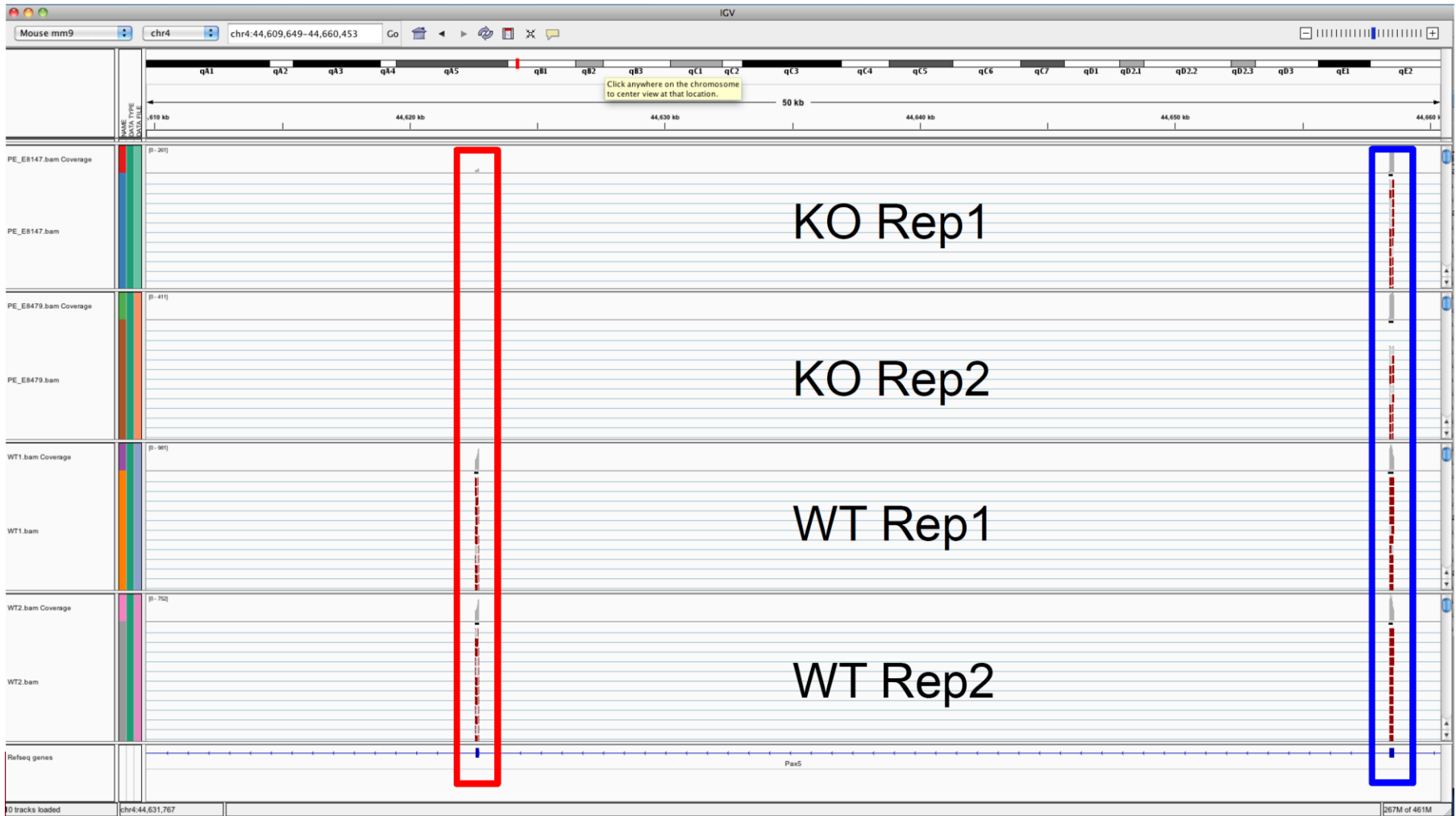
Bam file viewed with IGV

法则 5：质控组学数据

- 碱基和比对结果质量分的分布
- 特定基因（位点）的结果是否符合预期？
- 生物或技术重复样本是否聚类在一起？
- 与公共数据中的类似样本做比较
- 初步分析的生物学发现是否有文献支持？

比对结果是否符合预期?

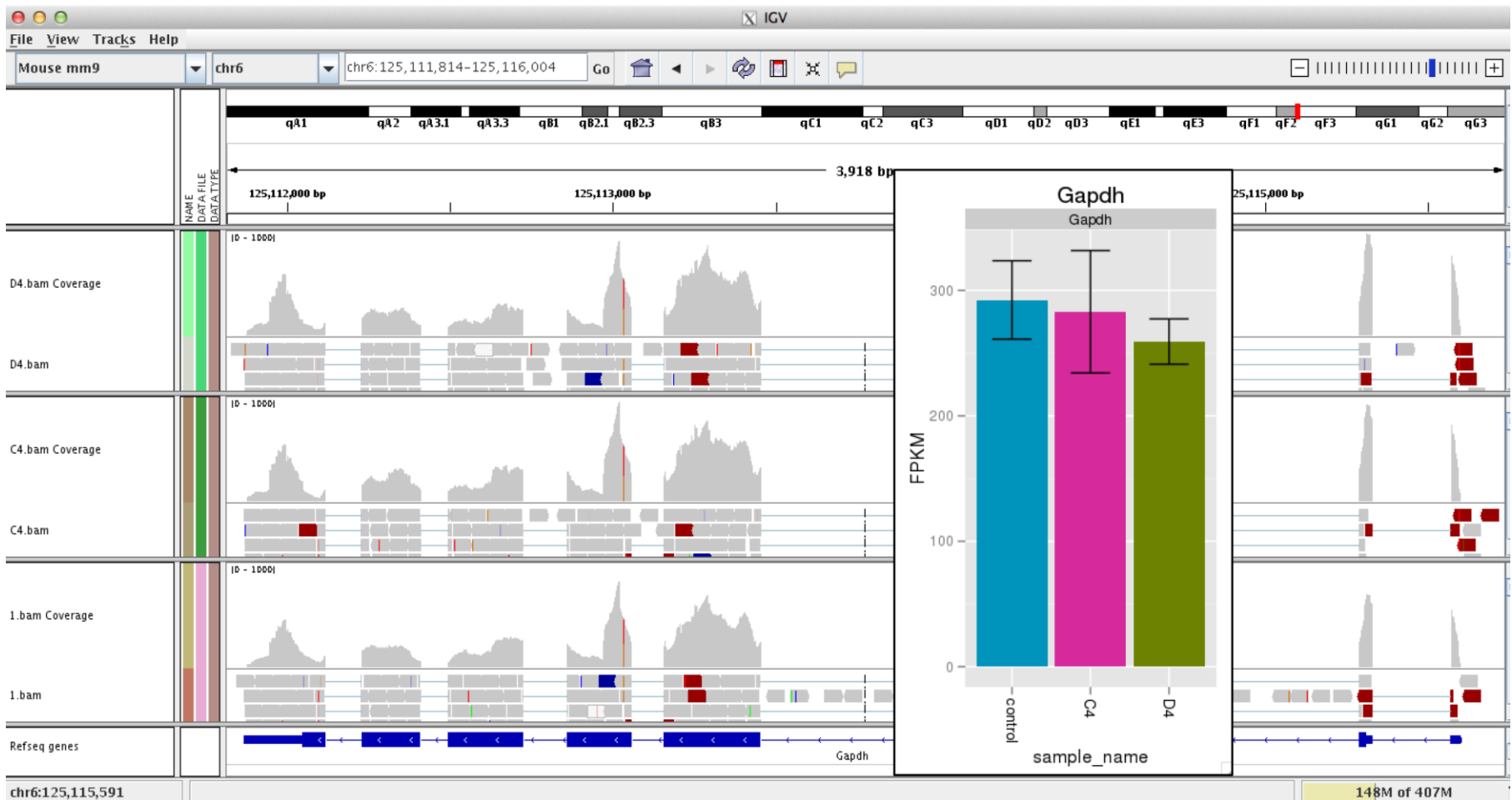
Example I: no reads mapped at knock-out site



Data Courtesy of Dr. Mike Farrar and Dr. Lynn Harris

比对结果是否符合预期?

Example II: Housekeeping genes should behave similarity across multiple samples



Data Courtesy of Dr. David Bernlohr and Dr. Ann Hertzell

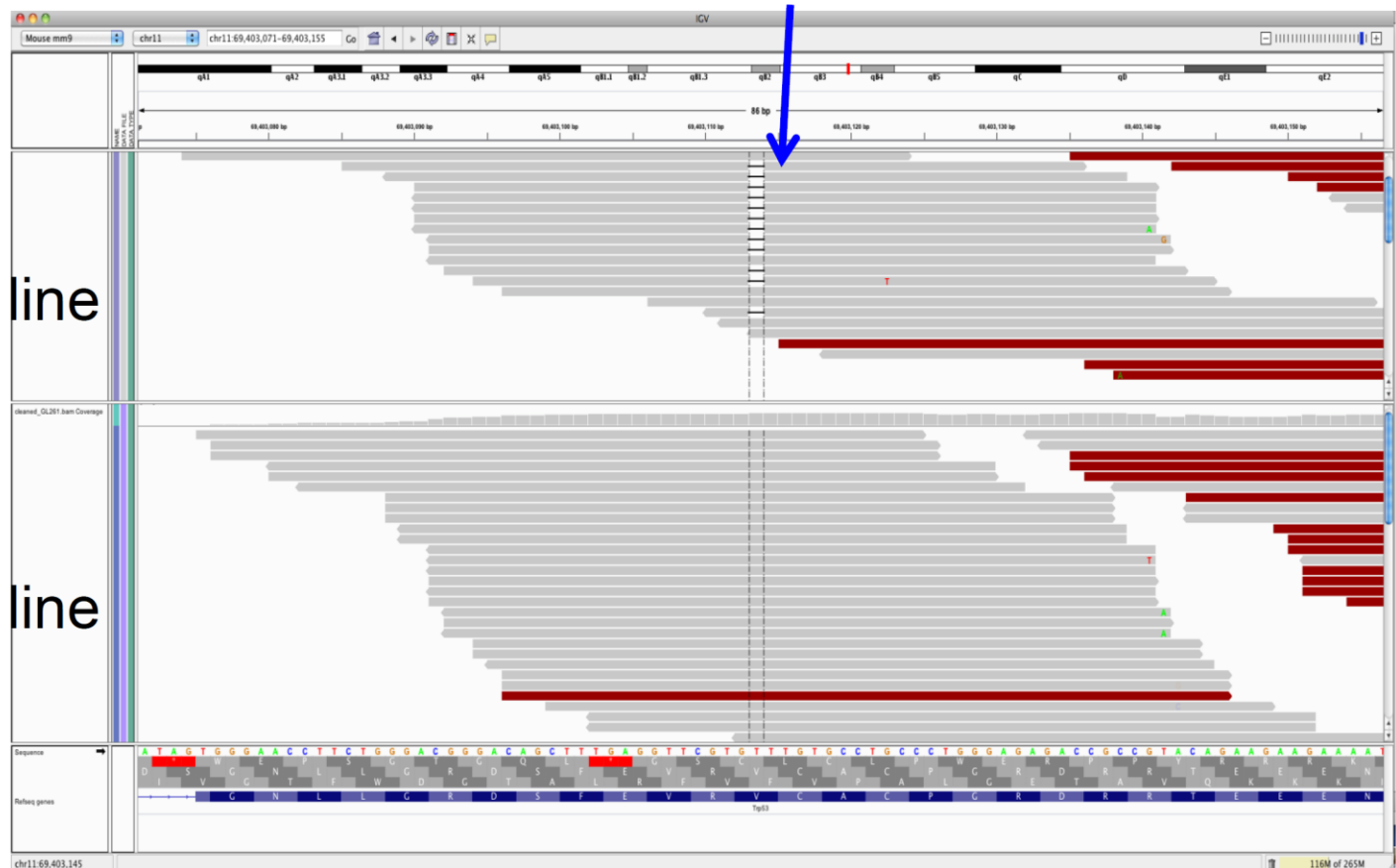
比对结果是否符合预期?

Example III: review of known biomarkers,
for example, known SNPs and indels

Heterozygous deletion of 'T' with 46% penetrance

Cancer cell line

Control cell line



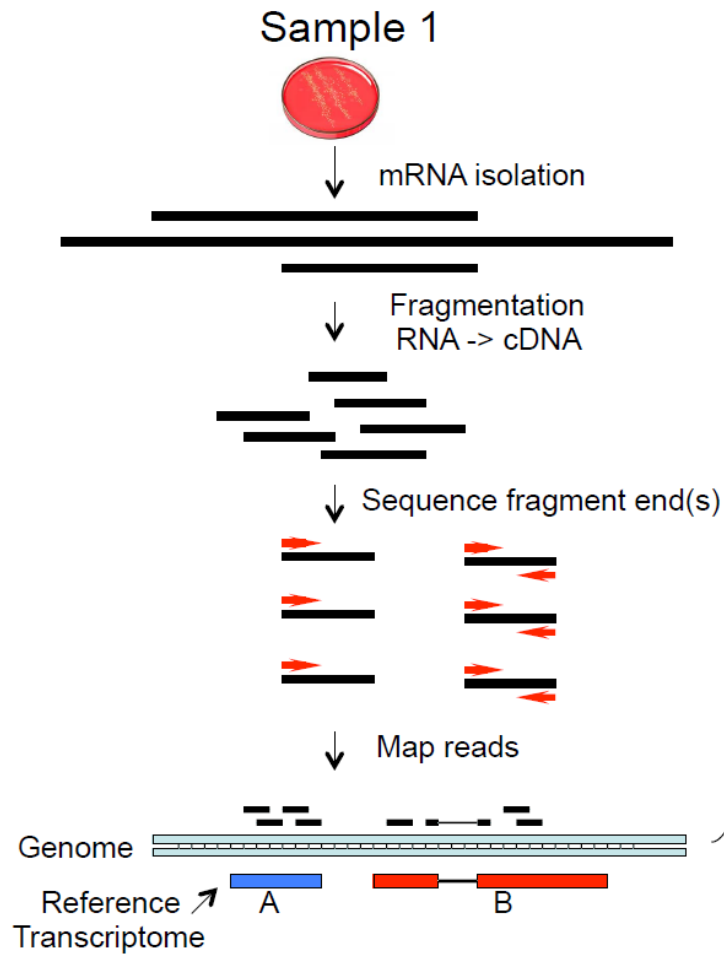
Data Courtesy of Dr. John Ohlfest and Dr. Flavia Popescu

Read counts of genes

EntrezGeneID	Length	MCL1-DG_	MCL1-DH_	MCL1-DI_B	MCL1-DJ_B	MCL1-DK_I	MCL1-DL_E
497097	3634	438	300	65	237	354	287
100503874	3259	1	0	1	1	0	4
100038431	1634	0	0	0	0	0	0
19888	9747	1	1	0	0	0	0
20671	3130	106	182	82	105	43	82
27395	4203	309	234	337	300	290	270
18777	2433	652	515	948	935	928	791
100503730	799	0	1	0	0	0	0
21399	2847	1604	1495	1721	1317	1159	1066
58175	2241	4	2	14	4	2	2
108664	1976	769	752	1062	987	995	903
18387	4707	0	0	0	0	0	0
226304	3692	2	0	0	0	0	0
12421	7046	1494	1412	1157	967	1603	1577
620393	858	0	0	0	0	0	0

GSE60450_Lactation-GenewiseCounts.txt

计算基因表达水平



Calculate transcript abundance

	Gene A	Gene B
Sample 1	4	4

of Reads

	Gene A	Gene B
Sample 1	4	2

Reads per kilobase of exon

	Gene A	Gene B	Total
Sample 1	4	2	6
Sample 2	7	5	12

Reads per kilobase of exon

	Gene A	Gene B	Total
Sample 1	.7	.3	6
Sample 2	.6	.3	12

Reads per kilobase of exon per million mapped reads

RPKM

基因表达水平：RPKM(FPKM)

- RPKM (FPKM): reads (fragments) per kb per million mapped reads

$$\text{RPKM} = \frac{\text{total exon reads}}{\text{mapped reads (millions)} * \text{exon length (KB)}}$$

Gene A 600 bases

Gene B 1100 bases

Gene C 1400 bases

$$\text{RPKM} = 12 / (0.6 * 6) = 3.33$$

$$\text{RPKM} = 24 / (1.1 * 6) = 3.64$$

$$\text{RPKM} = 11 / (1.4 * 6) = 1.31$$

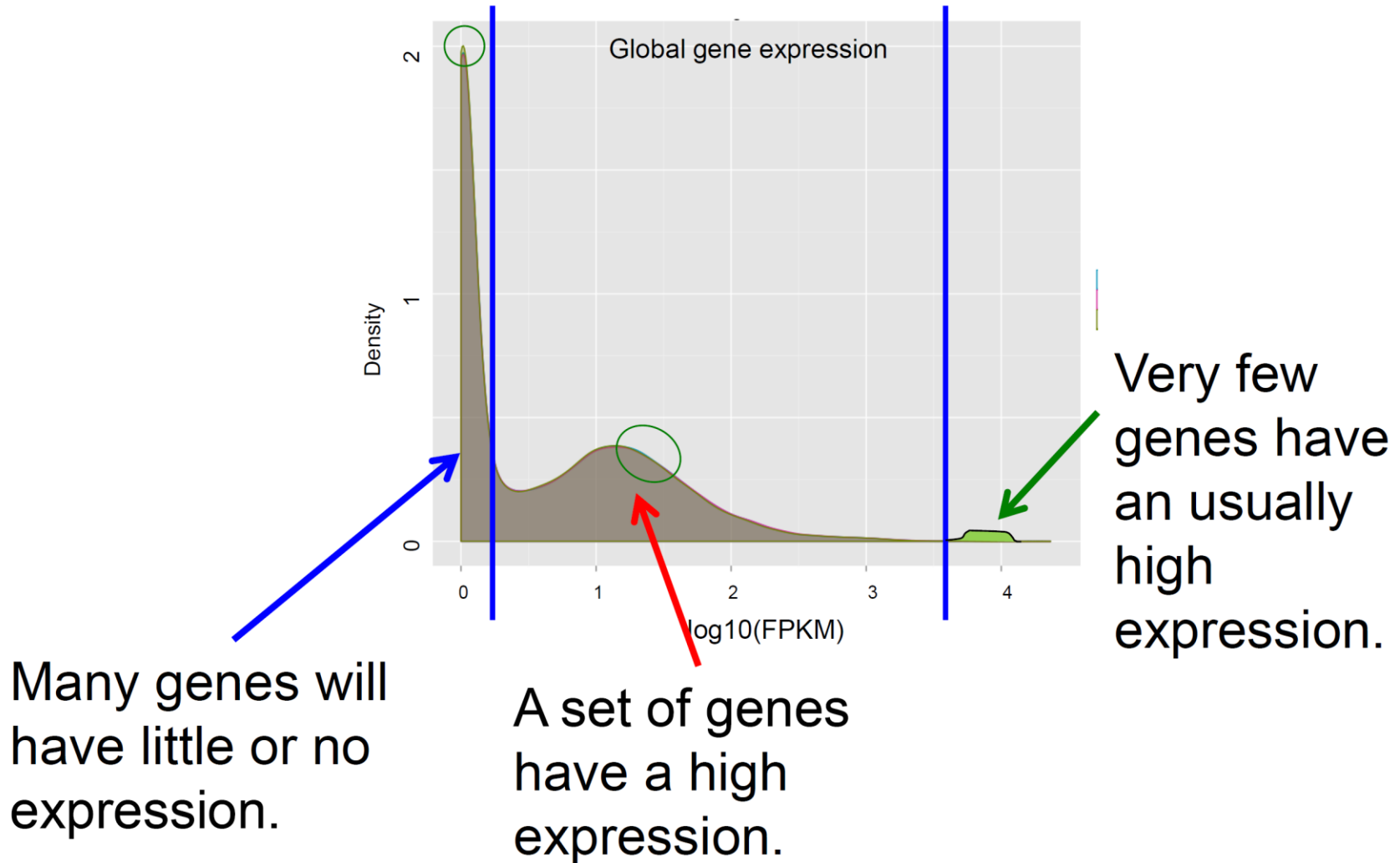


$$\text{RPKM} = 19 / (0.6 * 8) = 3.96$$

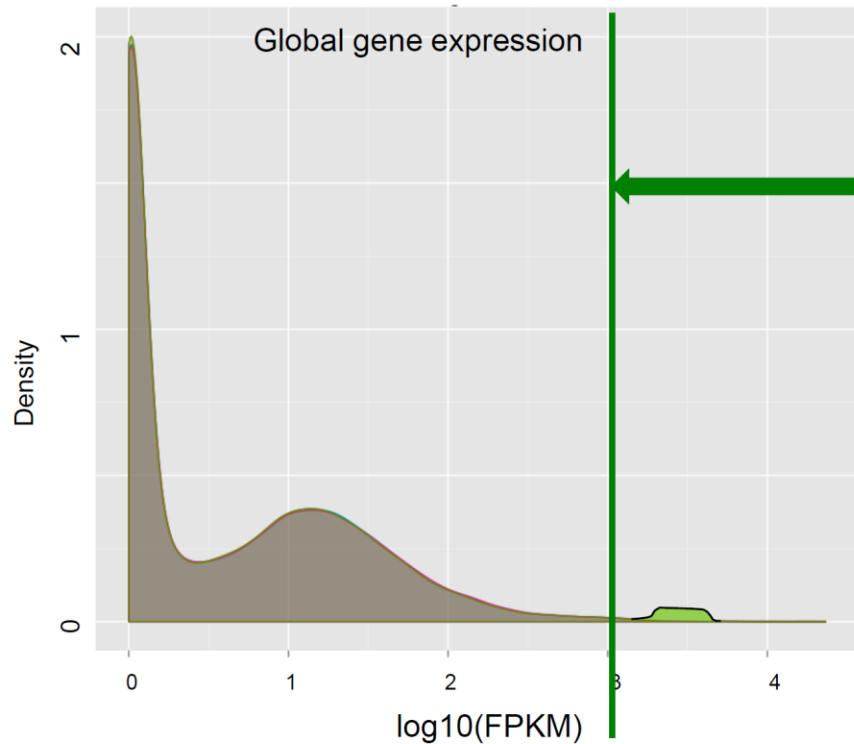
$$\text{RPKM} = 28 / (1.1 * 8) = 1.94$$

$$\text{RPKM} = 16 / (1.4 * 8) = 1.43$$

检查表达谱整体分布



检查表达谱整体分布（红细胞）



Exclude the highly-expressed genes for highly-unbalanced expression between conditions.
Set “yes” to “**Perform quartile normalization**”.



Perform quartile normalization:

Removes top 25% of genes from FPKM denominator to improve accuracy of differential expression calls for low abundance transcripts.

高表达基因的微小波动可能引发低表达基因的“差异表达”

差异表达分析

- Cuffdiff (Cufflinks package)
 - Pairwise comparisons
 - Differential gene, transcript, and primary transcript expression; differential splicing and promoter use
 - Easy to use, well documented
 - Input: transcriptome, SAM/BAM read alignments (abundance estimation built-in)
- EdgeR
 - Complex experimental designs using generalized linear model
 - Information sharing among genes (Bayesian gene-wise dispersion estimation)
 - Difficult to use R package
 - Input: raw gene/transcript read counts (calculate abundance using separate software)

RNA-seq分析经典流程

Differential gene and transcript expression analysis of RNA-seq experiments with TopHat and Cufflinks

Cole Trapnell^{1,2}, Adam Roberts³, Loyal Goff^{1,2,4}, Geo Pertea^{5,6}, Daehwan Kim^{5,7}, David R Kelley^{1,2}, Harold Pimentel³, Steven L Salzberg^{5,6}, John L Rinn^{1,2} & Lior Pachter^{3,8,9}

¹Broad Institute of MIT and Harvard, Cambridge, Massachusetts, USA. ²Department of Stem Cell and Regenerative Biology, Harvard University, Cambridge, Massachusetts, USA. ³Department of Computer Science, University of California, Berkeley, California, USA. ⁴Computer Science and Artificial Intelligence Lab, Department of Electrical Engineering and Computer Science, Massachusetts Institute of Technology, Cambridge, Massachusetts, USA. ⁵Department of Medicine, McKusick-Nathans Institute of Genetic Medicine, Johns Hopkins University School of Medicine, Baltimore, Maryland, USA. ⁶Department of Biostatistics, Johns Hopkins University, Baltimore, Maryland, USA. ⁷Center for Bioinformatics and Computational Biology, University of Maryland, College Park, Maryland, USA. ⁸Department of Mathematics, University of California, Berkeley, California, USA. ⁹Department of Molecular and Cell Biology, University of California, Berkeley, California, USA. Correspondence should be addressed to C.T. (cole@broadinstitute.org).

Published online 1 March 2012; corrected after print 7 August 2014; doi:10.1038/nprot.2012.016

Recent advances in high-throughput cDNA sequencing (RNA-seq) can reveal new genes and splice variants and quantify expression genome-wide in a single assay. The volume and complexity of data from RNA-seq experiments necessitate scalable, fast and mathematically principled analysis software. TopHat and Cufflinks are free, open-source software tools for gene discovery and comprehensive expression analysis of high-throughput mRNA sequencing (RNA-seq) data. Together, they allow biologists to identify new genes and new splice variants of known ones, as well as compare gene and transcript expression under two or more conditions. This protocol describes in detail how to use TopHat and Cufflinks to perform such analyses. It also covers several accessory tools and utilities that aid in managing data, including CummeRbund, a tool for visualizing RNA-seq analysis results. Although the procedure assumes basic informatics skills, these tools assume little to no background with RNA-seq analysis and are meant for novices and experts alike. The protocol begins with raw sequencing reads and produces a transcriptome assembly, lists of differentially expressed and regulated genes and transcripts, and publication-quality visualizations of analysis results. The protocol's execution time depends on the volume of transcriptome sequencing data and available computing resources but takes less than 1 d of computer time for typical experiments and ~1 h of hands-on time.

Nature Protocols 12 Differential gene and transcript expression analysis of RNA-seq experiments with TopHat and Cufflinks (可按照流程学习分析)

RNA-seq分析新流程

Transcript-level expression analysis of RNA-seq experiments with HISAT, StringTie and Ballgown

Mihaela Pertea^{1,2}, Daehwan Kim¹, Geo M Pertea¹, Jeffrey T Leek³ & Steven L Salzberg^{1–4}

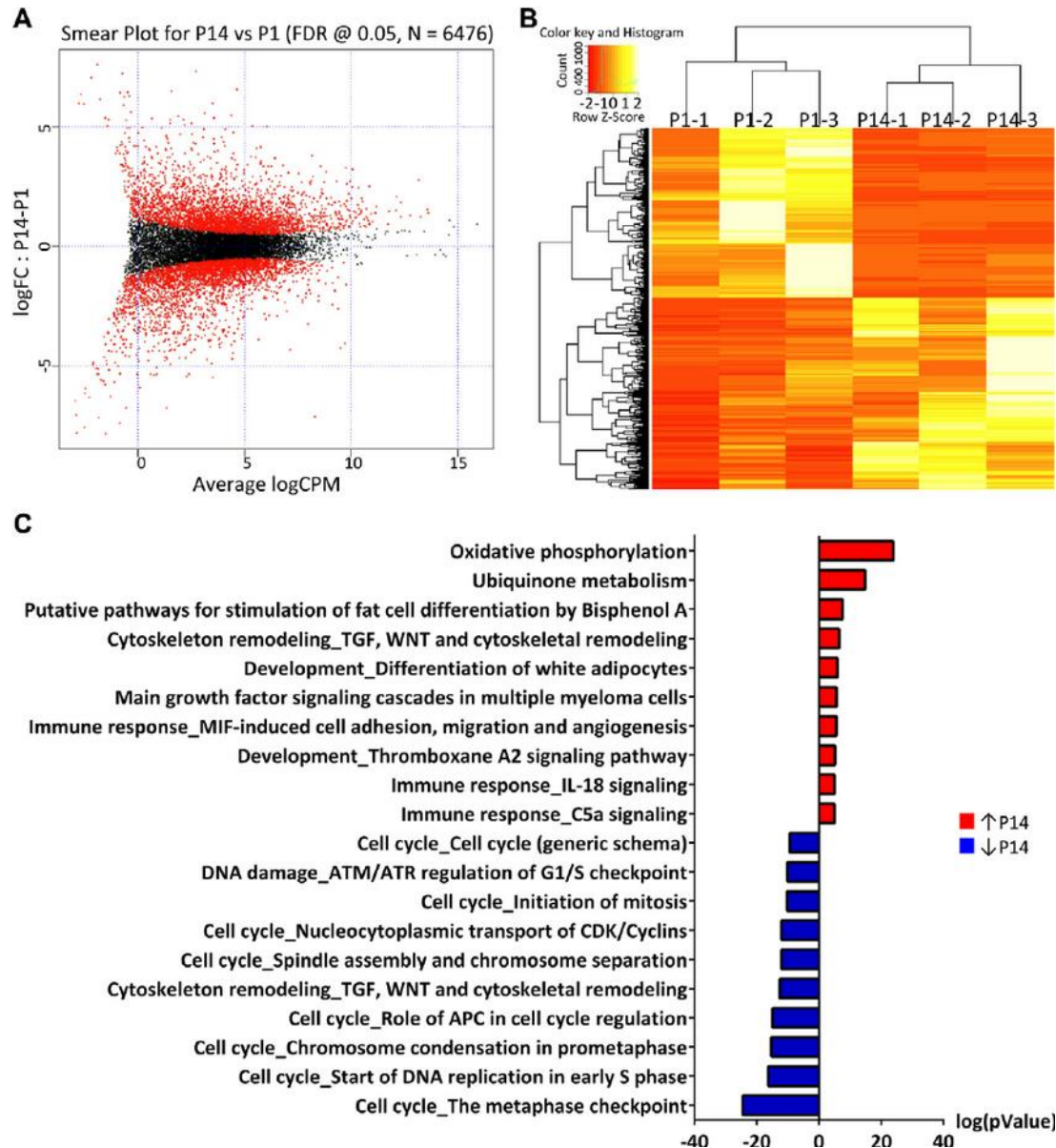
¹Center for Computational Biology, McKusick-Nathans Institute of Genetic Medicine, Johns Hopkins School of Medicine, Baltimore, Maryland, USA. ²Department of Computer Science, Whiting School of Engineering, Johns Hopkins University, Baltimore, Maryland, USA. ³Department of Biostatistics, Bloomberg School of Public Health, Johns Hopkins University, Baltimore, Maryland, USA. ⁴Department of Biomedical Engineering, Johns Hopkins University, Baltimore, Maryland, USA. Correspondence should be addressed to S.L.S. (salzberg@jhu.edu).

Published online 11 August 2016; doi:[10.1038/nprot.2016.095](https://doi.org/10.1038/nprot.2016.095)

High-throughput sequencing of mRNA (RNA-seq) has become the standard method for measuring and comparing the levels of gene expression in a wide variety of species and conditions. RNA-seq experiments generate very large, complex data sets that demand fast, accurate and flexible software to reduce the raw read data to comprehensible results. HISAT (hierarchical indexing for spliced alignment of transcripts), StringTie and Ballgown are free, open-source software tools for comprehensive analysis of RNA-seq experiments. Together, they allow scientists to align reads to a genome, assemble transcripts including novel splice variants, compute the abundance of these transcripts in each sample and compare experiments to identify differentially expressed genes and transcripts. This protocol describes all the steps necessary to process a large set of raw sequencing reads and create lists of gene transcripts, expression levels, and differentially expressed genes and transcripts. The protocol's execution time depends on the computing resources, but it typically takes under 45 min of computer time. HISAT, StringTie and Ballgown are available from <http://ccb.jhu.edu/software.shtml>.

Nature Protocols 16 Transcript-level expression analysis of RNA-seq experiments with HISAT, StringTie and Ballgown

RNA-seq analysis plots



Mark Ziemann

- **Bioconductor** provides **R packages for the analysis of genomic data**, tools for analysis of data from next-generation high-throughput sequencing methods
 - 1104 packages for bioscience data analysis (*as of March 10, 2016*)
 - active user community, open source and open development
 - two releases each year
- **Statistical analysis**: large data, technological artifacts, designed experiments; rigorous
- Comprehension: biological context, visualization, reproducibility
- High-throughput
 - **Sequencing**: RNASeq, ChIPSeq, variants, copy number, ...
 - Microarrays: expression, SNP, ...
 - **Flow cytometry**, proteomics, images, ...

```
source("http://bioconductor.org/biocLite.R")  
biocLite() # install core packages
```


《Bioconductor Case Studies》

Preface	v	5 Fold-Changes, Log-Ratios, Background Correction, Shrinkage Estimation, and Variance Stabilization	63
List of Contributors	xi	W. Huber	
1 The ALL Dataset	1	5.1 Fold-changes and (log-)ratios	63
F. Hahne and R. Gentleman		5.2 Background-correction and generalized logarithm	65
1.1 Introduction	1	5.3 Calling VSN	70
1.2 The ALL data	1	5.4 How does VSN work?	72
1.3 Data subsetting	2	5.5 Robust fitting and the “most genes not differentially expressed” assumption	74
1.4 Nonspecific filtering	3	5.6 Single-color normalization	78
1.5 BCR/ABL ALL1/AF4 subset	4	5.7 The interpretation of glog-ratios	79
2 R and Bioconductor Introduction	5	5.8 Reference normalization	81
R. Gentleman, F. Hahne, S. Falcon, and M. Morgan		6 Easy Differential Expression	83
2.1 Finding help in R	5	F. Hahne and W. Huber	
2.2 Working with packages	7	6.1 Example data	83
2.3 Some basic R	8	6.2 Nonspecific filtering	84
2.4 Structures for genomic data	11	6.3 Differential expression	85
2.5 Graphics	20	6.4 Multiple testing correction	87
3 Processing Affymetrix Expression Data	25	7 Differential Expression	89
R. Gentleman and W. Huber		W. Huber, D. Scholtens, F. Hahne, and A. von Heydebreck	
3.1 The input data: CEL files	25	7.1 Motivation	89
3.2 Quality assessment	28	7.2 Nonspecific filtering	90
3.3 Preprocessing	32	7.3 Differential expression	92
and filtering probe sets	33	7.4 Multiple testing	94
and preprocessing	40	7.5 Moderated test statistics and the limma package	95
4 Arrays	47	7.6 Gene selection by Receiver Operator Characteristic (ROC)	99
and Wolfgang Huber		7.7 When power increases	101
tion	47		
port	48		
ots	50		

Bioconductor的教材，可了解组学数据常用的分析方法和数据结构

Florian Hahne • Wolfgang Huber
Robert Gentleman • Seth Falcon

Bioconductor Case Studies

本节内容

- RNA-seq测序技术
- RNA-seq分析流程
- • RNA-seq分析举例
- 统计**模型**：数据归一化
- **助教演示**：Linux系统、RNA-seq分析

Data example

EGF-mediated induction of Mcl-1 at the switch to lactation is essential for alveolar cell survival

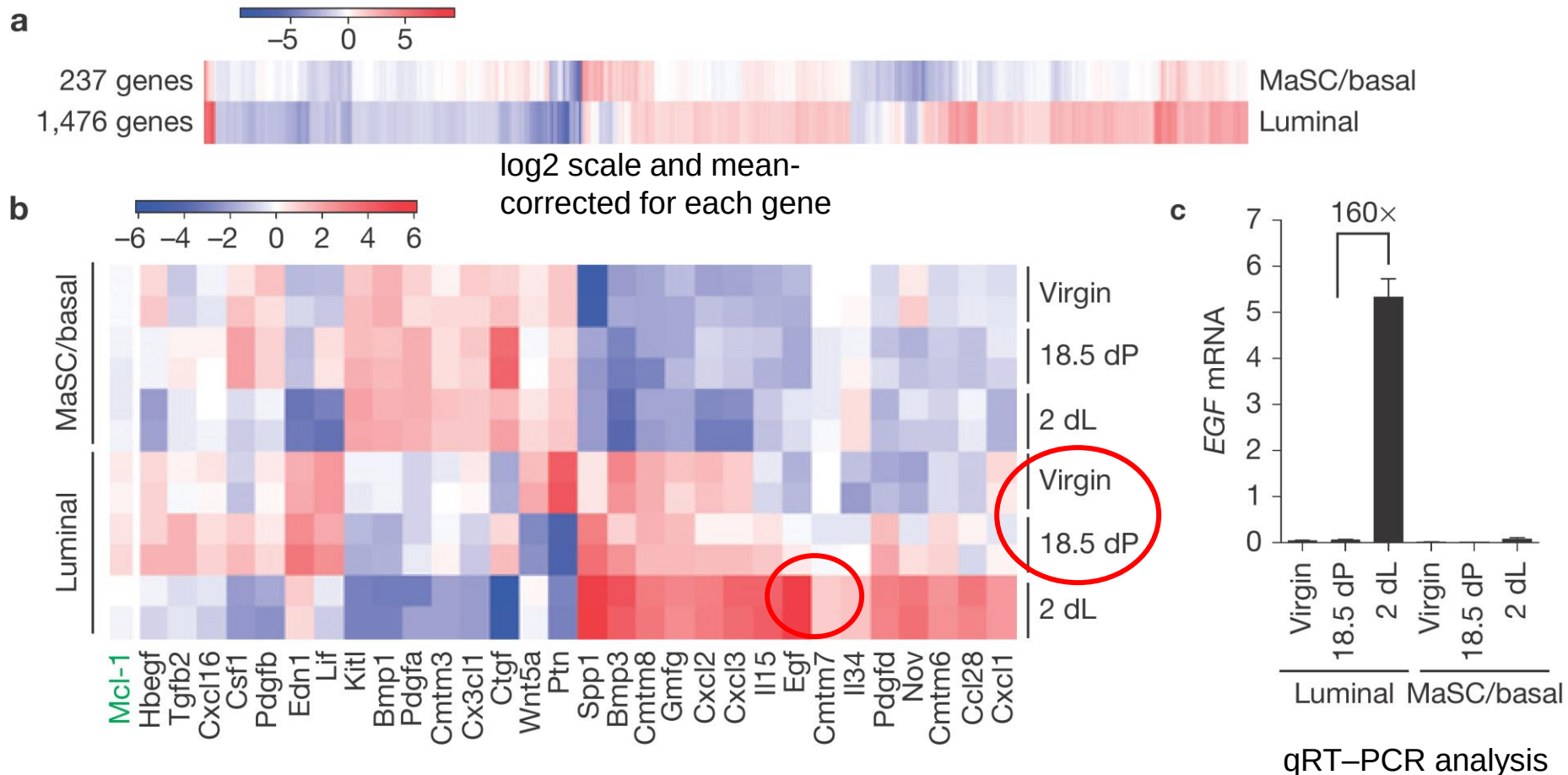
Nai Yang Fu^{1,2}, Anne C. Rios^{1,2,10}, Bhupinder Pal^{1,2,10}, Rina Soetanto³, Aaron T. L. Lun^{2,4}, Kevin Liu^{1,2}, Tamara Beck^{1,2}, Sarah A. Best^{1,2}, François Vaillant^{1,2}, Philippe Bouillet^{2,5}, Andreas Strasser^{2,5}, Thomas Preiss^{3,6}, Gordon K. Smyth^{4,7}, Geoffrey J. Lindeman^{1,8,9,10} and Jane E. Visvader^{1,2,10,11}

Expansion and remodelling of the mammary epithelium requires a tight balance between cellular proliferation, differentiation and death. To explore cell survival versus cell death decisions in this organ, we deleted the pro-survival gene *Mcl-1* in the mammary epithelium. *Mcl-1* was found to be essential at multiple developmental stages including morphogenesis in puberty and alveologenesis in pregnancy. Moreover, *Mcl-1*-deficient basal cells were virtually devoid of repopulating activity, suggesting that this gene is required for stem cell function. Profound upregulation of the Mcl-1 protein was evident in alveolar cells at the switch to lactation, and *Mcl-1* deficiency impaired lactation. Interestingly, *EGF* was identified as one of the most highly upregulated genes on lactogenesis and inhibition of EGF or mTOR signalling markedly impaired lactation, with concomitant decreases in Mcl-1 and phosphorylated ribosomal protein S6. These data demonstrate that Mcl-1 is essential for mammapoiesis and identify EGF as a critical trigger of Mcl-1 translation to ensure survival of milk-producing alveolar cells.

NCB 15 EGF-mediated induction of Mcl-1 at the switch to lactation is essential for alveolar cell survival

Data example

问题： We next interrogated potential molecular regulators that could mediate the marked induction of Mcl-1 expression that occurs on lactogenesis. (p.7)



GEO & SRA database



Gene Expression Omnibus

HOME | SEARCH | SITE MAP | GEO Publications | FAQ | MIAME | Email GEO

NCBI > GEO > **Accession Display**  Not logged in | [Login](#) 

Scope: Format: Amount: GEO accession:

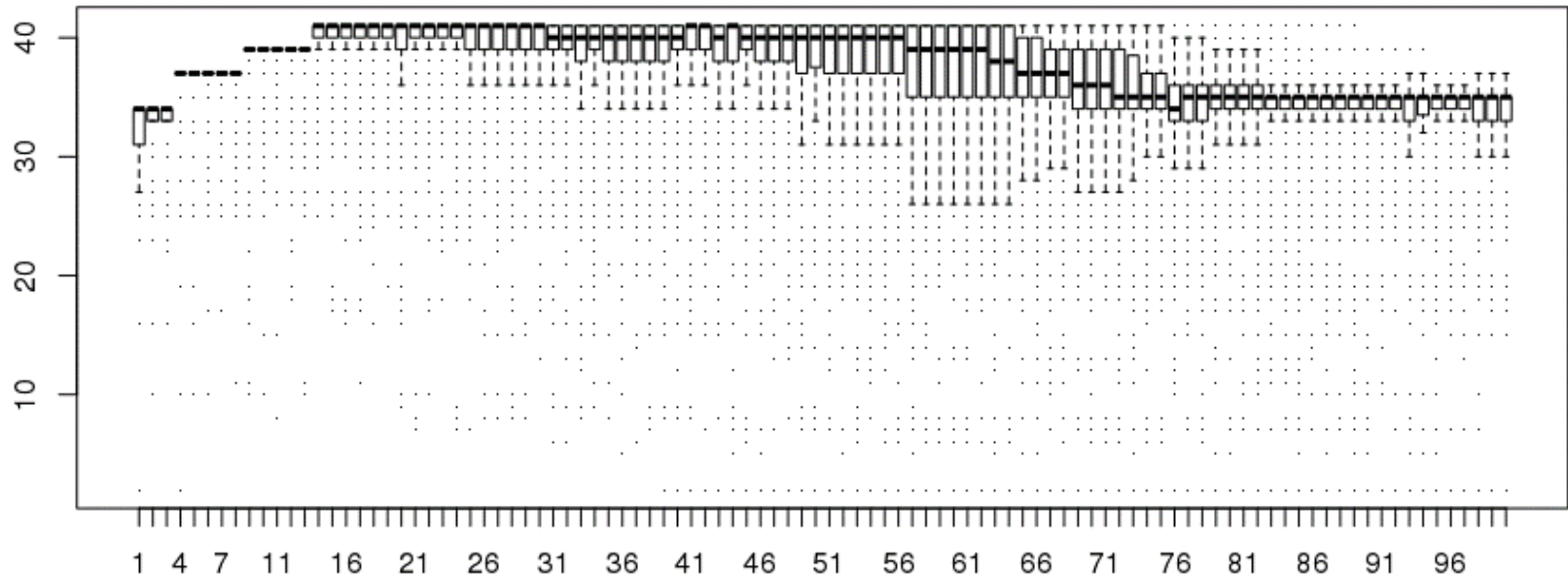
Series GSE60450

[Query DataSets for GSE60450](#)

Status	Public on Jan 19, 2015
Title	Transcriptome analysis of luminal and basal cell subpopulations in the lactating versus pregnant mammary gland
Organism	Mus musculus
Experiment type	Expression profiling by high throughput sequencing
Summary	To identify genes specifically expressed in lactating mammary glands, the gene expression profiles of luminal and basal cells from different developmental stages were compared.
Overall design	Comparison of gene expression in luminal and basal cells harvested from the mammary glands of virgin, 18.5 day pregnant and 2 day lactating mice (2 mice per stage).
Contributor(s)	Fu NY , Lun A , Smyth GK , Visvader JE
Citation(s)	Fu NY, Rios AC, Pal B, Soetanto R et al. EGF-mediated induction of Mcl-1 at the

<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE60450>
(可找到测序数据的SRA文件)

Basepair quality



X轴：碱基位置；Y轴：碱基质量值分布

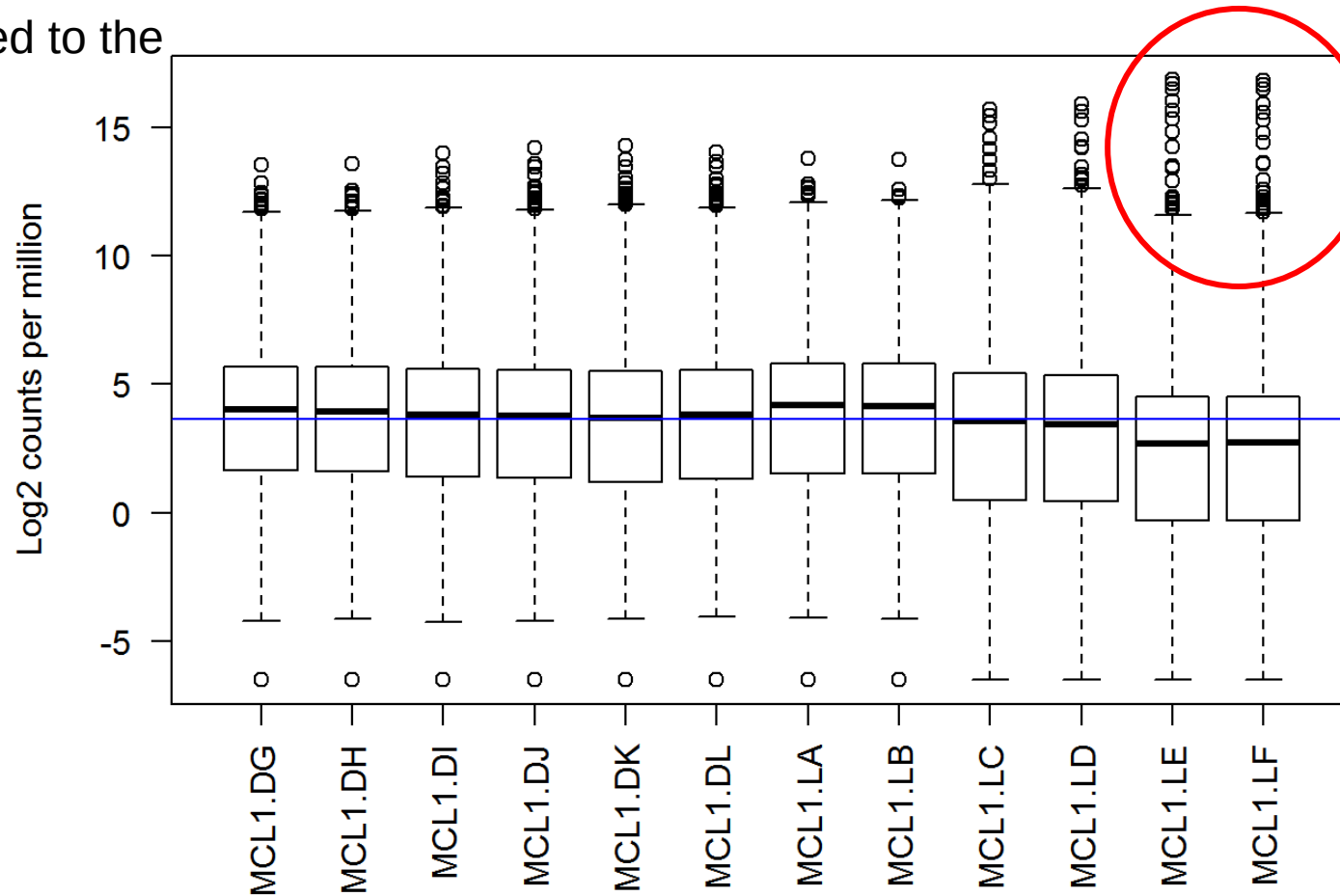
<http://combine-australia.github.io/RNAseq-R/07-rnaseq-day2.html>

课件：RNA-seq mapping.R

Count distribution

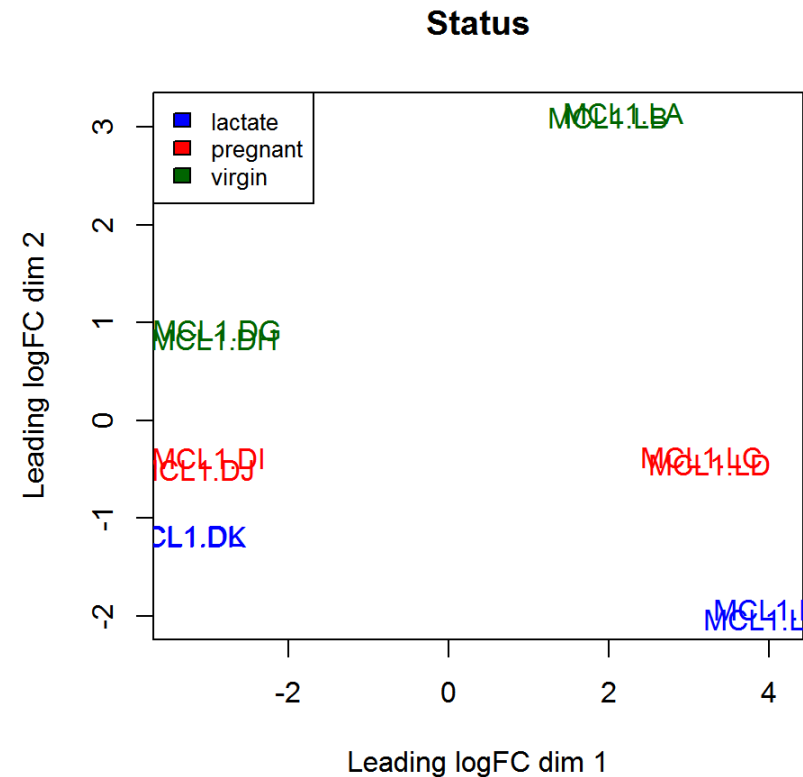
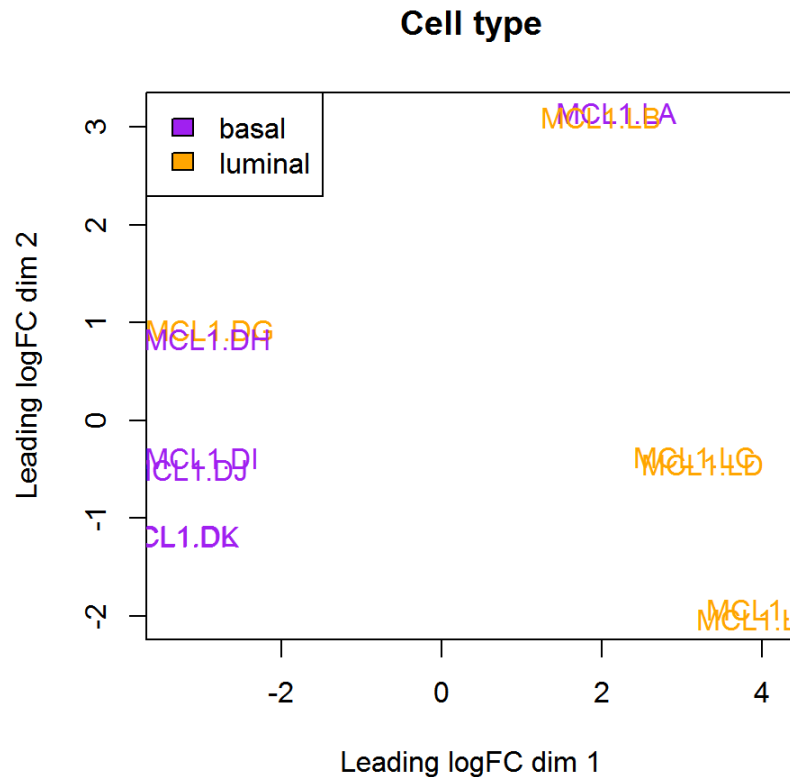
Do any samples appear to be different compared to the others?

Boxplots of logCPMs (unnormalised)



A few genes with large counts may bias the values of other genes

MDS (multi-dimensional scaling) 聚类

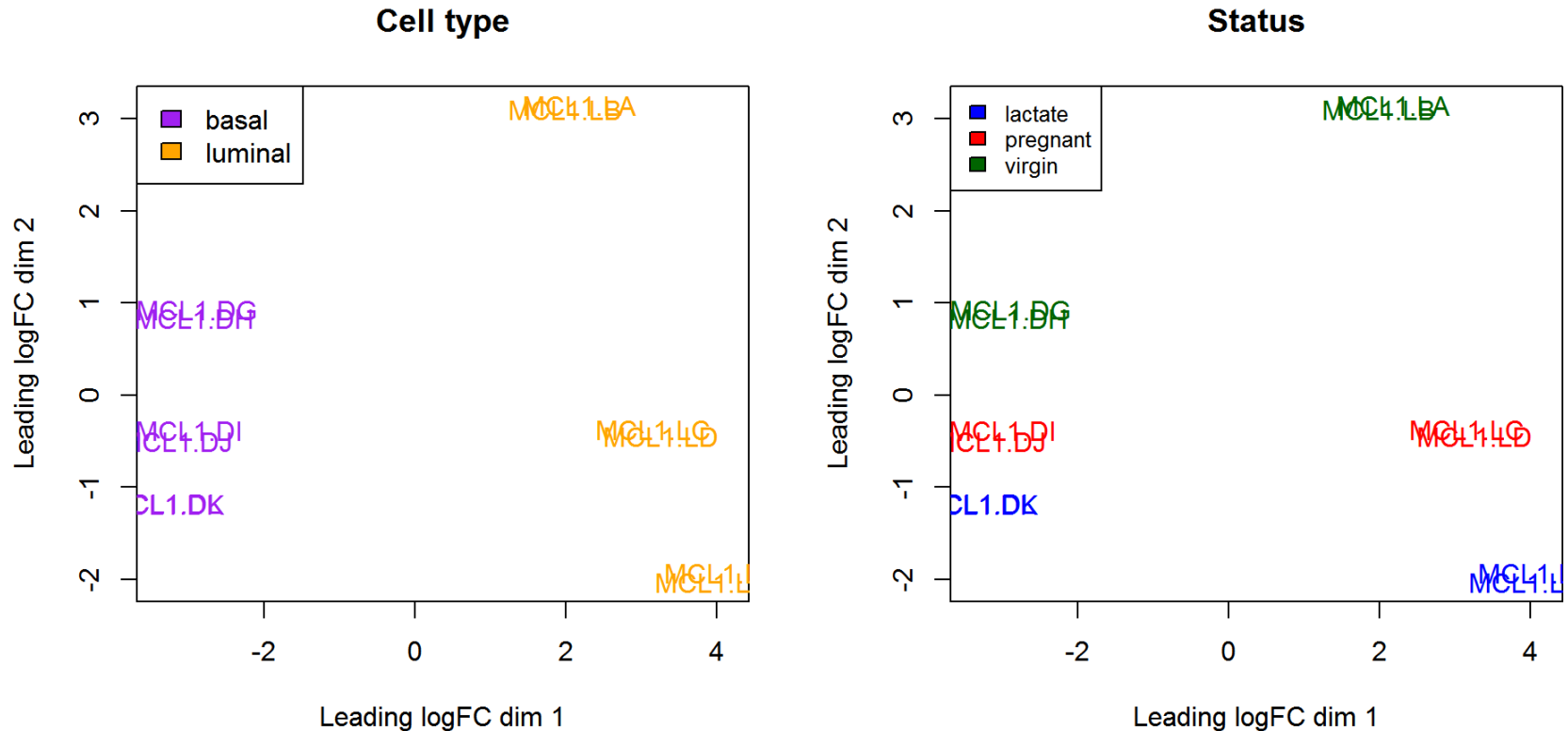


探索性数据
分析

Is there something strange going on with the samples? Identify the two samples that don't appear to be in the right place.

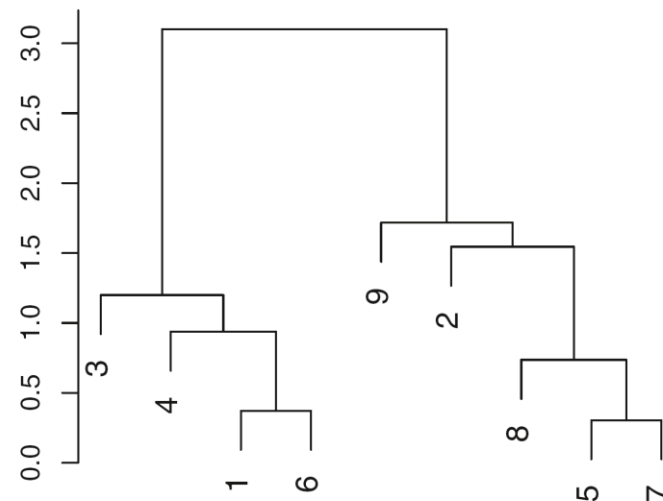
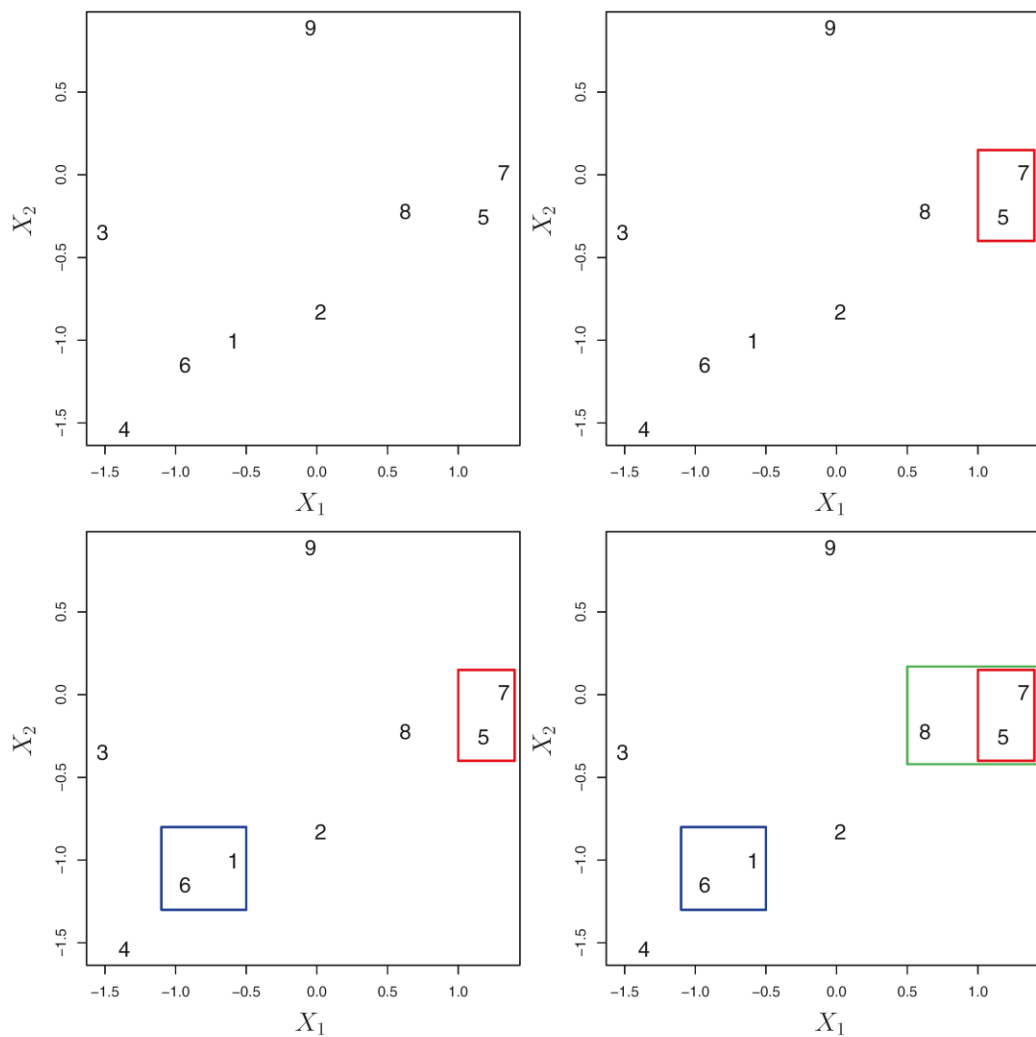
MDS clustering

Sample information corrected

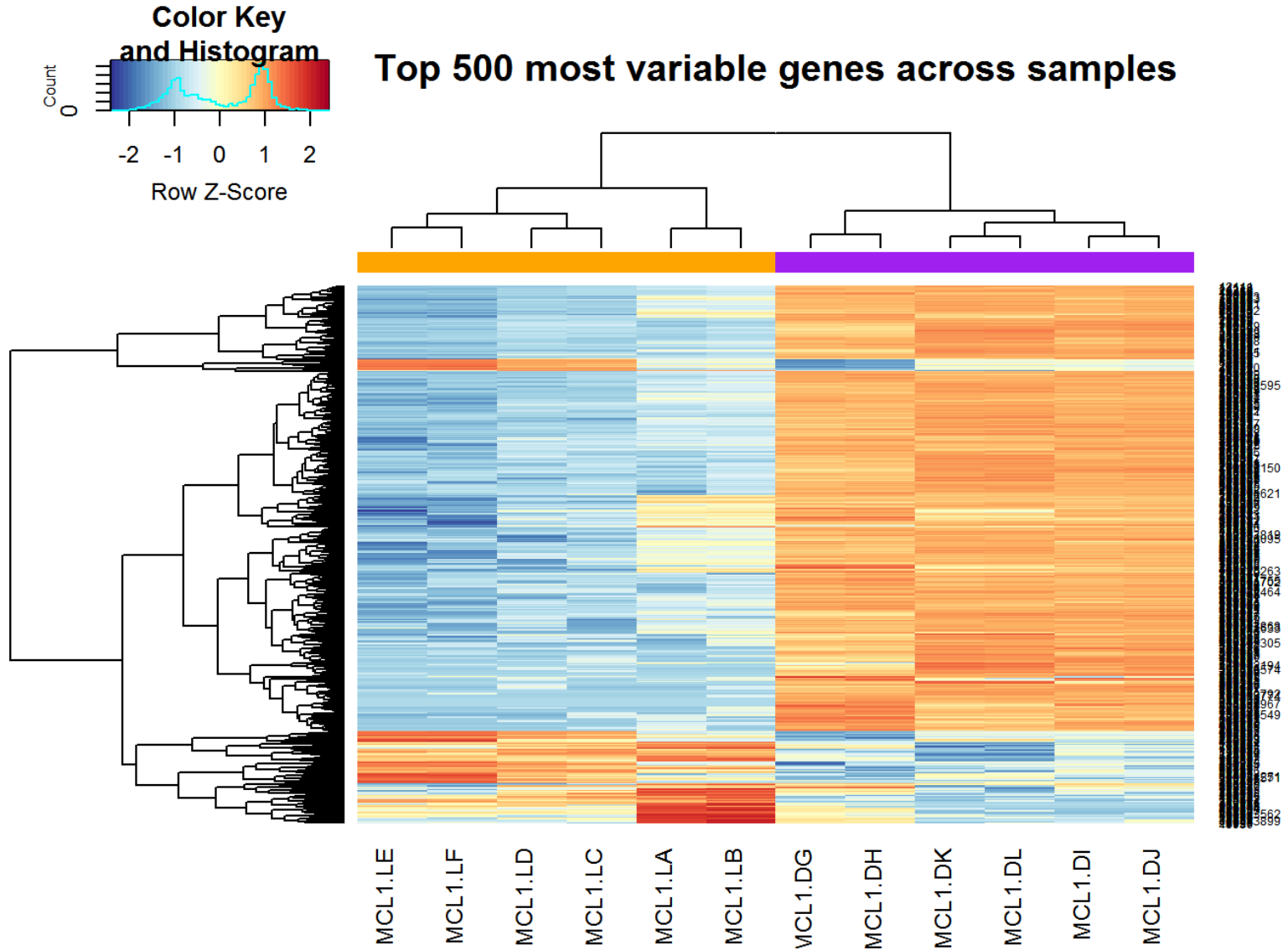


What is the greatest source of variation in the data (i.e. what does dimension 1 represent)? What is the second greatest source of variation in the data?

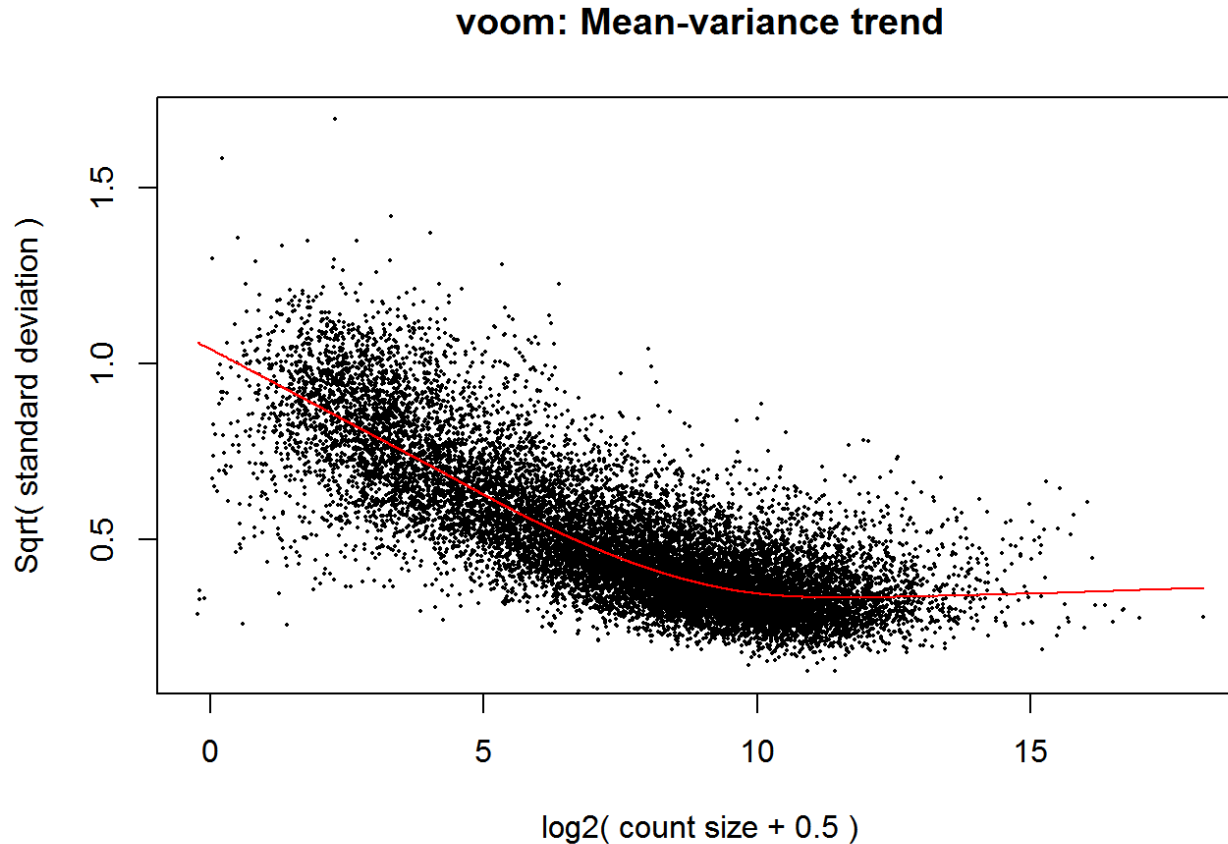
Hierarchical clustering (层次聚类) 算法



Hierarchical clustering with heatmaps



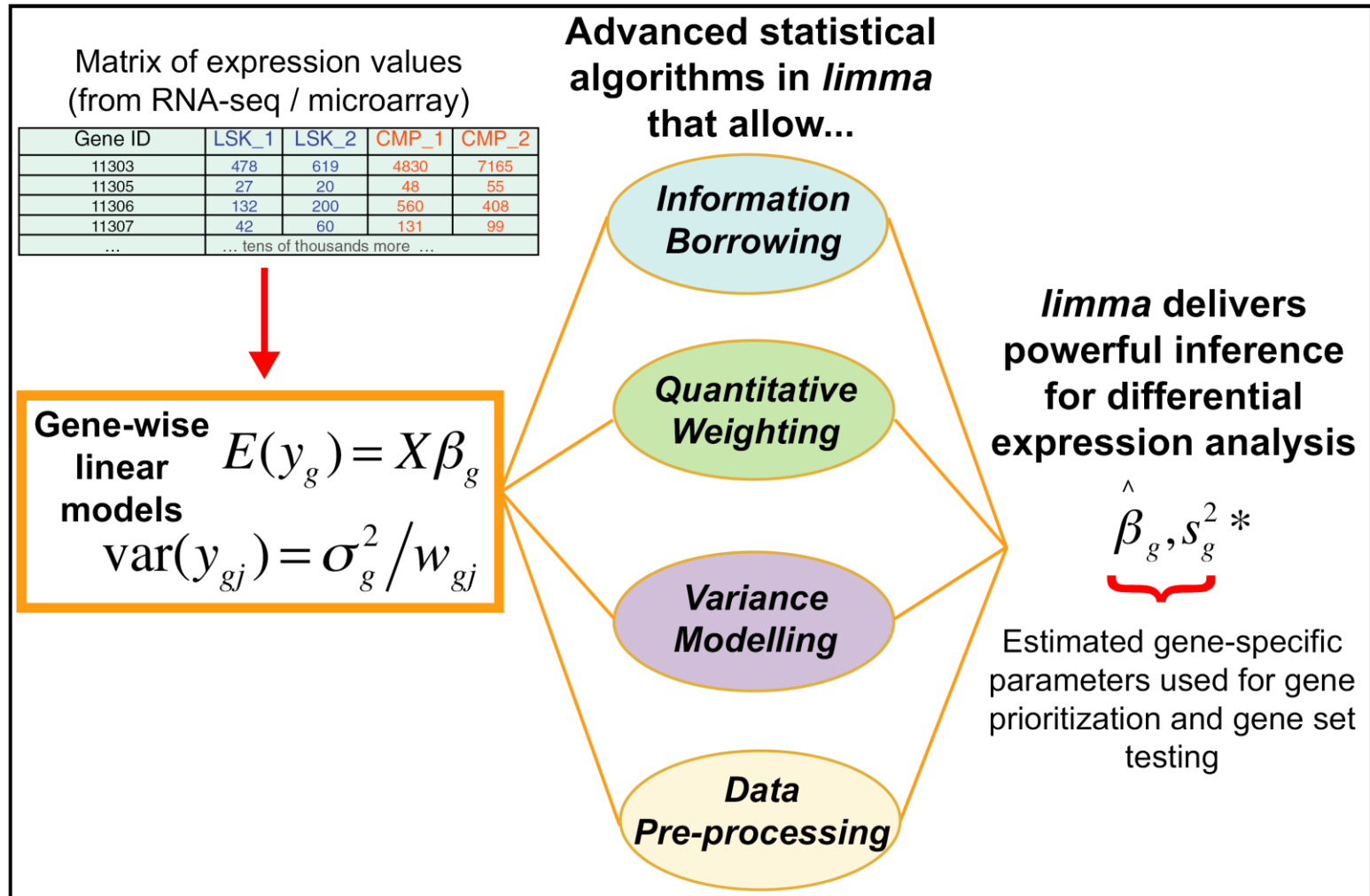
Gene variance vs. mean



Problem: 1 vs. 1 or 3 vs. 3 comparison: gene-wise t-test is not robust when using small sample size to estimate the variance of the mean

Bayesian methods: borrow variance information across genes

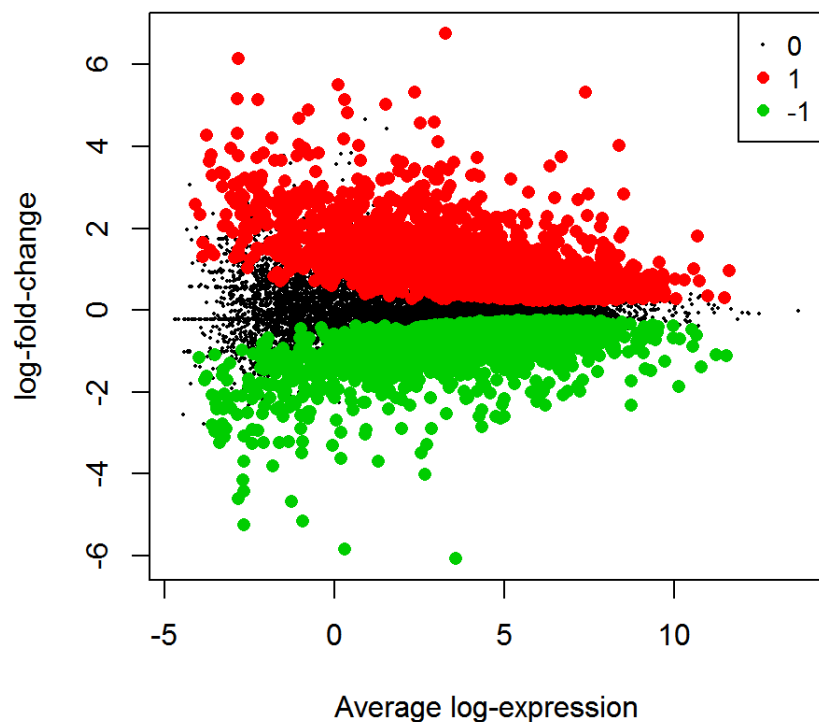
Limma analysis



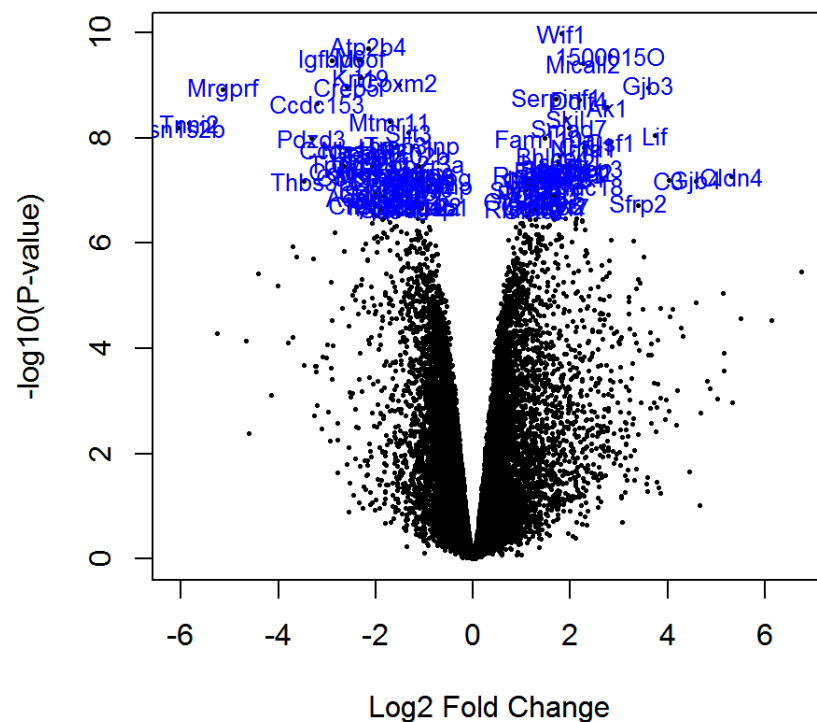
NAR 15 *limma* powers differential expression analyses
for RNA-sequencing and microarray studies

Differential expression plot

B.PregVsLac



M-A plot
(Min, Add)



火山图

分析结果

B.PregVsLacResults.csv - Excel										
文件 开始 插入 页面布局 公式 数据 审阅 视图 告诉我您想要做什么...										
A1										ENTREZID
	A	B	C	D	E	F	G	H	I	
1	ENTREZID	SYMBOL	GENENAME	logFC	AveExpr	t	P.Value	adj.P.Val	B	
2	24117	Wif1	Wnt inhibitory factor 1	1.819943	2.975545	20.1078	1.06E-10	1.02E-06	14.96977	
3	381290	Atp2b4	ATPase, Ca++ transpor	-2.14389	3.944066	-19.075	1.98E-10	1.02E-06	14.39556	
4	78896	15000150	RIKEN cDNA 15000150	2.807548	3.036519	18.54773	2.76E-10	1.02E-06	14.07416	
5	226101	Myof	myoferlin	-2.32974	6.223525	-18.2686	3.30E-10	1.02E-06	13.85802	
6	16012	Igfbp6	insulin-like growth fact	-2.89612	1.978449	-18.2152	3.41E-10	1.02E-06	13.46984	
7	231830	Micall2	MICAL-like 2	2.2534	4.760597	18.02627	3.86E-10	1.02E-06	13.676	
8	16669	Krt19	keratin 19	-2.31272	8.741892	-17.0794	7.26E-10	1.64E-06	13.17357	
9	55987	Cpxm2	carboxypeptidase X 2 (	-1.51547	2.834512	-16.6433	9.83E-10	1.72E-06	12.88782	
10	231991	Creb5	cAMP responsive elem	-2.5981	4.27593	-16.5363	1.06E-09	1.72E-06	12.6602	
11	14620	Gjb3	gap junction protein, b	3.600094	3.525281	16.46627	1.11E-09	1.72E-06	12.54071	
12	211577	Mrgprf	MAS-related GPR, men	-5.14627	-0.93683	-16.3657	1.20E-09	1.72E-06	10.44598	
13	20317	Serpinf1	serine (or cysteine) pep	1.71038	3.388835	15.7728	1.84E-09	2.36E-06	12.26586	
14	74747	Ddit4	DNA-damage-inducib	2.18037	6.864791	15.70145	1.94E-09	2.36E-06	12.22575	
15	270150	Ccdc153	coiled-coil domain cor	-3.21115	-1.34084	-15.5013	2.25E-09			
16	11636	Ak1	adenylate kinase 1	2.766745	4.303475	15.27694				
17	20482	Skil	SKI-like	1.887561	8.498925	14.6548				
18	194126	Mtmr11	myotubularin related p	-1.70897	2.508041	-14.6548				
19	21852	Tnni2	tropoin I skeletal fast	5.82789	0.202072					

生信思路：使用生物信息分析流程，容易很快产生结果。但需要注意质控、细节和积累分析经验，避免错误结果和解读

B.PregVsLacResults.csv

Gene set enrichment analysis

```
# Rerun goana with gene length information
go_length <- goana(fit.cont,coef="B.PregVsLac",species="Mm",covariate=gene_length)
topGO(go_length, n=10)
```

	Term Ont	N	Up	Down
GO:0003723	RNA binding	MF	1454	476 160
GO:1990904	ribonucleoprotein complex	CC	742	293 63
GO:0030529	intracellular ribonucleoprotein complex	CC	741	292 63
GO:0042254	ribosome biogenesis	BP	244	139 7
GO:0022613	ribonucleoprotein complex biogenesis	BP	361	178 24
GO:0022626	cytosolic ribosome	CC	107	80 1
GO:0006364	rRNA processing	BP	171	102 2
GO:0016072	rRNA metabolic process	BP	197	111 8
GO:0005840	ribosome	CC	212	107 5
GO:0022625	cytosolic large ribosomal subunit	CC	58	49 0

	P.Up	P.Down
GO:0003723	8.494584e-55	1
GO:1990904	8.278850e-54	1
GO:0030529	2.003347e-53	1
GO:0042254	1.651659e-48	1
GO:0022613	5.780765e-48	1
GO:0022626	4.702353e-44	1
GO:0006364	2.000100e-30	1

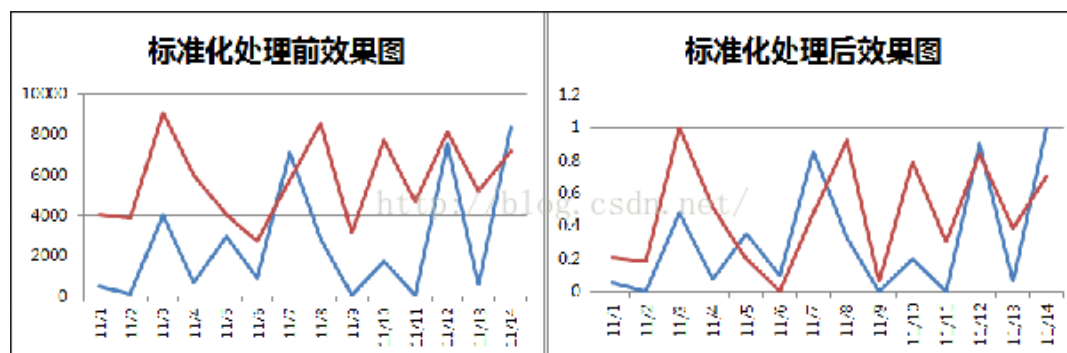
The N column represents the total number of genes that are annotated with each GO term. The Up and Down columns represent the number of differentially expressed genes that overlap with the genes in the GO term. The P.Up and P.Down columns contain the p-values for over-representation of the GO term across the set of up- and down-regulated genes, respectively. The output table is sorted by the minimum of P.Up and P.Down by default.

本节内容

- RNA-seq测序技术
- RNA-seq分析流程
- RNA-seq分析举例
- • **统计模型**：数据归一化
- **助教演示**：Linux系统、RNA-seq分析

归一化 (Normalization)

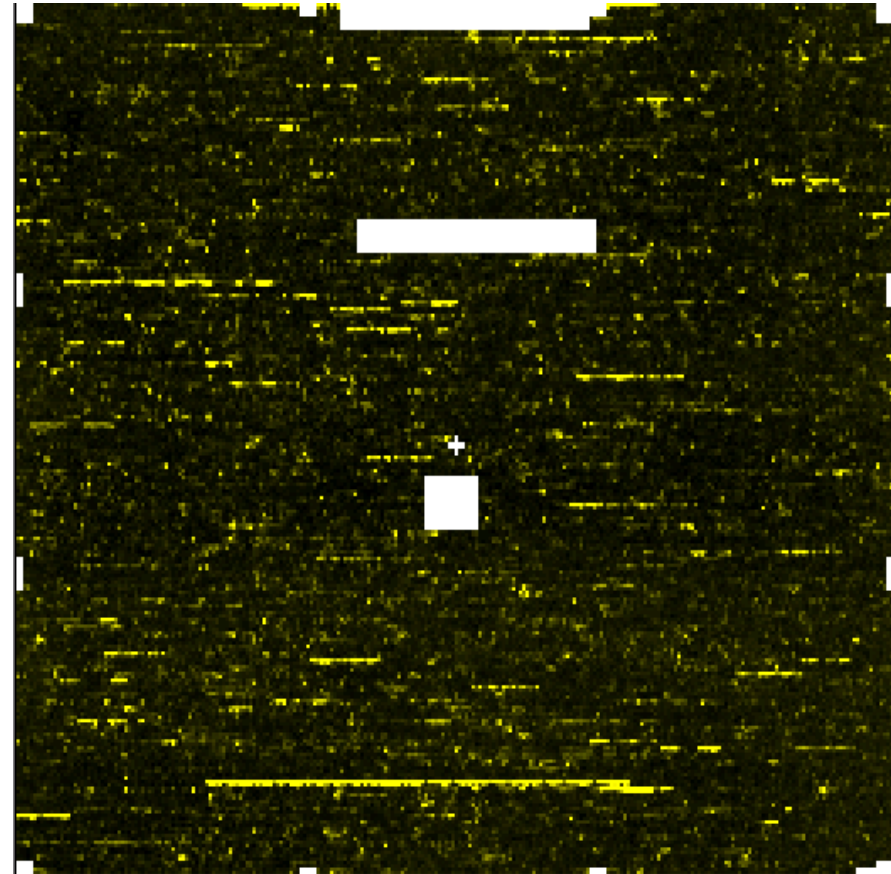
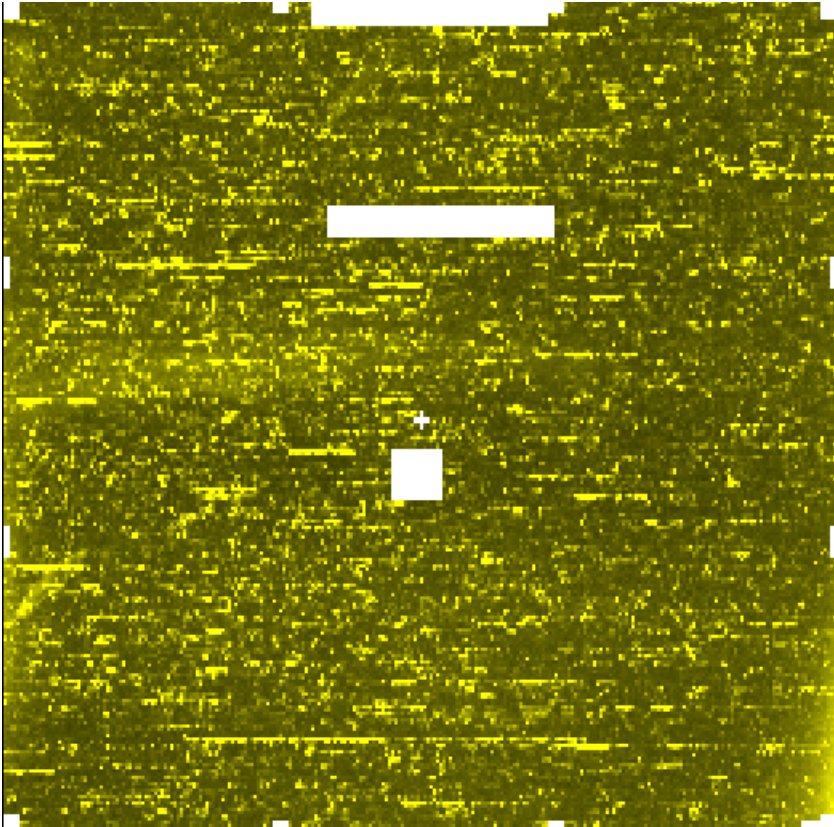
- 把数据变为 (0, 1) 之间的小数。主要是为了方便数据处理, 因为将数据映射到0~1范围之内, 可以使处理过程更加便捷、快速。
- 把有量纲表达式变换为无量纲表达式, 成为纯量。经过归一化处理的数据, 处于同一数量级, 可以消除指标之间的量纲和量纲单位的影响, 提高不同数据指标之间的可比性。
- 常用算法:
 - 线性转换, 即min-max归一化: $y = (x - \min) / (\max - \min)$
 - 对数函数转换: $y = \log_{10}(x)$



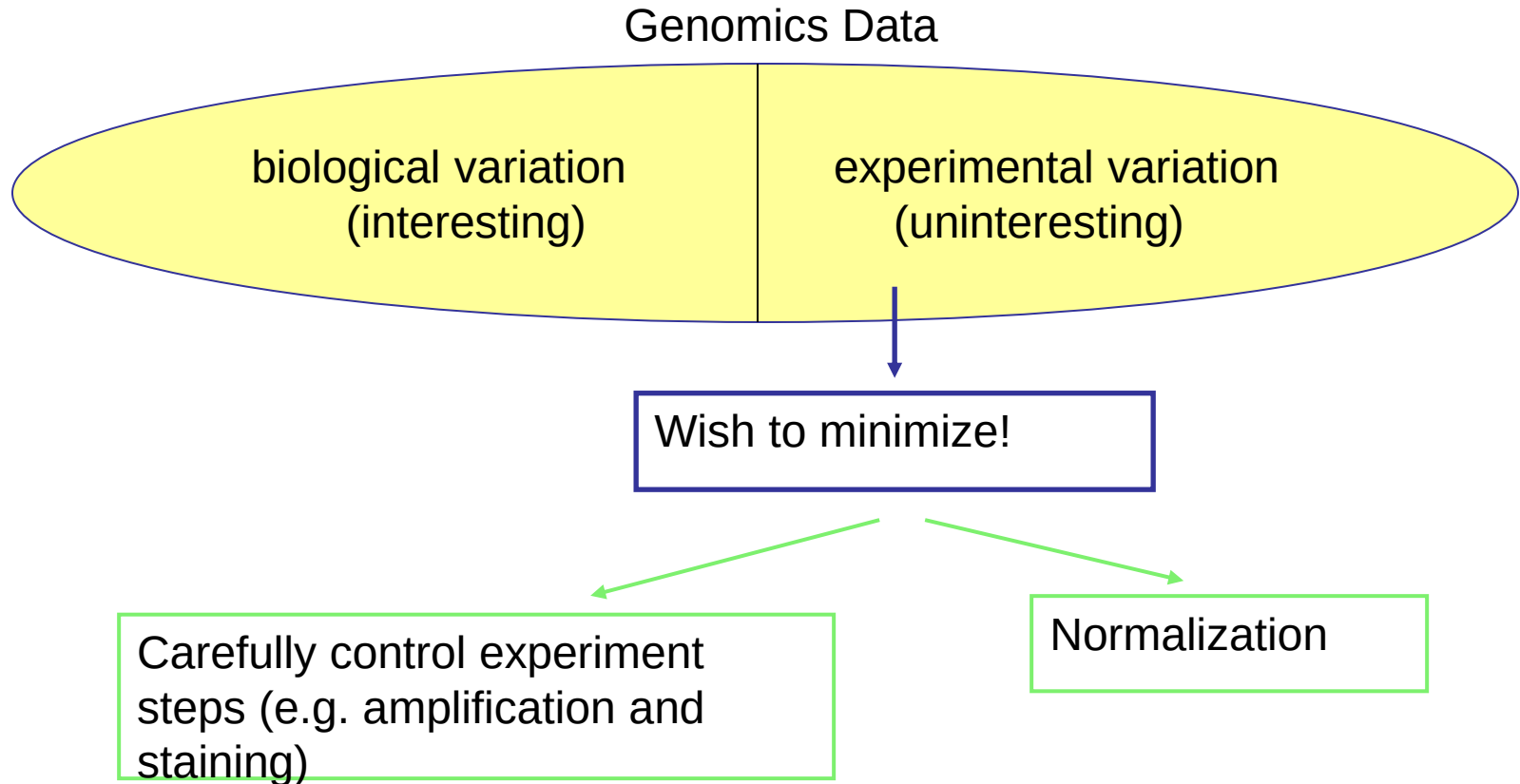
<https://blog.csdn.net/zyf89531/article/details/45922151>

<https://blog.csdn.net/pipisorry/article/details/52247379>

Normalization: Arrays may have different brightness



Normalization: Motivation



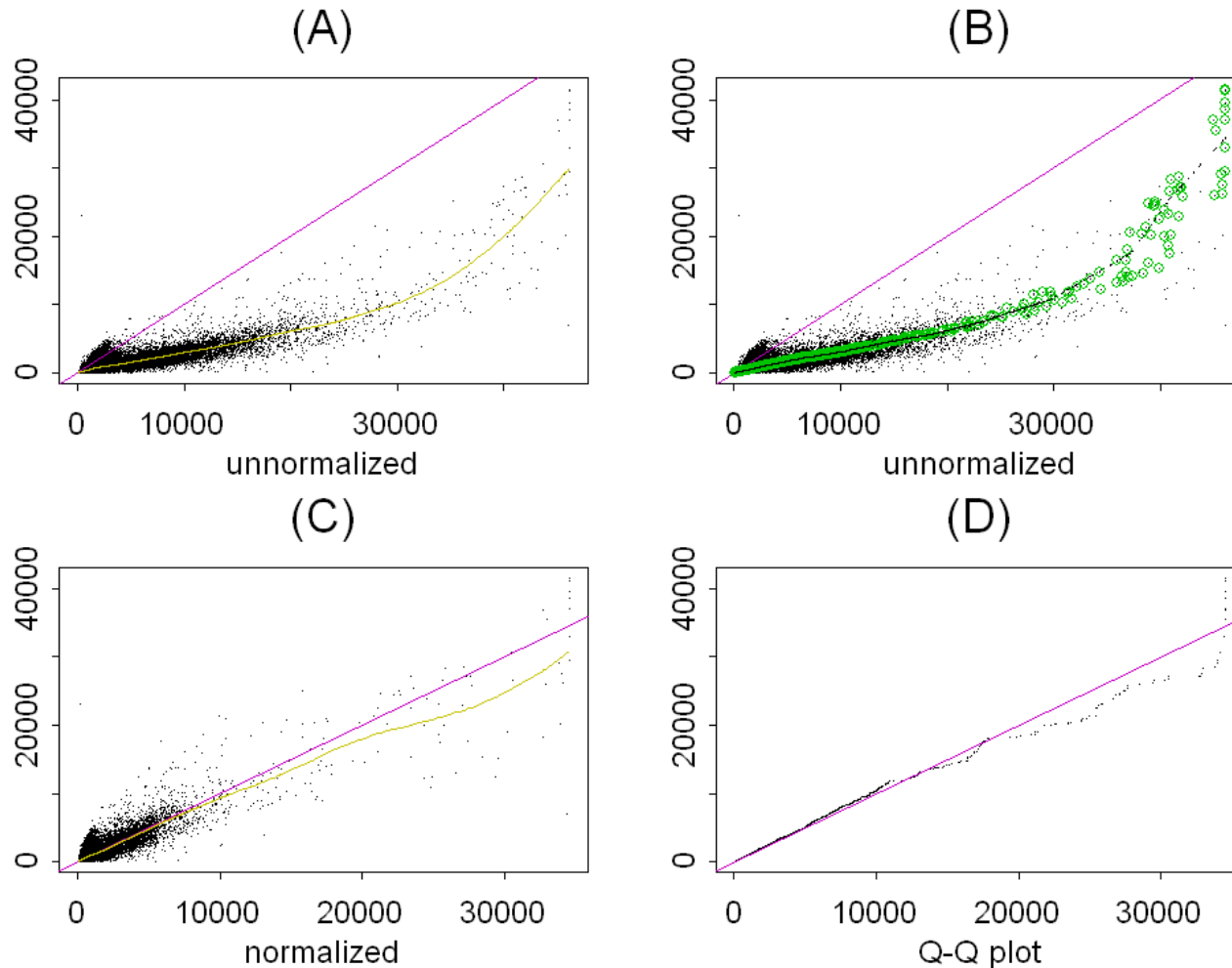
Normalization methods could have larger effect on analysis than the downstream steps such as group comparisons (Hoffmann et al. 2002 *Genome Biology*)

Normalization: Nonlinear methods, “Invariant Set”

Li and Wong 2001b, Tseng et al. 2001,
Schadt et al. 2001, Stuart et al. 2001

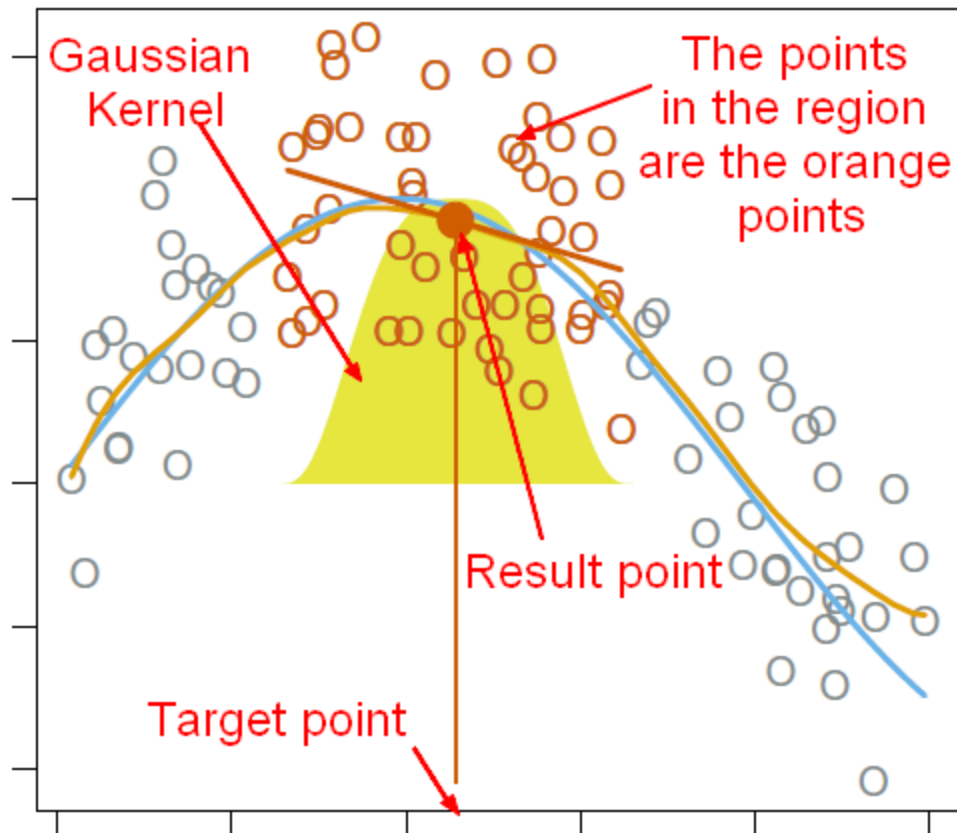
- It's not desired that normalization eliminates the real changes of differentially expressed genes
- Select a subset of probes or genes that are presumably non-differentially expressed as the basis for normalization
- Use absolute rank differences as iterative heuristics.
- Use nonlinear curve fitting to determine the normalization relation.

Normalization: Nonlinear methods, “Invariant Set”



Two different samples. The smoothing spline in (A) is affected by several points at the lower-right corner, which might belong to differentially expressed genes. Whereas the “invariant set” does not include these points when determining normalization curve, leading to a different normalization relationship at the high end.

LOWESS (locally weighted scatterplot smoothing)



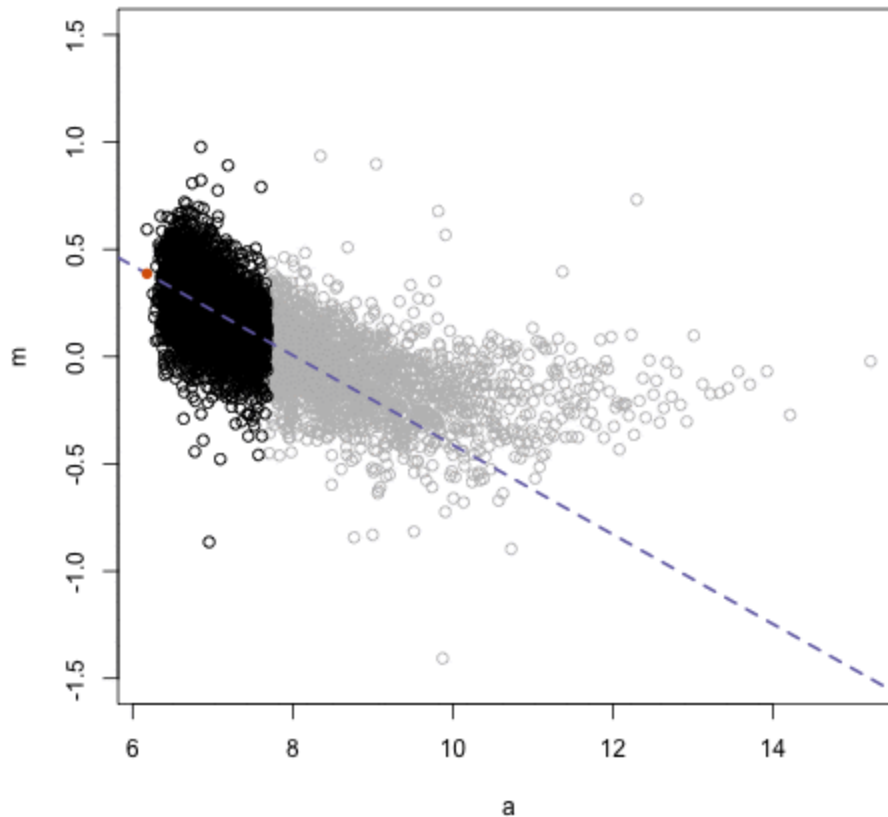
A linear function is fitted only on a local set of point delimited by a region. The polynomial is fitted using weighted least squares. The weights are given by the heights of the kernel (the weighting function).

We obtain then a fitted polynomial model but retains only the point of the model at the target point (x). The target point then moves away on the x axis and the procedure repeats and that traces out the orange curve.

https://gerardnico.com/data_mining/local_regression

R function **lowess()**: <https://www.rdocumentation.org/packages/stats/versions/3.6.1/topics/lowess>

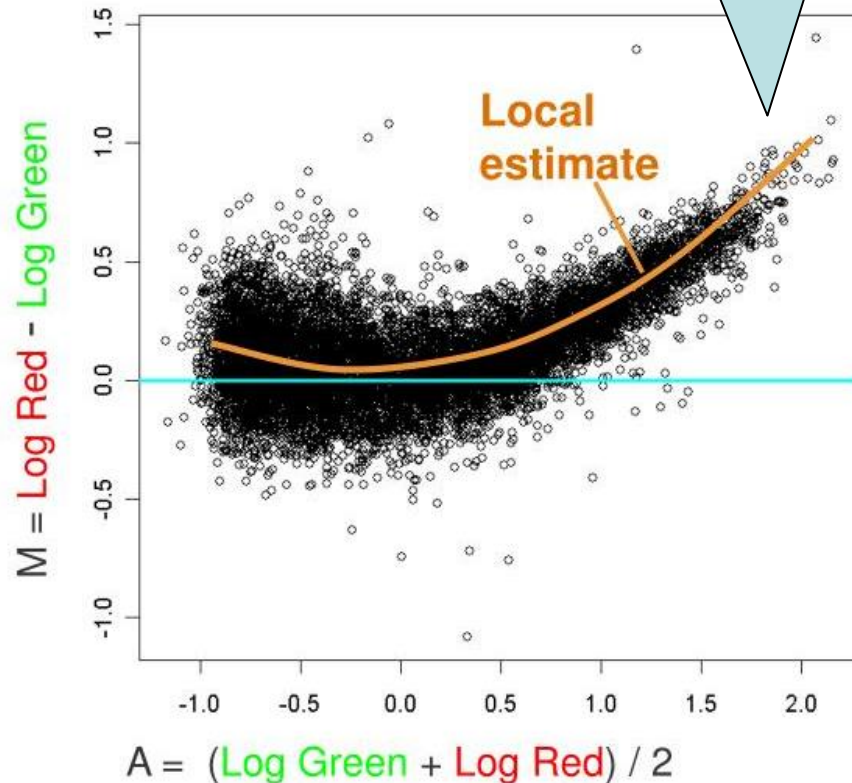
Loess explained in a GIF



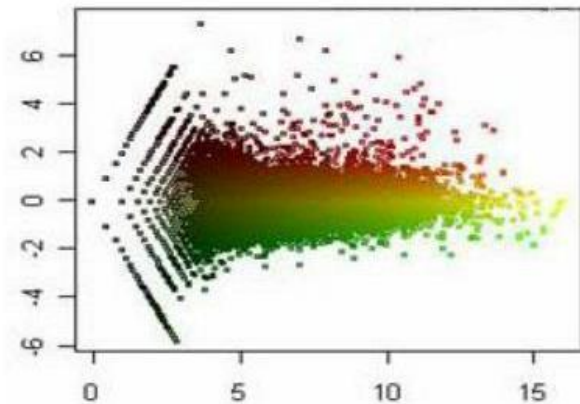
Rafael Irizarry: “Local regression (loess) is one of the statistical procedures I most use.”

Lowess normalization (M-A plot)

系统误差：表达量高的基因大都上调？



Use the estimate to bend
the banana straight



Quantile normalization

Raw data

2	4	4	5
5	14	4	7
4	8	6	9
3	8	5	8
3	9	3	5

在每个样本内部对
基因的表达量进行
排序
排序好的基因表达量，
就可以计算每一个基因
在所有样本的平均值。
得到平均值list

Order values
within each sample
(or column)

2	4	3	5
3	8	4	5
4	8	4	7
4	9	5	8
5	14	6	9

Average across rows
and substitute value
with average

3.5	3.5	3.5	3.5
5.0	5.0	5.0	5.0
5.5	5.5	5.5	5.5
6.5	6.5	6.5	6.5
8.5	8.5	8.5	8.5

把平均值按照
原来基因表达
量的排序情况
放回去即可。

这样所有的表
达量，就被归
一化成了平均
值啦！

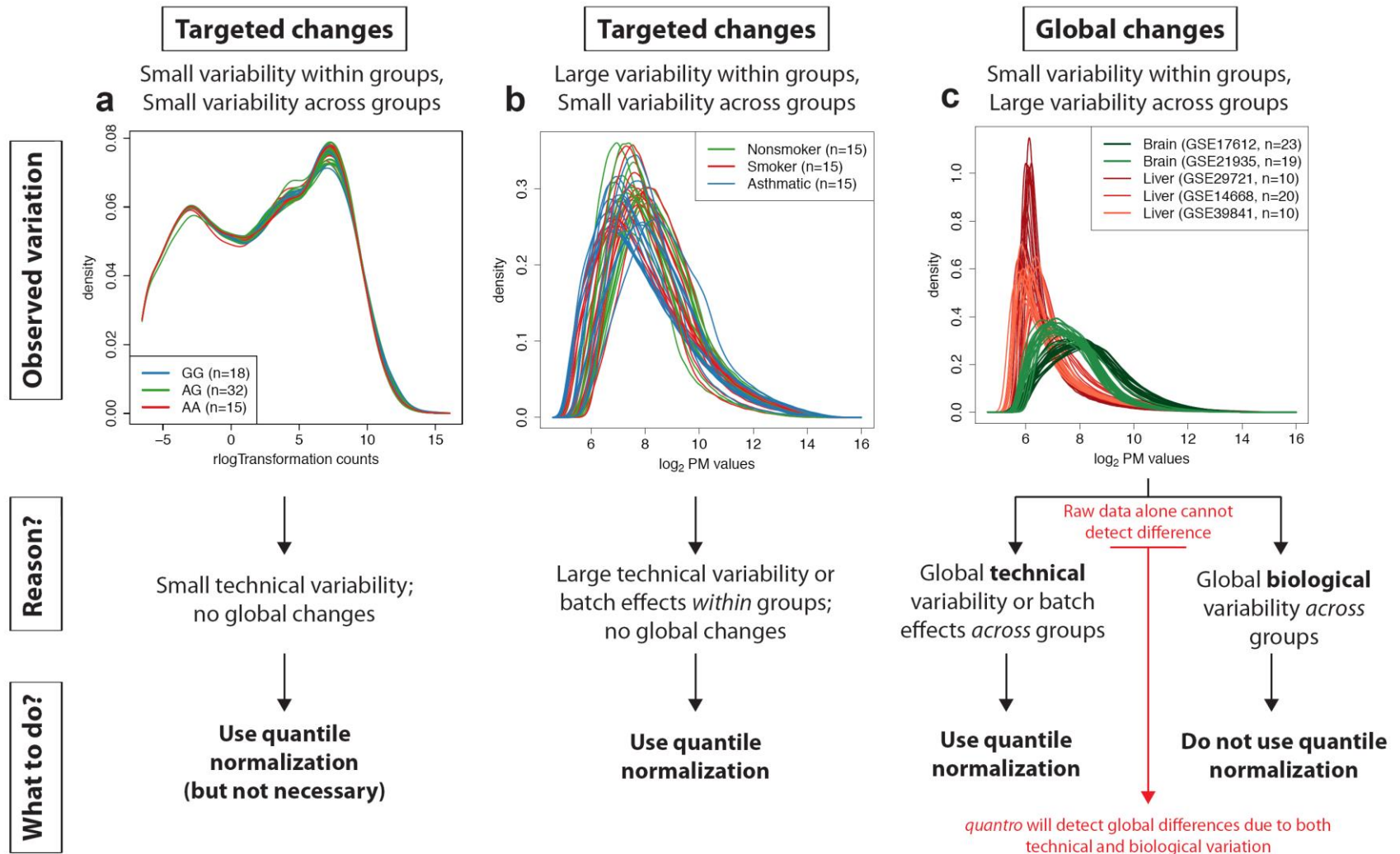
Re-order averaged
values in original
order

3.5	3.5	5.0	5.0
8.5	8.5	5.5	5.5
6.5	5.0	8.5	8.5
5.0	5.5	6.5	6.5
5.5	6.5	3.5	3.5

表达量矩阵的每一列是一个样本，每一行是一个基因（不同颜色），值是表达量

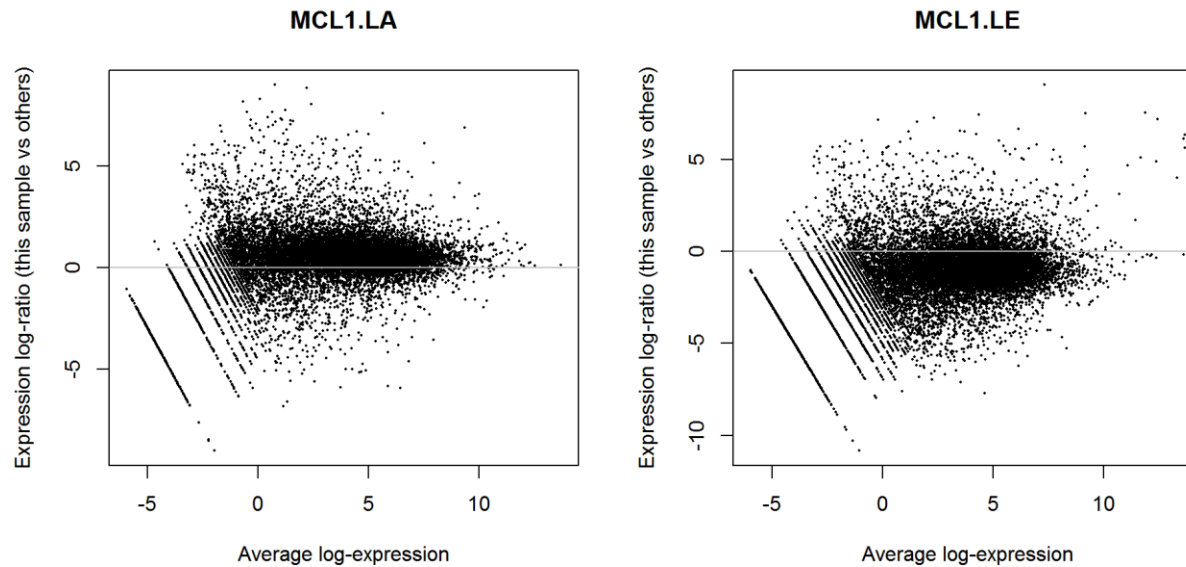
假设：每个样本中
所有基因的表达量
分布大致相同

When to use Quantile Normalization

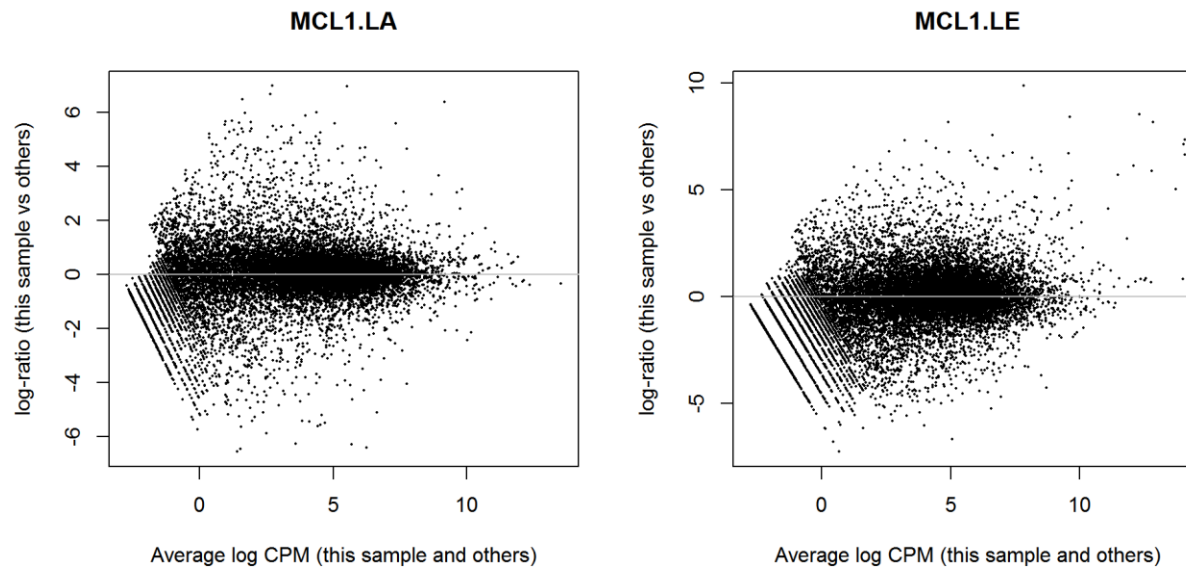


RNA-seq analysis with R: Normalization

Before
normaliation

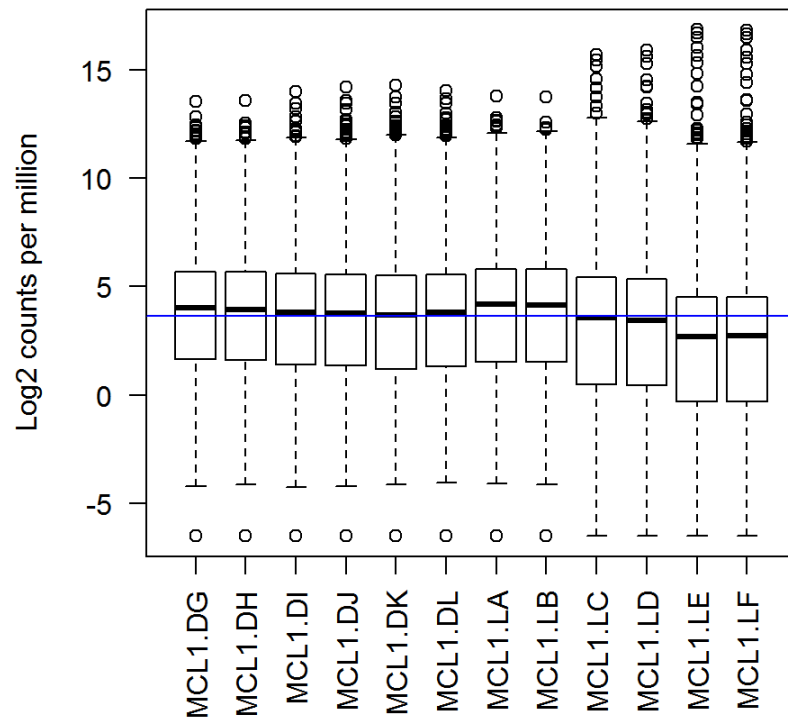


After
normaliation

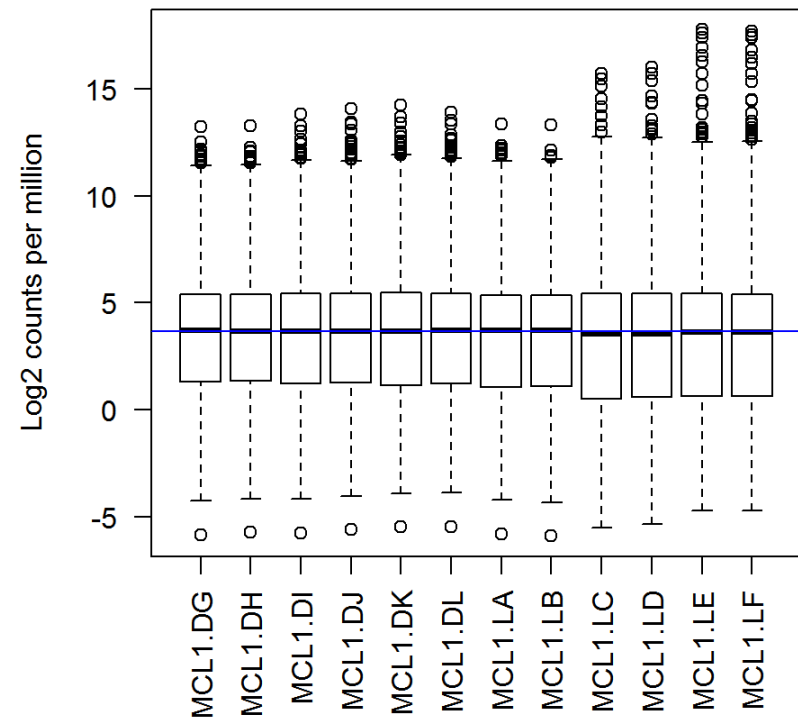


RNA-seq analysis with R: Normalization

Unnormalised logCPM

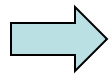


Voom transformed logCPM



本节内容

- RNA-seq测序技术
- RNA-seq分析流程
- RNA-seq分析举例
- 统计**模型**：数据归一化



- **助教演示**：Linux系统、RNA-seq分析

Linux 常用命令

1、ls命令

就是 list 的缩写，通过 ls 命令不仅可以查看 linux 文件夹包含的文件，而且可以查看文件权限(包括目录、文件夹、文件权限)□查看目录信息等等。

常用参数搭配：

`ls -a` 列出目录所有文件，包含以 . 开始的隐藏文件

`ls -A` 列出除 . 及 .. 的其它文件

`ls -r` 反序排列

`ls -t` 以文件修改时间排序

`ls -S` 以文件大小排序

`ls -h` 以易读大小显示

`ls -l` 除了文件名之外，还将文件的权限、所有者、文件大小等信息详细列出来



Linux中修改环境变量

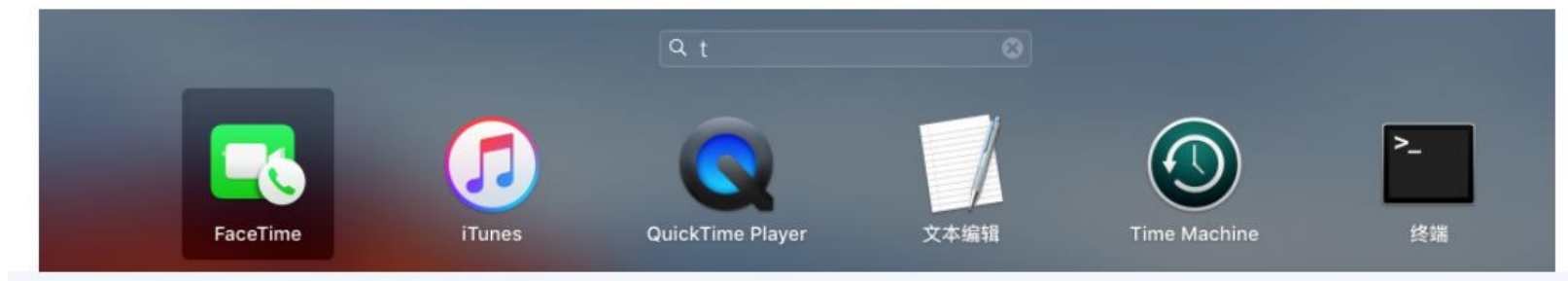
- 直接运行export命令定义变量【只对当前shell (BASH) 有效 (临时的)】
- 在shell的命令行下直接使用[export 变量名=变量值]定义变量，该变量只在当前的shell (BASH) 或其子shell (BASH) 下是有效的，shell关闭了，变量也就失效了，再打开新shell时就没有这个变量，需要使用的話还需要重新定义。
- 例如： export PATH=/usr/local/webserver/php/bin:\$PATH
- 环境变量的查看
 - (1)使用echo命令查看单个环境变量。例如：
echo \$PATH
 - (2)使用env查看所有环境变量。例如：
env echo
 - (3)使用set查看所有本地定义的环境变量。例如：
set

<https://blog.csdn.net/u011630575/article/details/49839893>

Mac下如何调出命令行窗口



调出的界面搜索terminal



常用命令: <https://www.jianshu.com/p/fdddcdf56e06>

云计算



通用型

CPU 1核
内存 1G
带宽 1Mbps
硬盘 50G云硬盘

```
tie2-2.4.2-linux-x86_64]# * Connection closed *
abished *
Oct 20 14:14:00 CST 2020 from 62.219.167.67 on ssh:notty
login attempts since the last successful login.
11:49:37 2020 from 111.230.154.177
ls
cd cheng
[root@VM-0-7-centos cheng]# ls
bowtie2-2.4.2-linux-x86_64 bowtie2-2.4.2-linux-x86_64.zip
[root@VM-0-7-centos cheng]# cd bowtie2-2.4.2-linux-x86_64/
[root@VM-0-7-centos bowtie2-2.4.2-linux-x86_64]# ls
AUTHORS bowtie2-build-s eg1.sam MANUAL
bowtie2 bowtie2-build-s-debug example MANUAL.markdown
bowtie2-align-l bowtie2-inspect lambda_virus.1.bt2 NEWS
bowtie2-align-l-debug bowtie2-inspect-l lambda_virus.2.bt2 README.md
bowtie2-align-s bowtie2-inspect-l-debug lambda_virus.3.bt2 scripts
bowtie2-align-s-debug bowtie2-inspect-s lambda_virus.4.bt2 TUTORIAL
bowtie2-build bowtie2-inspect-s-debug lambda_virus.rev.1.bt2
bowtie2-build-l BOWTIE2_VERSION lambda_virus.rev.2.bt2
bowtie2-build-l-debug doc LICENSE
[root@VM-0-7-centos bowtie2-2.4.2-linux-x86_64]#
```

腾讯云试用: <https://cloud.tencent.com/act/free>

腾讯云的“Linux基础入门”在线实验课

三 菜单 | 腾讯云

00:43:42

腾讯云大学 | 在线课堂 | 学习路径 | 培训认证 | 高校合作 | 企业培训 | 人才培养 | 动手实验室

文件浏览器

bin

boot

data

dev

etc

home

lib

lib64

lost+found

media

mnt

opt

proc

Linux 基础入门

1. 目录操作

创建目录

使用 mkdir 命令创建目录

```
mkdir $HOME/testFolder
```

下一步

切换目录

移动目录

删除目录

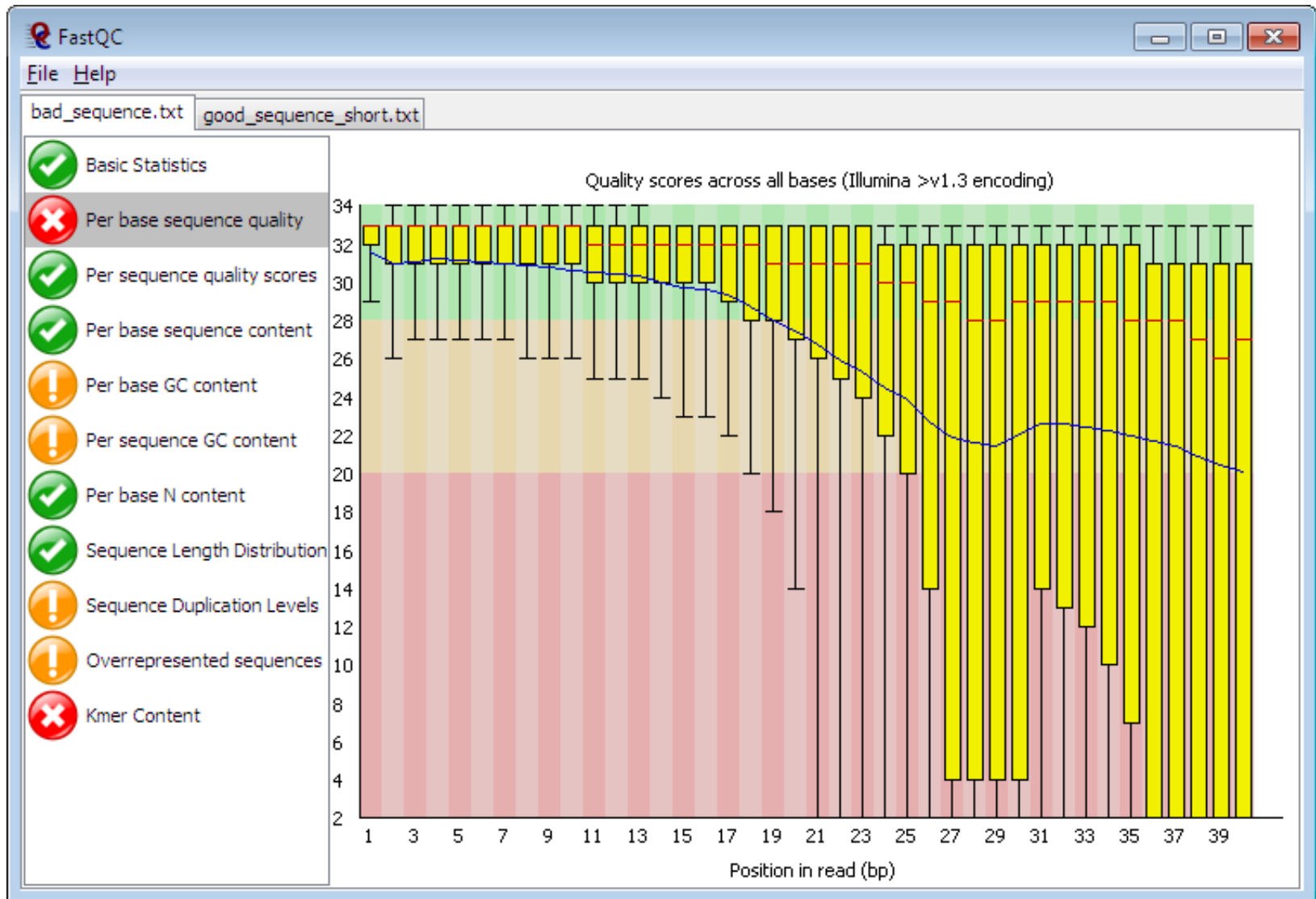
查看目录下的文件

终端

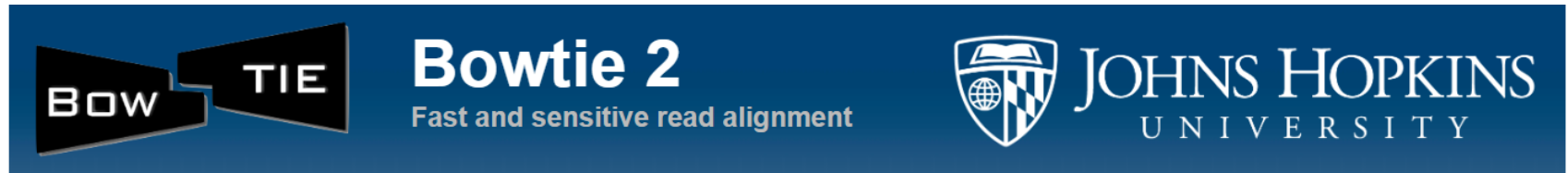
```
[root@VM-180-41-centos ~]# ls
[root@VM-180-41-centos ~]# mkdir $HOME/test
[root@VM-180-41-centos ~]# ls
test
[root@VM-180-41-centos ~]#
```

<https://cloud.tencent.com/developer/labs/gallery>

测序数据质控：FastQC



序列比对软件: Bowtie2



Introduction

[How is Bowtie 2 different from Bowtie 1?](#)

[What isn't Bowtie 2?](#)

Obtaining Bowtie 2

[Building from source](#)

[Adding to PATH](#)

[The bowtie2 aligner](#)

[bowtie2-2.4.2-sra-linux-x86_64.zip](#)

[s local alignment
mple](#)

[bowtie2-2.4.2-linux-aarch64.zip](#)

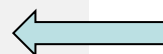
[ar
re example](#)

[bowtie2-2.4.2-source.zip](#)

[mple
exceed the minimum](#)

[bowtie2-2.4.2-macos-x86_64.zip](#)

[bowtie2-2.4.2-linux-x86_64.zip](#)



`uname -a` 命令可以直接显示 Linux 系统架构的命令

Site Map

[Home](#)

[News archive](#)

[Manual](#)

[Getting started](#)

[Frequently Asked Questions](#)

[Tools that use Bowtie](#)

Latest Release

install with [bioconda](#)

[Bowtie2 v2.4.2](#)

10/05/20

Please cite: Langmead B, Salzberg S. Fast gapped-read alignment with Bowtie 2. *Nature Methods*. 2012, 9:357-359.

在本地和云服务器之间传输文件

bowtie2-2.4.2-linux-x86_64 - root@188.131.143.175 - WinSCP

本地(L) 标记(M) 文件(F) 命令(C) 会话(S) 选项(O) 远程(R) 帮助(H)

同步 传输选项 默认

root@188.131.143.175 × 新建会话

C:\Windows 查找文件

上传 编辑 属性 新建

下载 编辑 属性 新建

C:\方程\课程\A04-RNA测序\数据\bowtie2-2.4.2-linux-x86_64\

名字	大小	类型	已改变
..		上级目录	2020.10.19 下午 11:11
example		文件夹	2020.10.06 上午 11:11
scripts		文件夹	2020.10.06 上午 11:11
doc		文件夹	2020.10.06 上午 11:11
eg1.sam	3,282 KB	SAM 文件	2020.10.19 下午 11:11
lambda_virus.rev.2.bt2	12 KB	BT2 文件	2020.10.19 下午 11:11
lambda_virus.rev.1.bt2	4,113 KB	BT2 文件	2020.10.19 下午 11:11
lambda_virus.2.bt2	12 KB	BT2 文件	2020.10.19 下午 11:11
lambda_virus.1.bt2	4,113 KB	BT2 文件	2020.10.19 下午 11:11
lambda_virus.4.bt2	12 KB	BT2 文件	2020.10.19 下午 11:11
lambda_virus.3.bt2	1 KB	BT2 文件	2020.10.19 下午 11:11
bowtie2-align-l-deb...	32,149 KB	文件	2020.10.06 上午 11:11
bowtie2-align-s-deb...	32,157 KB	文件	2020.10.06 上午 11:11

/root/cheng/bowtie2-2.4.2-linux-x86_64/

名字	大小	已改变	权限
..		2020.10.20 下午 12:31:38	rw-r--r--
example		2020.10.06 上午 11:19:40	rw-r--r--
scripts		2020.10.06 上午 11:19:39	rw-r--r--
doc		2020.10.06 上午 11:19:39	rw-r--r--
eg1.sam	3,282 KB	2020.10.20 下午 12:48:33	rw-r--r--
lambda_virus.rev.2.bt2	12 KB	2020.10.19 下午 11:24:22	rw-r--r--
lambda_virus.rev.1.bt2	4,113 KB	2020.10.19 下午 11:24:22	rw-r--r--
lambda_virus.4.bt2	12 KB	2020.10.19 下午 11:24:21	rw-r--r--
lambda_virus.3.bt2	1 KB	2020.10.19 下午 11:24:21	rw-r--r--
lambda_virus.2.bt2	12 KB	2020.10.19 下午 11:24:21	rw-r--r--
lambda_virus.1.bt2	4,113 KB	2020.10.19 下午 11:24:21	rw-r--r--
bowtie2-align-l-deb...	32,149 KB	2020.10.06 上午 11:19:20	rw-r--r--
bowtie2-align-s-deb...	32,157 KB	2020.10.06 上午 11:19:15	rw-r--r--

<https://cloud.tencent.com/document/product/213/2131>

Bowtie 2: Lambda phage example

Indexing a reference genome

To create an index for the [Lambda phage](#) reference genome included with Bowtie 2, create a new temporary directory (it doesn't matter where), change into that directory, and run:

```
$BT2_HOME/bowtie2-build $BT2_HOME/example/reference/lambda_virus.fa lambda_virus
```

Aligning example reads

Stay in the directory created in the previous step, which now contains the `lambda_virus` index files. Next, run:

```
$BT2_HOME/bowtie2 -x lambda_virus -U $BT2_HOME/example/reads/reads_1.fq -S eg1.sam
```

Using SAMtools/BCFtools downstream

Use `samtools view` to convert the SAM file into a BAM file. BAM is the binary format corresponding to the SAM text format. Run:

```
samtools view -bS eg2.sam > eg2.bam
```

Use `samtools sort` to convert the BAM file to a sorted BAM file.

```
samtools sort eg2.bam -o eg2.sorted.bam
```

<http://bowtie-bio.sourceforge.net/bowtie2/manual.shtml#getting-started-with-bowtie-2-lambda-phage-example>

Galaxy生信分析网站

The screenshot displays the Galaxy web interface with a dark blue header. The 'Galaxy' logo is on the left, and a 'Using 0%' status bar is on the right. The main content area is divided into three panels. The left panel, titled 'Tools', lists various bioinformatics tools with descriptions: 'samtools mpileup' (multi-way pileup of variants), 'Samtools stats' (generate statistics for BAM dataset), 'Samtools depth' (compute the depth at each position or region), 'Samtools sort' (order of storing aligned sequences), 'Samtools merge' (merge multiple sorted alignment files), 'Samtools fixmate' (fill mate coordinates, ISIZE and mate related flags), and 'BedCov' (calculate read depth for a set of genomic intervals). The middle panel shows the configuration for the 'Samtools sort' tool. It includes a 'Versions' button, an 'Options' dropdown, and a 'Favorite' button. The 'BAM File' section has a file input field containing '1: eg2.bam'. The 'Primary sort key' is set to 'coordinate'. The 'Job Resource Parameters' are set to 'Use default job resource parameters'. The 'Email notification' section has 'Yes' and 'No' buttons. The right panel, titled '历史' (History), shows a search bar for datasets. Below it, the 'Unnamed history' section displays a list of jobs. The first job, '4: Samtools sort on data 1', is highlighted in green. It shows a size of 2.5 MB, a format of 'bam', and a database of '?'. The job description includes '[bam_sort_core] merging from 0 files and 5 in-memory blocks...'. Below the description are links to 'display with IGV local', 'display in IGB View', and 'display at bam.iobio bam.iobio.io'. The job type is listed as 'Binary bam alignments file'.

Galaxy Using 0%

Tools ☆ ↑

samtools ✕

Show Sections

samtools mpileup multi-way pileup of variants

Samtools stats generate statistics for BAM dataset

Samtools depth compute the depth at each position or region

Samtools sort order of storing aligned sequences

Samtools merge merge multiple sorted alignment files

Samtools fixmate fill mate coordinates, ISIZE and mate related flags

BedCov calculate read depth for a set of genomic intervals

Samtools Versions Options Favorite

sort order of storing aligned sequences (Galaxy Version 2.0.3)

BAM File

1: eg2.bam

Primary sort key

coordinate

Job Resource Parameters

Use default job resource parameters

Email notification

Yes No

历史 ↺ + 📄 ⚙️

搜索数据集 ? ✕

Unnamed history

4 shown

5.38 MB

4: Samtools sort on data 1 👁️ ✎️ ✕

2.5 MB

格式: bam, 数据库: ?

[bam_sort_core] merging from 0 files and 5 in-memory blocks...

📄 ⓘ ↺ 📊 ? 📁 💬

display with IGV local

display in IGB View

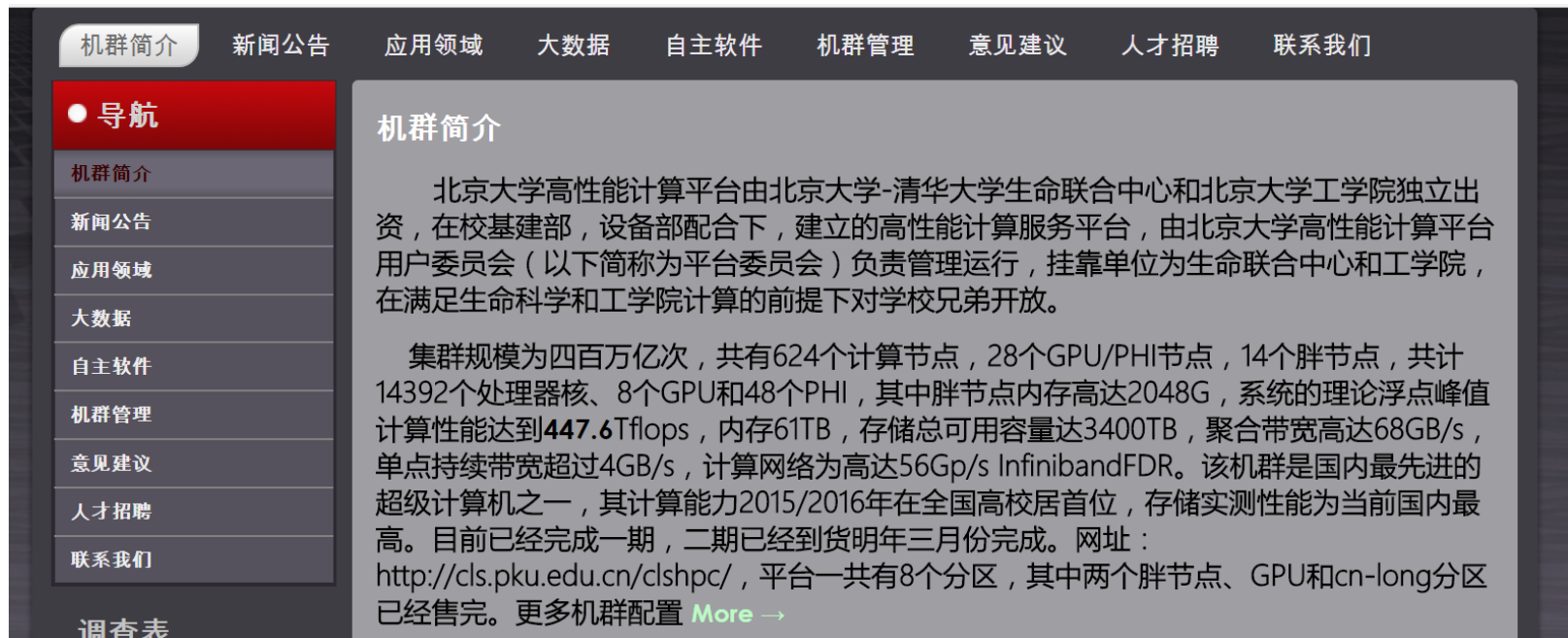
display at bam.iobio bam.iobio.io

Binary bam alignments file

<https://usegalaxy.org/>

北京大学“北极星”高性能计算平台

<http://www.aais.pku.edu.cn/clshpc/>



陈芳进、裴剑锋老师《定量生物学计算与软件使用》课程

- 交叉学院/秋季/3学分，研究生课程号08402070
- 陈芳进老师，2020年9月：“定量生物学计算与软件使用”课程是一门理论与实践结合的课程。课程讲授的主要内容为Matlab基础和使用、Linux基础和使用、Python基础和使用、生物网络分析、机器学习、深度学习、以及其他定量生物学计算方法和资源等。其中linux基础和使用包括Linux shell，文件传输，GPU加速，任务并行，北极星集群使用和任务提交。上课地点：老化学楼东配楼101教室，时间为：星期二7-9节课，9月22日开始。需要设备：自带笔记本。

课后练习

1. 阅读文献

《NCB15 EGF-mediated induction of Mcl-1 at the switch to lactation is essential for alveolar cell survival》

思考作者为什么做RNA-seq实验（图6），它的数据分析结果对课题的推进和结论有哪些帮助？

2. 数据分析练习

a) 安装和运行“演示”部分的软件和流程

遇到安装、运行错误，可以先上网搜索，但如果花费超过15分钟尚未解决，及时在微信群提问

b) 结合课堂内容和《RNAseq-R教程》（<http://combine-australia.github.io/RNAseq-R/>），运行课件中的R代码，理解所用函数的功能和输出结果。比较得到的结果和（1）中文献分析结果的异同。

延伸阅读

- 用R做RNA-seq分析的其他教程
 - <https://bioinformatics-core-shared-training.github.io/RNAseq-R/>
 - <http://bioconductor.org/help/workflows/rnaseqGene/>
- 选择课件目录《文献阅读》中感兴趣的文献阅读。